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Real-Time Product Structural Validation for Fused Filament Fabrication

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Real-time product structural validation for fused filament fabrication

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Dedication

I would like to dedicate this work with gratitude to the University of South Carolina.

Thank you.

Acknowledgments

Time flies, and in the blink of an eye, four years have passed, and my academic journey as a Ph.D. student is ending. Looking back on the journey, I still remember the initial trepidation and anticipation when I first stepped into the laboratory, the joy and happiness when my first paper was published, and the moments at school and in the laboratory. I know this page is insufficient to express my sincere appreciation to my family, advisor, colleagues, and friends, who have shown care, provided help, and offered guidance throughout the years of my studying here.

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Abstract

Fused filament fabrication (FFF) is one of the fastest-growing, most promising, and widely-used Additive manufacturing (AM) technologies, as its capabilities of essentially reducing material waste, feasible and flexible designing complicated structure or geometry product. Except for the advantages mentioned above, the unique affordability has made FFF a popular technique for small-scale 3D printing. However, due to the process's intrinsic uncertainties, its widescale adaption for industrial applications still faces significant challenges, especially for Industry 4.0. Even a trivial bonding gap or material void dramatically influences the final product's structural quality and mechanical properties during the printing process. Therefore, product structural quality validation is essential for the FFF printing process to guarantee consistent quality and mechanical properties.

Extensive defect detection systems utilizing machine learning (ML) for FFF printing have been built to detect faults, including bonding quality problems and infill defects. However, the ultimate goal of defect detection is to guarantee product quality rather than find the defect. In addition, defects' effects on functional product quality are often more critical than easy-to-spot surface defects. Therefore, effective online defect detection integrated with product structure quality validation for FFF is needed. The proposed system should not only detect visible defects but also infer the defect's impact on functional qualities caused by variations in the printing process.

Finite Element Analysis (FEA) can help identify potential weaknesses, stress concentrations, or areas with defects that may require reinforcement, allowing manufacturers to optimize the product's performance and mitigate potential risks associated

with structural failures. Substantial research focused on the numerical simulation with the FEA method to evaluate the AM product structural quality. The existing research studies in FEA simulation focus on good quality specimen quality evaluation but do not consider defects in the actual printing process. However, the manufacturing process is inherently complex and prone to printing deficiencies. Obtaining stochastic defects in the samples and updating the relevant defect information in the FEA model poses significant challenges. To address the limitation, a comprehensive and innovative approach is required for product structural quality validation. Such an approach should automatically identify observable defects, incorporate them into the FEA model, and assess their individual and combined effects on the product's functional properties to ensure end-use quality.

This dissertation proposes two different real-time structural validation approaches for FFF capable of inferring structural quality: (1) a real-time machine learning-based defect detection and evaluation system and (2) a real-time finite element analysis system for structural validation. The proposed online real-time structural validation systems for FFF can provide a reliable and efficient way to guarantee product quality during manufacturing. This work advances the state-of-the-art real-time structural validation for FFF while providing new insights into defect detection, FEA, and structural validation for the other AM technologies.

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Chapter 1

Introduction

1.1 Background and motivation

Additive manufacturing (AM) capabilities of vastly reducing material waste, feasible and flexible designing complicated structure or geometry, have attracted extensive attention from industry, academics, and even hobbyists. According to the Wohlers Report 2021, despite impacting by the global Covid-19 pandemic, the 2020 AM industry still has a 7.5% growth rate, which is nearly $ 12.8 billion. Fused filament fabrication (FFF) is one of the fastest-growing, most promising, and widely-used AM technologies. Except for AM's advantages, the unique affordability has made FFF a popular technique for small-scale 3D printing. However, its wide-scale adaption for industrial applications still faces significant challenges, especially for Industry 4.0, due to the process's intrinsic uncertainties. Typical defects, such as corner warpage, bonding gap, layer crack, infill defect, and geometry error as shown in Fig. 1.1, can arise during the 3D printing process. Some small defects, such as the bonding gap or misprinting hole, in a sensitive position can dramatically influence the final product's structural quality and mechanical properties during the printing process. Therefore, a product structural quality validation system for the FFF printing process to reduce part-to-part variations and assure product mechanical properties is essential.

Extensive defect detection systems utilizing machine learning (ML) for the FFF printing process have been built to detect various faults, including bonding quality problems and infill defects. For example, Jin et al. utilized an optical cam-

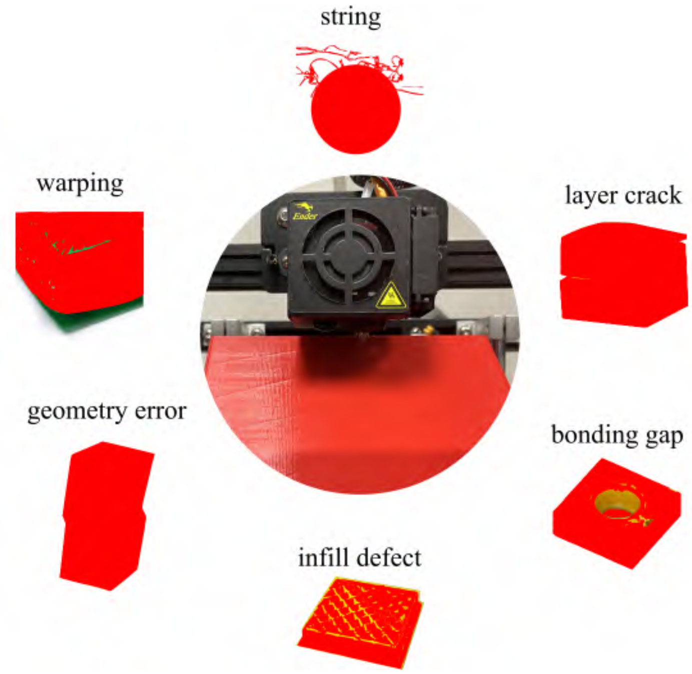

Figure 1.1: The typical defects in the fused filament fabrication process.

era integrated with a material extrusion printer to capture diverse product quality images representing conditions such as under-extrusion, optimal quality, and over-extrusion [139]. These captured images formed the basis for constructing a dataset, which was then employed to train a ResNet 50 architecture convolutional neural network classification algorithm. They implemented real-time monitoring by feeding these images into the pre-trained classification model to assess the current printing condition. Banadaki et al. established a robust online quality monitoring system employing a deep convolutional neural network algorithm [17]. Following the training of the model with images acquired at various stages, the proposed online model can accurately classify five distinct quality grades stemming from overfilled or underfilled conditions, achieving an average accuracy of 94%. Wu et al. introduced an approach to detect malicious infill defects introduced into the 3D printing process using a classification-based ML method for analyzing images captured by a camera [325]. The achieved accuracies for infill defect prediction were 85.26% with a naive Bayes classifier and 95.51% with a J48 decision tree, respectively. Holzmond et al. presented the FFF product quality assurance system, employing a three-dimensional digital image correlation system to capture the geometry and subsequently comparing it with the computer model to achieve real-time detection of printing errors [121]. A validation experiment was conducted on a FFF printer, demonstrating the in situ detection of both localized and global defects. Hossain et al. investigated in situ infrared temperature sensing for real-time defect detection in fused deposition modeling [160]. They applied a statistical method to detect temperature deviations caused by deliberately introduced defects. Khan et al. employed a convolutional neural network algorithm to detect real-time malicious defects in the printing process, focusing on feature extraction of geometrical anomalies occurring in infill patterns [143]. Charalampous et al. introduced a vision-based real-time monitoring system for FFF, designed for automatic manufacturing error detection [51]. This method compared high-resolution point cloud data of the printed part with the corresponding digital 3D model at various phases of the printing process to automatically monitor and detect defects. The effectiveness of the proposed methodology was validated through several cases. However, the ultimate goal for defect detection is to guarantee product quality rather than find the defect. In addition, defects' effects on functional product quality are often more critical than easy-to-spot surface defects. Therefore, effective online defect detection integrated with product structure quality validation for FFF is needed. The proposed system should detect not only visible defects but also diagnose defects in functional product qualities (e.g., structural performance) caused by variations in the printing process (e.g., abnormal extrusion).

Finite Element Analysis (FEA) can help identify potential weaknesses, stress concentrations, or areas with defects that may require reinforcement, allowing manufacturers to optimize the product's performance and mitigate potential risks associated with structural failures. Substantial research focused on the numerical simulation with the FEA method to evaluate the AM product structural quality. For example, Rahman et al. proposed an FEA method to analyze the behavior of continuous carbon fiber-reinforced AM [236]. By examining four different configurations of fiber reinforcement volumes, they demonstrated that the simulation method reliably predicts the behavior of end-use components, particularly when dealing with fiber volume fractions of approximately 11%. Cao et al. developed an advanced FEA model that integrates the equivalent boundary condition method and dynamic mesh method to accelerate thermo-mechanical simulations in laser powder bed fusion printing [44]. Their model accurately predicts the thermo-mechanical behavior of multiple powder bed fusion deposits across various process parameters. The results exhibited excellent agreement with experimental data, particularly regarding in-situ temperature and residual stress. Yang et al. introduced a three-dimensional thermo-elastic-plastic model employing the FEA method to accurately forecast the thermo-mechanical response during the laser-engineered net shaping process of Ti-6Al-4V [336]. This model effectively replicates computed thermal history and mechanical deformations, demonstrating strong concurrence with experimental measurements. To enhance the simulation capabilities of FEA, Wang et al. introduced an innovative Gaussian process-constrained general path model [308]. This model aims to estimate high-fidelity FEA simulation results by utilizing low-fidelity results voxel-by-voxel, thereby improving the accuracy of the simulations. The existing research studies in FEA simulation focus on good quality specimen quality evaluation but do not consider defects in the actual printing process. However, the manufacturing process is inherently complex and prone to printing deficiencies. Obtaining stochastic defects in the samples and updating the relevant defect information in the FEA model poses significant challenges. To address the limitation, a comprehensive and innovative approach is required for product structural quality validation. Such an approach should automatically identify observable defects, incorporate them into the FEA model, and assess their individual and combined effects on the product's functional properties to ensure end-use quality.

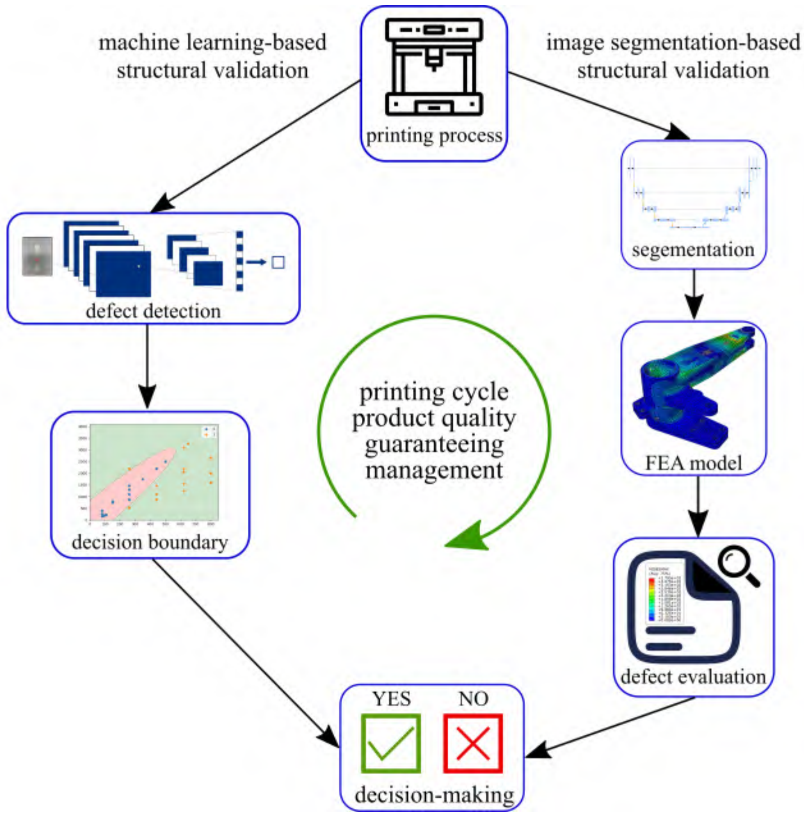

Figure 1.2: The research object and proposed approaches for real-time FFF product structural quality validation.

Ensure end-use product quality with structural validation, which signifies the manufacturer's commitment to meeting rigorous quality standards while also subjecting the product to a comprehensive validation process to guarantee its structural integrity and performance. This guarantee is particularly crucial for products necessitating robust structural attributes. AM products are frequently utilized in the aerospace industry due to their lightweight characteristics and exceptional structural reliability. Nevertheless, the iterative nature of AM introduces inherent uncertainties. Any defects in critical areas can significantly jeopardize the overall structural quality of the product, potentially resulting in severe consequences. Structural validation offers an effective solution to this challenge by subjecting the product to a battery of tests, simulations, and analyses. These evaluations assess the product's ability to withstand various loads, stresses, vibrations, and environmental factors it may encounter during its intended operational use. The ultimate aim is to verify that the product's design, materials, and construction adhere to safety and performance standards. In the realm of AM, the ultimate objective of real-time product structural quality validation is shown in Fig. 1.2, which is to ensure the instant, worry-free utilization of the finished product, characterized by impeccable structural integrity and top-notch performance, with no quality concerns arising post-printing. In other words, when the part is pulled off, it is ready to go without any quality issues.

To achieve the research objective, two real-time product structural quality validation approaches are proposed, which are shown in Fig. 1.2. The machine learning-based real-time structural validation is described in Chapter 3. The proposed image segmentation-based real-time FEA structural validation approach, which is presented in Chapter 4.

1.3 Outline

Chapter 1 introduce the research background, motivation and objectives of the dissertation.

Chapter 2 reviews the current research states for FFF in situ monitoring in this work. From the review, many defect detection systems utilizing ML for the FFF printing has been developed; however, current research only focuses on defect detection, while the ultimate goal is to guarantee product quality. Therefore, new novel methodologies for FFF product quality guarantee should be proposed, which is capable of inferring structural quality rather than just detecting defects. That is the core idea throughout the whole research process.

Chapter 3 describes a real-time defect detection and product structural validation approach for FFF. The layer-wise image from an optical camera will be fed into the fully trained CNN (convolutional neural network). The defect information will be outputted from CNN. The developed system links the defect information with the product structural quality decision boundary to evaluate defect impact. Moreover, the proposed real-time defect detection multi-dimensional accumulation-threshold-based decision-making system can make the right decision according to the different input defect information to alert the users to continue or cancel printing.

In the Chapter 4, efforts are being placed on developing a new and better model-based structural validation approach links defect segmentation with finite element analysis (FEA) performed online in real-time to guarantee product quality. An advanced image segmentation-based real-time automatic FEA product structural quality validation system. A case study on a consumer-grade 3D printer validates the proposed system. The defect information is segmented from the layer image with a fully trained U-net. Defect information is applied to update the layer property in the Abaqus Python Script. The simulation process runs automatically after the Abaqus Python script updating is completed.

Chapter 5 makes a summary of contributions to the dissertation and gives suggestions for future works.

Chapter 2

In situ monitoring for fused filament fabrication process: A review[1]

2.1 Abstract

Real-time monitoring of the additive manufacturing process offers the promise of guaranteeing product quality and increasing the efficiency of the printing process. This paper summarizes research results for the in situ monitoring of the printing process for the fused filament fabrication process. To have a systematic and comprehensive summary, different methods, devices, and achievements in a range of monitoring systems for 3D printing are described. Sensor types and devices used in the literature for printer health-state monitoring and printing process product quality monitoring are summarized. Discussion of current and future research directions concludes the review.

2.2 Introduction

Additive manufacturing (AM) is actively used in a variety of industries. According to the Wohlers Report on additive manufacturing in 2020 it shows enormous economic potential [125,127]. The revenue is expected to rise to $23.9 billion in 2022 for all AM products and services worldwide. It is further predicted to climb to $35.6 billion in 2024 [42]. The printing process of AM is typically stacking material layer-by-layer to

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build three-dimensional (3D) products. Due to the advantages of low production lead time and the ability to create complicated geometries and shapes, the process has been employed for various industrial applications. One of the fastest-growing and widely-used AM technologies is fused filament fabrication (FFF) [112]. During the FFF process, the thermoplastic filament is fed into a hot end that melts the material and extrudes it from the nozzle onto a build platform, which forms a thin cross-sectional layer or track. This process is repeated until the whole object is fully printed. The most common FFF thermoplastic filaments are acrylonitrile butadiene styrene (ABS) and polylactic acid (PLA) [196], while polycarbonate (PC), polyetherimide (Ultem), polyphenylsulfone (PPSF), nylon, and polyether ketone ketone (PEKK) are also used in industrial settings [313]. FFF has been utilized for various applications, including biomedical [110], civil [345], and medical [191, 300], in addition to being explored by hobbyists and educators [83, 127, 286].

Despite its advantages, FFF has lower reliability than other manufacturing processes, including other AM processes such as stereolithography (SLA) [131], selective laser sintering (SLS) [24], and laminated object manufacturing (LOM) [80]. Previous research estimates a 20% printing failure rate for FFF by unskilled users [316]. These reliability statistics arise from challenges related to material runout, nozzle clogging, excessive vibration, over-extrusion or under-extrusion, and defects associated with temperature validation. In addition, the FFF process is very sensitive to environmental conditions [94, 223, 303]. Better temperature settings, more uniform filament quality, and novel filament feed system could improve the printing process [81, 192, 304]. Furthermore, robust closed-loop control of the FFF process could mitigate process variations [53, 173]. Active control of the printing process will become critical as the size and complexity of fabricated parts increase.

Table 2.1: Sensors and devices for the in situ monitoring of the fused filament fabrication process

Devices	Printer's health state monitoring	Printing product quality monitoring
Thermocouple	[11, 228, 238] [20, 58, 200, 203]	[171, 190, 319]
Pressure sensot	[11, 22]	
Rheometer	[57, 58, 228]	
Acoustic emission sensor	[174, 320, 321] [203, 322, 337]	[149, 174, 321] [161, 162, 174] [323]
Vibration sensor	[20, 169]	[169]
Accelerometer sensor	[22, 39, 95] [238, 290]	[171, 319]
Rotary sensor	[22]	
Temperature and humidity sensor	[22]	
Current sensor	[291]	[147]
Force torque sensor	[63]	
Piezo sensor		[28]
Thermal camera	[238]	[59, 79, 164] [171, 190, 319] [186, 230, 233] [117, 256, 257] [123]
Optical camera	[21, 108]	[47, 53, 200] [65, 95, 139] [209–211] [215, 271, 324] [213, 214, 219] [74, 156, 274] [128, 273, 275] [216–218] [75–77] [17, 118, 121] [126, 138, 152] [187, 235, 248] [311, 326, 343] [357]
Microscope		[173]
Borescope		[20, 238]
Laser scanning		[71, 165, 293] [146, 172, 224] [235, 237, 265]
Flatbed scanner		[29]
Encoder	[200, 267]	
Magnetometer	[22, 95]	
Ultrasonic device		[61, 330]
Physics-based compressive sensing		[178–180]
Coherent gradient sensing		[163]

In general, there are two forms of quality monitoring for the FFF process: (1) monitoring of printer health state; and (2) detecting product defects during the printing process. Ensuring high printer health state is key to guarantee printing efficiency and product quality [277]. Researchers have investigated the use of sensors, such as vibration sensors, acoustic emission (AE) sensors, accelerometers, infrared thermometers, and visual cameras for monitoring printer health state [243]. Beyond monitoring the printer's health state, in situ printing process defect monitoring is indispensable for detecting product quality defects [204, 324]. Here, the vast majority of research has focused on combining images taken during the printing process with image analysis, machine learning (ML), or other defect detection algorithms. Table 2.1 is a compilation of sensors and devices from the literature which are used for monitoring the 3D printer's health state, monitoring the printing process product quality, or both.

The purpose of this paper is to review and summarize the research that has been done on monitoring of the FFF printing process, examine the impacts of these monitoring systems, and discuss the current and future research direction. In this paper, the background is described in section 2.2, and section 2.3 presents different ways and devices used for monitoring 3D printer health conditions. In section 2.4, research that focuses on monitoring the product quality during the printing process is discussed, the conclusion and future trends are provided in section 2.5. The interested reader is referred to [41, 64, 99, 113, 135, 137, 148, 159, 166, 193, 206, 225, 234, 292, 318, 332] for a more detailed review on AM techniques in general or [48, 115, 196, 231] for FFF and similar extrusion processes.

2.3 Monitoring and analysis of 3D printer health state in the printing process

For a consumer-grade FFF machine, Fig. 2.1a gives a generalized overview of the main components. In FFF, the extruder cold end drives the thermoplastic filament

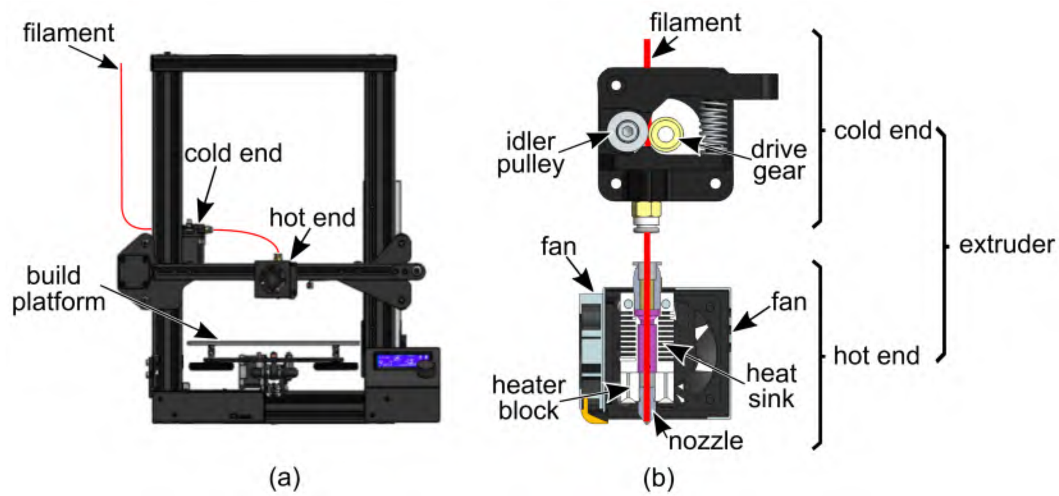

Figure 2.1: The profile of the most common consumer-grade 3D printer: (a) the 3D printer's key components; and (b) the layout of the extruder's cold end and hot end.

to the hot end, where it is heated until molten, then fed through the nozzle to bond it layer-by-layer on the heated build platform until the product is completed. In some cases, the platform is not heated and referred to as just a build platform [82]. By monitoring a 3D printer's health state during the printing process, machine downtime and material loss can be avoided. For the majority of FFF machines used by individuals and industry, the extruder and heated build platform are the two essential parts that, to a large extent, determine the success of printing [297]. Specific examples of conditions/parameters to be monitored include nozzle clogging, nozzle temperature, filament runout, and heated build platform temperature. The extruder is made up of two different parts: the cold end and the hot end. The details for these two components can be seen from Fig. 2.1b. The cold end drives the filament into the extruder's hot end and has two important components: the stepper motor and drive gear or hobbed bolt that drives the filament and the idler pulley shaft that holds the filament in the correct place during the extrusion process. When the filament arrives at the hot end, the components in the hot end (i.e., heater block, heat sink, heater cartridge, and thermocouple) work together to accurately heat the filament to a liquid state before extruding it from the nozzle [183]. Generally speaking, there are two extrusion systems available: direct extrusion and Bowden extrusion [7]. These vary by the location of the extruder's cold end. As the name implies, a direct extrusion system positions the cold end directly above the hot end. For Bowden extrusion, the cold end is most often mounted on the printer's body or frame, so that the filament feeds to the hot end through a Bowden tube [2]. Each of the extrusion type has pros and cons [109]. In direct extrusion, a benefit is the finer extrusion and retraction control due to the extruder's motor position directly over the hot end. Moreover, as less torque is needed to push the filament from the motor, smaller motors can be utilized. But with the extruder mounted adjacent to the print head, the total weight of the moving portion of the print head is increased. This extra weight adds constraints to printing speed, and it also causes wobbling and a potential loss of accuracy in the X and Y axis. For Bowden extrusion, since the extruder is fixed on the printer's frame, less weight is on the printing header. This results in higher printing accuracy and faster print speeds. However, the Bowden extruder needs to pull and push the filament through the long tube; therefore, greater friction exists in this process. Greater friction means a larger motor is required to supply the greater torque to drive the filament. And controlling the filament through the long tube also increases the response time. Moreover, some flexible and abrasive material is easily tied-up in or may wear by the long Bowden tube.

In Fig. 2.1, a Bowden extruder style machine is used to illustrate the critical components of the 3D printer. The extruder's hot end is the most common location for monitoring and is typically used for gathering information related to nozzle clogging, nozzle temperature, filament melting condition, and nozzle movement. The second most common monitoring position is the cold end. Information related to filament feeding conditions and filament slippage is typically measured here. Thirdly, the heated build platform is used for collecting information about the printing process, including heated build platform temperature and vibration data. In addition to these locations, the filament itself can be monitored to collect filament quality and condition data (e.g., filament diameter variation and filament breakage) at a variety of locations either on or off the printer.

In the section that follows, a discussion of extruder state monitoring, which includes extruder's cold end, hot end, and nozzle state monitoring is presented in section 2.3. Next, section 2.3 discusses the state-of-the-art in filament state monitoring relates to filament breakage, filament runout, and filament property monitoring. Lastly, section 2.3 covers build platform state monitoring.

Extruder state monitoring

The extruder component plays the most crucial role in a 3D printer, whose condition directly determines whether the printing is successful. As mentioned above, it includes two parts, the cold end and hot end. The following text is organized as follows: section 2.3 presents the work that has been done on the cold end state monitoring, hot end condition monitoring is described in section 2.3. Section 2.3 reports research on nozzle monitoring. Although the nozzle is part of the hot end, its significant impact on the printing process (e.g., layer resolution and printing speed) necessitates a separate section.

Extruder cold end state monitoring

The cold end consists of a stepper motor, hobbed bolt or gear, spring loaded idler to hold the filament, and tubing to guide the filament. In the cold end, the stepper motor drives the filament towards the hot end. The stepper motor is able to impart full torque at low speeds and make small movements at relatively high level of precision. As the cold end module applies the force required to push the filament smoothly through the heat block and nozzle, monitoring the extruder's cold end condition is essential. Most of the fault that occurs on the cold end module are filament transport slippage and abnormal filament feed rate. Also, various research on filament breakage and filament runout are detected at the cold end. For the sake of clarity, these are discussed in section 2.3.

The most common approach to identifying different cold end states is by analyzing vibration signals obtained from the sensor mounted on the cold end module. Liu et al. demonstrated a method to recognize and classify different extruder states by combining extracted and reduced features from the AE signals with the unsupervised clustering fast search and find of density peaks (CFSFDP) algorithm [174]. This method provides an accurate and reliable real-time system for monitoring the 3D printer's health and can classify the cold end states such as filament loading or unloading and filament runout. Wu et al. proposed a non-intrusive monitoring method by integrating AE sensing technology with support vector machines (SVM) for classification [320]. This method is capable of recognizing three different extruder cold end conditions, which are material loading, normal extruding, and filament runout. Bukkapatnam et al. developed a two degrees of freedom lumped mass model [39]. By extracting the printer's vibration feature with this model, uncommon cold end conditions such as fast feed and slow feed could be identified. Wu et al. developed a more versatile algorithm using a hidden semi-markov model (HSMM) to diagnose the extruder states [322]. This new method reduces the input AE data's dimensions and sizes while achieving more than 90% accuracy with the time resolution of 0.1 s. The experimental result shows that the HSMM algorithm could achieve multi-state (filament loading and unloading, filament runout, idle, and normal extruding) identification simultaneously. Furthermore, Li et al. demonstrated that through utilizing a least squares support vector machine (LS-SVM) and combining the vibration data extracted from the extruder and heated build platform, a filament jam state could be effectively recognized with an accuracy of over 90% [169]. Fig. 2.2 shows in situ monitoring and diagnosing details for the FFF printing process developed by Li et al. [169]. In the flowchart, BPNN is the back-propagation neural network, LS-SVM is the least squares support vector machine, and DAQ is the data acquisition system. By analyzing the vibration signal, the system can diagnose the filament jam failure caused by spring fatigue (i.e. weakening of the spring). Under normal condition, the spring keeps the drive wheel aligned with the cold end filament connector. When the spring fatigues, the drive wheel displaces, thus increasing friction force between the cold end filament connector and the filament.

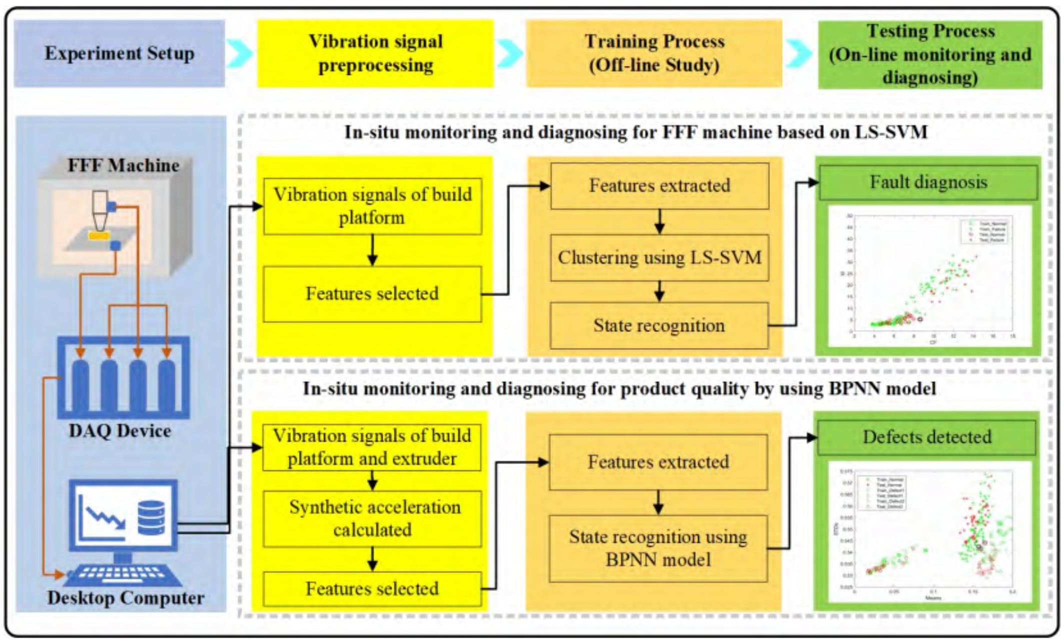

Figure 2.2: Flowchart for combined in-situ monitoring and diagnostic of a FFF machine and product quality, showing, (top) the flowchart for in-situ monitoring and diagnosing 3D printer healthy state in the FFF process, which uses the build platform vibration data as input data, after the offline training, the LS-SVM algorithm diagnoses the printer fault online, and (bottom); the flowchart for in-situ monitoring and diagnosing FFF product quality, which utilizes the vibration data from build platform and extruder as input, after the offline training process, the BPNN model detects product defects online.

Accurate knowledge of the filament feed rate is necessary to guarantee product quality during printing. However, filament slippage can occur between filament and feed gear, which leads to product functional and dimensional error. Greeff et al. combined an image processing system with a microscope video camera to explore the speed difference between filament and extruder feed gear speed [108]. The study shows that feed rate and extrusion temperature are the two main factors that influence the slippage rate, but adding closed-loop control to the extruder can decrease the slippage rate considerably. Go et al. explored the slippage rate limits in the FFF printing process [103]. Part of the research shows the relationship between filament shear failure and extrusion force. Fiedler et al. designed a new extruder drive bolt's tooth geometry and evaluated the filament grip force generated from it [81]. A smart FFF 3D printer was built by Moretti et al. with heterogeneous sensing (e.g., camera, thermocouple, positional encoders, and filament encoder) [200]. By mounting an encoder on the back of the drive motor to track the filament advancement, which can indirectly evaluate the extruder material feed rate. This encoder can detect a 7.5 μm minimum filament advance.

Extruder's hot end state monitoring

As hot end conditions directly affect the inter-layer bonding, monitoring the extruder's hot end condition is essential in determining the 3D printer's state [59,199]. Generally, the extruder hot end condition can be divided into three states: normal-extruding, semi-blocked, or completely-blocked. Unhealthy hot end conditions result in reduced product performance such as fatigue, flexural, or strain properties [70, 297, 309].

Researchers have demonstrated various successful methods for monitoring extruder hot end condition. One of the common methods used for inferring the hot end conditions is using AE. AE sensors detect the release of acoustic energy caused by the rapid release of localized stress energy in a material (e.g., the polymer being printed). Acoustic events can be caused by a variety of abnormal phenomena. It is sensitive to any of the extruder's hot end states, and as such advanced signal processing techniques must be applied to the acquired data to detect anomalous extruder states. An approach to detect and classify different extruder states by utilizing the features from the AE signals as input data was proposed by Liu et al. [174]. The unsupervised clustering fast search and find of density peaks (CFSFDP) algorithm is trained by the features from the AE signals. This approach is shown to be useful for the real-time monitoring of the 3D printer's health, especially for the extruder's hot end health state. Fig. 2.3 shows the established extruder condition monitoring platform developed by Liu et al., which consists of an AE sensor, a preamplifier, a data acquisition card, and a desktop computer. A non-intrusive monitoring system was proposed by Wu et al. to classify the extruder hot end states [320]. This approach combines AE sensing technology with the SVM algorithm, which can recognize semi-blocked and blocked hot end conditions.

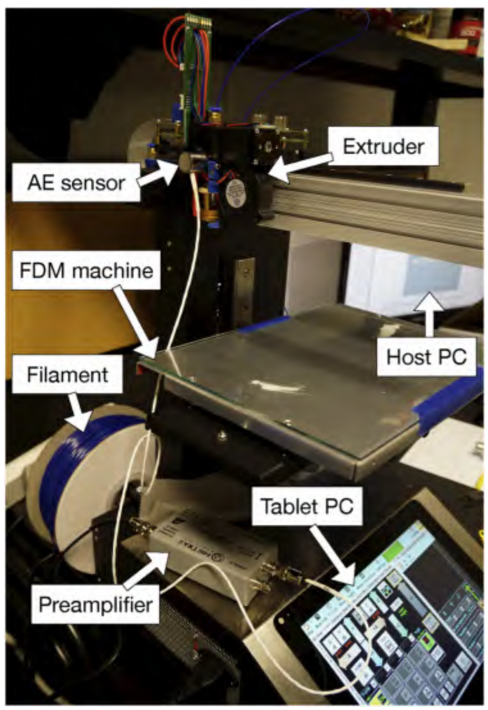

Figure 2.3: AE sensor-based condition monitoring platform.

Another method for monitoring the extruder hot end state is to monitor the vibration signals by either vibration or acceleration sensors [249]. The model (two degree of freedom lumped mass model) for cold end monitoring by Bukkapatnam et al. [39] was also used to recognize the overflow, underflow, fast feed, and slow feed anomalous conditions for the hot end. Moreover, Rao et al. built an online real-time quality monitoring system using heterogeneous sensors, including accelerometers mounted on the heated build platform and extruder head to measure the vibration [238]. The system could identify the normal state, abnormal state due to insufficient filament extrusion, and failure to extrude state cause by nozzle clog.

Images or videos captured by cameras also have been used to monitor various hot end conditions, such as blockages and missing material flow. Extruder hot end state can be quantified using either a person-in-the-loop, advanced image processing algorithms, or a combination of the two. If the system is automated, it can achieve real-time extruder hot end state monitoring. Rao et al. utilized a borescope fixed on the printer's extruder to get a real-time video of the extrusion process, allowing people to visually monitor the extruder's status during printing [238]. Another real-time monitoring system for advanced manufacturing processes is called the online sparse estimation-based classification proposed by Bastani et al. [20]. By applying the Bayesian estimation method to analyze sensor signals (i.e., thermocouples, vibration sensors, and infrared melt pool temperature sensor), the system can detect the printing process drift (process state change from normal to abnormal and finally completely stops due to the clogged nozzle) with an accuracy of 90% (F1-score). Baumann et al. proposed an auxiliary thresholding algorithm, which can convert digital images captured from the camera fixed on the print-head into binary images to monitor the extruder state [21]. The system can get an accurate distinction between the object and object's background by utilizing this algorithm. Moreover, this system can detect abnormal extruder states such as missing material flow due to a clogged nozzle, air-bubble in the extruder, or jammed material.

Beyond the hot end module, other components also impact product quality, and have invoked the researcher's interest. Fans on the extruder's hot end also play an important role in obtaining excellent product quality, especially the heat sink cooling fan. The heat sink cooling fan is required as it keeps heat from transferring up into the other components. Note that the fans for cooling down the printed layer are optional [173]. To avoid improper fan speed impact on the product quality. Gao et al. utilized acoustic information to predict the fan speed, which could prevent the defects caused by improper fan speed [95]. Kim et al. built a data-driven in situ FFF monitoring and diagnosing system by utilizing an AE sensor and an accelerometer [149]. The author designs a bolt loosening error to drive the nozzle head abnormal motion, which produces interlayer shifting. After the signal processing and feature extraction process, the SVM-based algorithm is used to diagnose whether the printing process is healthy or faulty. The result showed the non-linear SVM-based model had an accuracy better than 87.5%.

Nozzle state monitoring

Variations in nozzle temperature, nozzle distortion, and nozzle clogging can occur in the heated nozzle. Of these, nozzle clogging is one of the most frequent printing errors. Given the persistent challenge of nozzle clogging and its prominent effect on the product's geometrical accuracy and mechanical properties, a majority of research has focused on this area [122]. He et al. proposed an online defect monitoring approach for the FFF process based on the product's temperature field variation [117]. To get the product's layer temperature, an infrared (IR) thermal camera is utilized. After feature extraction, the difference between normal and abnormal printing features are calculated. This proposed approach not only detects the defects but also monitors the nozzle working state. The detection of nozzle blockage can also be monitored by tracking melt pool temperature data. Rao et al. demonstrated that normal, abnormal, and failure process states showed a recognizable pattern by analyzing the melt pool temperature data obtained from the non-contact IR temperature sensor [238]. By applying the nonparametric Bayesian and Dirichlet process mixture model and evidence theory (ET) with this data, this defect detection method can identify the abnormal FFF process such as nozzle clog with high accuracy (85% average F1-score). Some corrective actions can be applied to avoid nozzle clogging by detecting the temperature drift process from normal to abnormal. Nuchitprasitchai et al. built an open-source low-cost real-time monitoring system with a variety of cameras that can detect clogged nozzles, filament runout, or object geometry error using a stereo calibration algorithm [209–211]. Kim et al. studied a methodology to detect material deposition status [147]. The research demonstrates that the nozzle's extrusion pressure affects the current draw from the feed motor, which shows the potential to detect nozzle states using this methodology. A dynamic current-based model was proposed by Tlegenov et al. to monitor nozzle conditions [291]. By modeling the relationship between the current draw, extruding torque, and effective nozzle diameter, nozzle clogging conditions could be identified. This work is validated experimentally by monitoring the extruding stepper motor's current draw with a current-based monitoring system. Moreover, Tlegenov et al. proposed a method to predict the nozzle condition by analyzing the acceleration's amplitudes recorded by an accelerometer fixed on the extruder [290]. The setup used by Tlegenov et al. is shown in Fig. 2.4. Results from both the theoretical modeling and experimental study verifies that the amplitude of vibration shows a non-linear increase as the nozzle becomes clogged for both direct and Bowden extruders.

Monitoring the filament flow rate and nozzle melting temperature are also commonly carried out for the nozzle conditions. Anderegg et al. developed a method that used pressure transducers and thermocouple insertion devices, so the nozzle temperature and pressure distribution during the printing process can be measured, which gives vital insights for monitoring and improving the printing process [11]. Coogan et al. designed an in-line FFF rheometer by incorporating the pressure transducer and thermocouple into a modified nozzle, which could provide accurate filament viscosity,

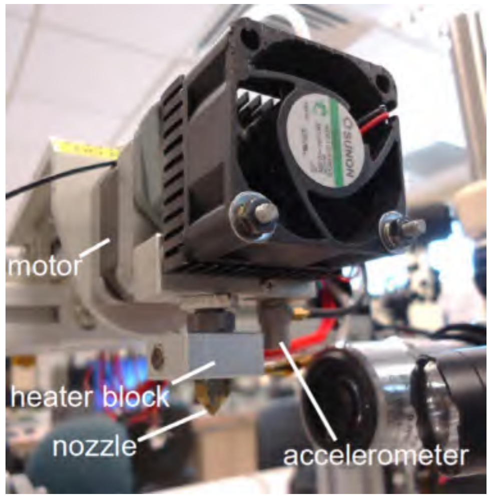

Figure 2.4: Accelerometer sensor for detecting nozzle state.

filament feed rate, as well as melt temperature during the printing process [57]. Balani et al. optimized the FFF printing conditions by exploiting the printing parameters and PLA's physical properties [16]. A series of results are achieved in this research, such as filament's maximum inlet velocity in the liquefier, the PLA's rheological behavior, and the distribution of the velocity along the nozzle radius. Filament flow and temperature history in material extrusion were investigated by Peng et al. [228]. Filament temperature evolution is investigated by embedded thermocouples. Filament flow behavior is achieved by an introduced dye marker within the filament. The experimental results show that the shear rate near the center of the filament is lower, which results in a more blunted velocity profile, and the material extrusion is a highly non-isothermal process. Moreover, Coogan et al. built a model to predict the material extrusion product's interlayer strength, which combines a model to predict interlayer contact with another model for predicting polymer chain diffusion [58]. The in-line rheometer added to the nozzle can directly measure the melt temperature and pressure. The heat transfer analysis is achieved by measuring the critical environmental temperatures using thermocouples. Interlayer contact prediction is based on in-line pressure data, whereas polymer chain diffusion prediction is based on in-line viscosity and temperature measurements. A case study using this model demonstrates applicability for real-time defect detection for material extrusion AM processes.

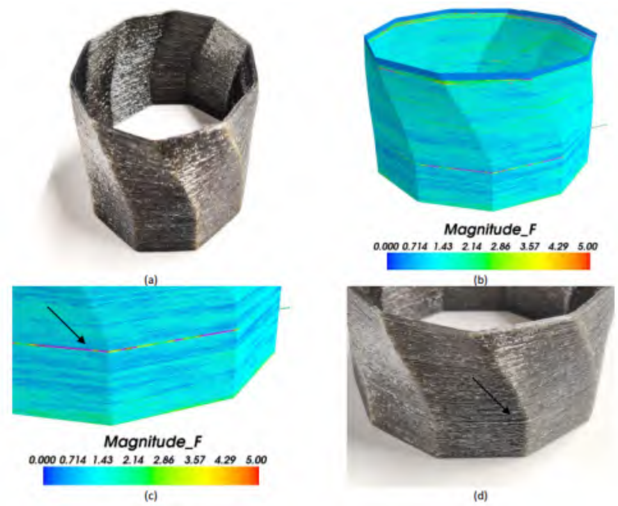

Figure 2.5: Twisted decagon sample for defect detection: (a) printed part; (b) the force sensed during the printing; (c) a zone with high measured tension forces; and (d) the resulting defect on the part.

Force sensors mounted between the print head and printer frame have also been used to monitor the nozzle's condition, where the printing head and the build platform has a material connection through the melted filament. If the print head's speed is not matched with the extrusion speed, printing defects will occur. DeBacker et al. proposed a printing force and torque monitoring solution for the FFF printing process [63]. In this solution, a force-torque sensor is mounted between the printing head and gantry system frame and used for recording and timing the printing force and torque data. By applying an algorithm to the force-torque data, a force and torque intensity geometry map is made. This force and torque intensity geometry map can be used for capturing printing failures by highlighting the product area where the recorded force or torque values are greater than a set threshold. Fig. 2.5 displays one defect detection sample by using the force and torque intensity geometry map. Fig. 2.5a shows a twisted decagon product used for verifying defect detection, and in Fig. 2.5b is the force and torque intensity geometry map. The detail in Fig. 2.5c shows an anomaly at the top of the product that is detected by monitoring the force history. As shown in Fig. 2.5d, the real defect occurs at the same location on the printed product.

Filament state monitoring

Monitoring filament condition during the printing process helps to guarantee product quality. Consistency in filament parameters such as diameter, moisture, and pigment also have an influence on the final print in terms of product quality and mechanical strength [349, 350]. For filament condition monitoring, the majority of research has focused on four cases: filament breakage, filament runout, material classification, and material quality monitoring.

Filament breakage leads to compounding malfunctions such as surface roughness error, geometrical misalignments, or printing failures. Most filament breakage is caused by inhomogeneous filament material, which can not sustain the required pulling force. Yang et al. detected filament breakage with AE sensors, showing that the distribution of the AE signal changed after filament breakage occurred. The signal changes are quantified using two measurable indicators: instantaneous skewness and relative similarity [337]. Between these indicators, the relative similarity is shown to outperform instantaneous skewness. Moreover, the relative similarity is capable of separating the breakage state from the steady state. Fig. 2.6 shows the AE sensor mounted on the extruder's shell to detect the filament condition. Fig. 2.7 displays how the amplitude spectrum of AE signals changes under different manufacturing conditions.

Filament runout occurs during printing. Wu et al. developed a method that uses AE hits as an indicator and applies the SVM with the radial basis function kernel to classify the printer running out of filament condition [320]. Liu et al. presented

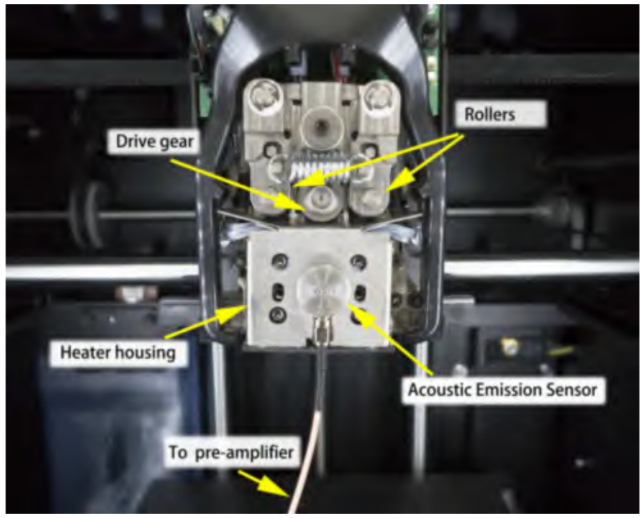

Figure 2.6: AE sensor mounts on the extruder shell for detecting filament breakage.

a method to recognize and classify different extruder states, including when printer is running out of material, using features extracted from the AE signals [174]. In addition to AE sensors, advanced image processing algorithms are widely used for detecting filament runout. Nuchitprasitchai et al. built a real-time monitoring system with different numbers of cameras by utilizing the stereo calibration algorithm [209–211]. Baumann et al. proposed an auxiliary threshold algorithm to detect filament fault [21]. By converting digital images captured from the camera fixed on the print-head into binary images, this system could detect missing material flow due to filament runout condition.

Filament property varies from type to type, so identifying material properties (e.g., type and color) has also been of interest to researchers from both the print quality and cybersecurity perspective [101, 315]. Straub et al. proposed an approach for the filament assurance process, which can compare the filament that is being used with the right one based on the pre-calibration process [275]. The pre-calibration process considers filament type, lighting condition, and other controlling factors compiled by the user and stored in a database. This method can detect the inadvertent loading of incorrect filament and filament contamination during printing.

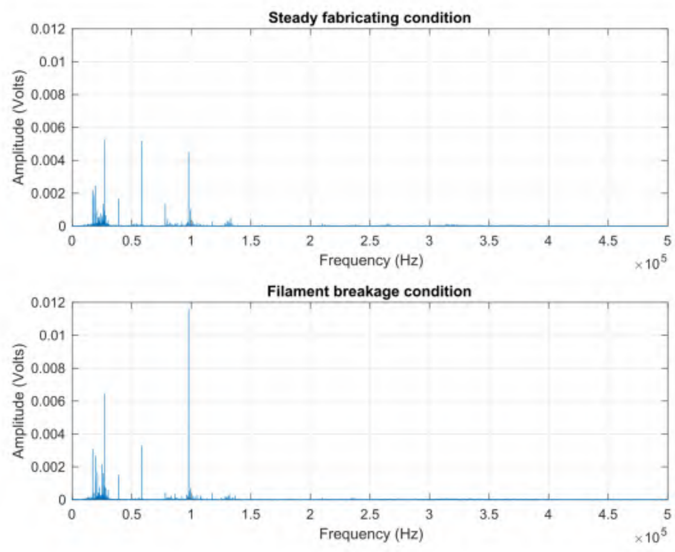

Figure 2.7: Amplitude spectrum of AE signals under different manufacturing conditions.

Polymer filaments with inconsistent diameters will affect final print quality if the variation in diameter is high enough. Inconsistencies in filament diameters may cause slippage, extruder jamming, or bad product quality. Cardona et al. researched the filament diameter tolerances effects in FFF [46]. An array of digital calipers and an optical comparator are utilized to measure the filament diameter. The experimental result indicates that using a low extrusion rate (4.23 mm/s) and moderate extrusion temperature (180 °C) combination makes it possible to get high diameter conformity and low diameter tolerance (0.01 mm). Combining mechanical and load cell sensors with different encoders, Soriano Heras et al. developed an advanced filament detection system [267]. The system can detect filament diameter variation and moving conditions while adjusting the extruder's gear speed to match the real printing speed.

Build platform state monitoring

Monitoring of the heated build platform for undesired temperature variations is essential, as deviations in the temperature of the heated build platform may warp or misalign printing products [25, 268, 310]. The two most common parameters considered when monitoring the heated build platform are vibration (by acceleration sensors) and temperature (by thermocouples or IR cameras). To a lesser extent, AE sensors have also been utilized for monitoring the heated build platform state. Li et al. demonstrated that vibration data extracted from the extruder and heated build platform could be used to detect filament jam states with the LS-SVM algorithm [169]. Baumanna et al. designed a sensor array including acceleration sensors and thermocouples [22]. Gao et al. built an online process monitoring system to defend against cyber-physical attacks that uses the heated build platform vibration data as one of the indicators to monitor the printing process [95]. Another online real-time quality monitoring system using heterogeneous sensors was developed by Rao et al., which utilizes both thermocouples and accelerometers to monitor the heated build platform [238]. Lastly, Wu et al. combined an AE sensor affixed to the heated build platform with K-means clustering to build a real-time failure monitoring system that could identify the failure time and mode by analyzing the time-domain and frequency-domain AE signal [321].

In the printing process, ensuring that the first layer adheres well to the build platform is the first step in producing a good quality product. However, an unlevel build platform significantly impacts the bonding process and may cause defects such as layer shifting and misalignment. Nam et al. proposed a data-driven approach to diagnosing a mis-leveled build platform [203]. The author uses different sensors, including accelerometers, AE sensors, and thermocouples, to collect the printer's health state information. The flawed process designed in this experiment is the build platform leveling fault, which results in a warpage and bad adhesion. The result verified that the non-linear SVM-based model with the printer frame acceleration root mean square information has the best testing accuracy.

2.4 Monitoring and analysis of the product quality in the printing process

Monitoring and analyzing the printing process is an enabling technology for real-time quality control as it can diagnose printing abnormalities and defects. In addition to monitoring 3D printer health condition, another main task of a monitoring system is the analysis of product quality. A variety of aspects influence the FFF product quality, including surface roughness, printing infill defects, and dimension or geometry accuracy [26, 244]. To achieve the objective of monitoring the printing process and product quality, researchers have proposed various methods that will be discussed in section 2.4. Section 2.4 describes the monitoring of temperature variations during the printing process. In section 2.4, a discussion of in-process printing abnormality is presented, while product surface roughness defects detection is discussed in section 2.4. Finally, infill defect detection (section 2.4) and product geometry error detection (section 2.4) are also presented and discussed.

In-process printing temperature variations monitoring

As the FFF process deposits individual tracks to form layers, it is critical to maintain thermal energy to ensure there is proper inter-layer bonding [30, 199]. When the thermoplastic filament is being extruded from the nozzle's hot end, it is melted and sticky, allowing the material to fuse with the tracks and layers previously deposited on the build platform [13, 31]. However, this bonding mechanism is no longer present once the filament temperature falls below the glass transition point. As expected, the lack of inter-layer bonding can affect part performance and geometrical accuracy [14, 353]. Therefore, monitoring product temperature during the printing process is being actively investigated as a means to ensure the quality of printed components.

Monitoring the temperature during deposition is important in understanding the filament weld formation. Seppala and Migler utilized IR imaging to measure the polymer's surface temperature profile in the printing process [257]. The crucially important weld temperature temporal profile can be obtained from the printed layer and sublayers temperature profiles. The result shows that the weld temperature drops at a rate of approximately 100 °C/s, with temperature staying above the glass transition for approximately 1 s under typical printing conditions. To quantify the inter-layer weld strength from the polymer inter-diffusion perspective, Seppala et al. built an experimental framework comprising thermography, rheology, and fracture mechanics [256]. They also developed an equivalent isothermal weld time which was then tested in relationship to the fracture energy. Costa et al. illustrated an analytical solution to the transient heat conduction during the filament deposition process [59]. The analysis process is coupled with an algorithm that can activate the related boundary condition taking into account the deposition sequence. The analysis can predict the temperature and adhesion with time evolution by considering the main printing process parameters. The experimental result validates that the analysis prediction agrees with the measurement for both temperature and adhesion.

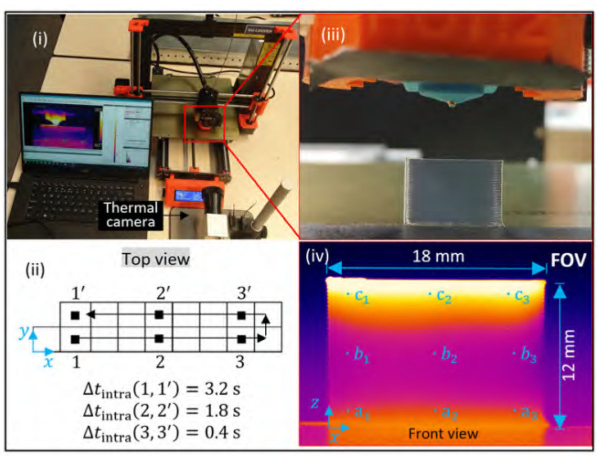

Figure 2.8: IR camera for monitoring the printing product temperature: (i) picture of the thermal camera set-up for FFF printing process monitoring; (ii) the product geometry, filament deposition sequence, and intra-layer time for elements located on given x-y planes from the top view; (iii) picture of the printed product; and (iv) thermogram of the printed part.

IR cameras are the most commonly used method for monitoring the temperature of 3D printed components [360]. Dinwiddie et al. used an IR camera set up on a FFF printer to get layer-by-layer thermography images of ABS filament printing process, its thermal behavior and thermal evolution [69]. Malekipour et al. researched an ABS product's layer temperature distribution and average temperature condition by combining the layer-based temperature profile plot with the temporal temperature plot [186]. An IR-based system capable of capturing spatial and temporal temperature variations during the progressive printing process was presented by Ferraris et al. [79]. Li et al. developed a new framework by combining physics-based and data-driven methods to predict the component-scale layer-to-layer thermal field [164]. Fig. 2.8 shows a temperature variation monitoring system proposed by Ferraris et al. with an IR camera facing the 3D printer's nozzle [79]. The system captured the printing temperature variation from the product's front $x-z$ plane.

Physics-based compressive sensing (PBCS) technology can be used to obtain high-fidelity information by using low-fidelity measurements. This technology has also been applied to measure temperature distribution. Lu et al. proposed a new compressive sensing method by applying a physics-based approach to efficiently and precisely monitor the printing process temperature distribution [178]. The PBCS accuracy is assessed by comparing the reconstructed temperature distribution using grayscale thermal images obtained from an IR camera with the temperature distribution getting from the PBCS. This proposed physics-based compressive sensing method has great potential to improve monitoring accuracy and efficiency with lower costs by reducing the number of sensors used in the monitoring system. Furthermore, Lu and Wang proposed an efficient transient temperature distribution monitoring approach for the FFF process by using PBCS [179, 180]. The printing process can be monitored by reconstructing 3D temperature distributions using the sparse samplings in temporal and spatial domains.

In-process printing abnormalities detection

The term "abnormal printing process" refers to conditions where the printer is in a perceived healthy condition, but printing failures still occur. These abnormalities include poor bonding quality between the filament and build platform or inter-layers, shrinkage and warpage, and product position error. All these abnormalities have a direct or indirect effect on the printing product quality.

First layer bonding error detection

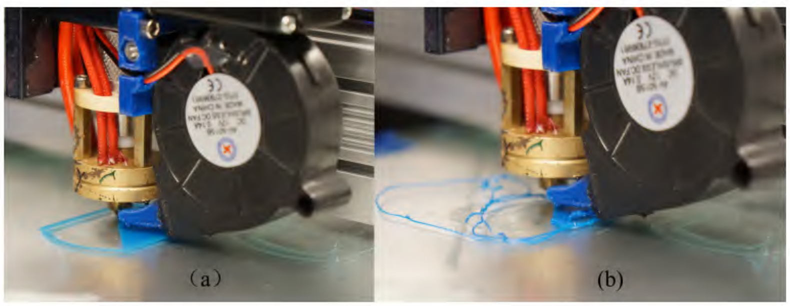

Figure 2.9: Image of normal and abnormal printing process: (a) normal FFF printing status; and (b) abnormal FFF printing status.

Poor bonding quality between the first layer and build platform leads to the material peeling off from the build platform, resulting in mechanical contact (i.e., scratching) between the nozzle and distorted material [222]. These failures directly influence the quality of printing parts. The normal and abnormal printing processes are shown in Fig. 2.9 [321]. To expand, Fig. 2.9a shows a normal printing process where the material is deposited by the nozzle smoothly and the filament is correctly attached to the heated build platform. A failed process is shown in Fig. 2.9b, where the filament does not adhere to the heated build platform.

Various sensors have been shown to be capable of detecting abnormal bonding processes such as filament peeling and dragging. Wu et al. set up an AE system attached to the upper surface of the heated build platform [321]. AE events vary significantly when the material is peeling, warping, or dragged by the nozzle as compared to the normal printing process. The research result shows that abnormal printing failures could be automatically identified by K-mean clustering. Wu et al. proposed an on-line data-driven monitoring approach based on the AE sensor for the FFF process to diagnose the process failure [323]. An unsupervised ML algorithm self-organizing map (SOM) is utilized to cluster different failure modes. The proposed method could detect three types of failure: scratching and hitting between the distorted filament and extruder, filament peeling off from the build platform, and material rubbing by nozzle or gliding on the build platform. Bhavsar et al. presented a methodology to detect first layer bond quality for the FFF printing process by utilizing piezoelectric sensor and the discrete wavelet energy approach [28]. The research shows that the first layer's good or bad bond quality can be detected by analyzing the piezoelectric polyvinylidene difluoride sensor's vibroacoustic signals.

Inter-layer bonding error detection

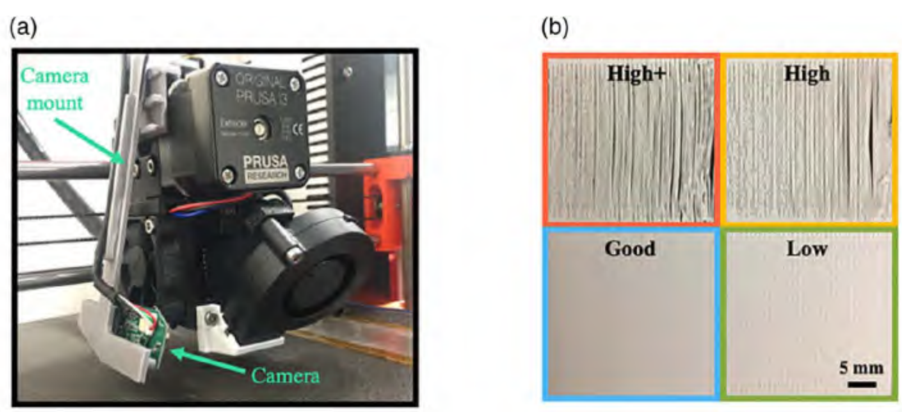

Figure 2.10: Delamination detection system setup and product quality images: (a) the monitoring system with a camera mounted on the extruder; and (b) four cases product quality images, which represent the four nozzle height condition.

The interfacial bonding quality has a significant impact on the product's mechanical property. Delamination is a printing failure, which is usually caused by improper fusion between the previous layer and the new layer. While it is a local defect, it can affect the entire printed product.

Researchers have developed various approaches for delamination detection. Jin et al. built an automated real-time interlayer defect detection and prediction system for the FFF process. By combining real-time camera images with the CNN algorithm, this system is able to detect delamination, which is caused by the improper spacing between nozzle and build platform. Accuracy on validation data is 97.8%, shown in Fig. 2.10 [138]. In Fig. 2.10a, a camera connected over USB is mounted on the left side of the nozzle to capture the printing images. Fig. 2.10b shows product quality images for four different representing four nozzle spacing conditions. The "High" in Fig. 2.10b means the nozzle height between the filament and build platform is high, which leads to poor adhesion quality. Furthermore, a "High+" nozzle height makes the delamination even worse. On the other hand, the "Low" nozzle height results in a nonuniform surface quality due to the restricted space under the nozzle. Cummings et al. presented an in-process ultrasonic inspection for AM product [61]. Four piezoelectric transducers are bonded on the build platform to inspect the product periodically in the printing process. By comparing the normalized frequency response with the experimentally determined ideal response, the system can find defects such as delamination in recently printed layers.

Other methods have also been explored for monitoring and assessing the interlayer bonding quality. Xu et al. presented a real-time AM process monitoring system by applying phononic crystal artifacts [330]. This research is based on ultrasonic wave propagation in phononic coupons consisting of repeating substructures, which can be used to monitor and finally assess product bonding quality and uniformity. Product structure periodicity results in the dispersion of the wave, which is easily affected by geometric or materials properties and irregularities. To predict the filament connection quality between lines in different printing parameters, Jiang et al. proposed a deep learning ML model to predict the filament bonding quality for the FFF process [136]. Filament extrusion speed, print speed, layer height, and line distance are chosen as the deep neural network's input, and the layer connection quality is the output. The prediction accuracy of this ML model can be as high as 83%.

Shrinkage and warpage error detection

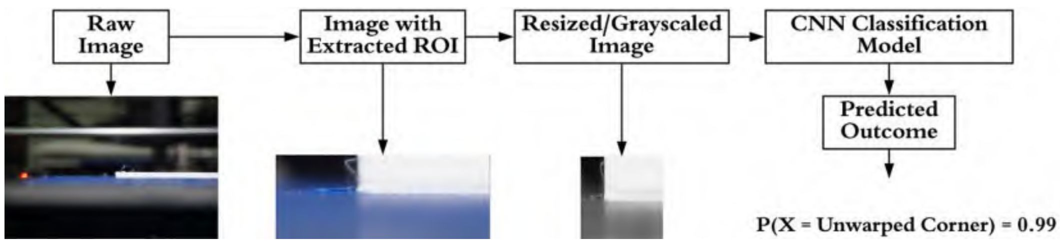

Figure 2.11: Image acquisition and analysis pipeline for the warpage classification.

Detecting shrinkage and warpage in the printing process is important to control the geometry tolerance. During the printing process, when the printed polymer cools down, its volume decreases, even though its temperature is still above the glass transition temperature. When polymeric parts cool at different speeds or in an anisotropic way at diverse positions, the printed product is prone to warpage [269]. As a common printing error, detecting product shrinkage and warpage are essential to getting a good printing product.

Researchers have investigated the relationship between warpage and printing parameter settings. Panda et al. built a warp performance evaluation model to evaluate the process's warping characteristics [222]. The proposed evolutionary system identification (SI) method quantifies the warping by the printing process parameters, including line width compensation, extrusion velocity, filling velocity, and layer thickness. Analysis results show the layer thickness and extrusion velocity affects warping the most. To study how the printing process variables affect the product warpage in FFF, Armillotta et al. selected three variables related to product's geometry (i.e., length, width, and height) and the layer thickness as influence factors [12]. The research identifies the influence factors individual and interaction effects on the warpage generation.

A series of shrinkage and warpage detection methods have been introduced by researchers. Hu et al. illustrated a FFF printing fault detection method, which explores the classification of faults produced by the temperature filed variation [123]. With the SVM algorithm, the method can recognize different defects (i.e., insufficient filling, warping, and serious fault printing). Saluja et al. built a closed-loop FFF in-process warping detection system [248]. They extract product corner features as input into the CNN model, while the probability of a product corner being warped is output. The experimental result shows the mean accuracy yield from this model to be 99.3%. Fig. 2.11 shows the process proposed by Saluja et al. to detect the warpage for the FFF printing process [248]. Li et al. proposed a product status recognition method for the FFF process by utilizing the AE sensor [161]. Two algorithms, HSMM and SVM, are applied to recognize the product normal, looseness, and curl status. The result shows the HSMM performs a bit better than the SVM. Jin et al. built an automated real-time interlayer defect detection and prediction system for the FFF process [138]. By fixing a strain gauge on the build platform, this system could detect product warping. Li et al. demonstrated that by utilizing a back-propagation neural network (BPNN) and combining acceleration data and vibration data extracted from the extruder and heated build platform, warpage product defects and abnormal filament leakage could be detected [169]. The result shows the diagnosis accuracy by BPNN is over 95%.

Product positioning error detection

Product position discrepancy influences product quality. Straub et al. developed an identification and decision-making system integrated with quality control process for identifying the object position discrepancy [274]. The system can compare the object's silhouette taken from CAD files with actual product positions to identify the matching extent. Baumann et al. developed a vision-based error detection system [21]. The system uses cameras to obtain the printing process video and then applies the developed image processing algorithm to analyze them. As the normal printing product only grows vertically, a suddenly increasing horizontal pixel indicates horizontal movement (i.e., product detachment from the heated build platform). The algorithm achieves detachment detection by calculating image differentiation between two consecutive images. Results show that the system could detect product detachment in a short response time. By integrating multiple heterogeneous sensors, a smart FFF 3D printer was built by Moretti et al., to monitor the printing process and product quality [200]. The nozzle path can be reconstructed by the X, Y, Z position information from the exes positional encoders. The positioning error resulting in a visible misalignment can be detected by comparing the measured nozzle path with the nominal G-code path.

Product surface roughness quality monitoring

Surface roughness is a significant factor in part quality as it influences critical product precision, especially its functionality or assembly at mechanical interfaces [283]. When the extruder is in an under-extrusion or over-extrusion condition, product surface roughness changes drastically during the printing process [230,256,329]. Guaranteeing that the product's surface roughness is within specifications is a critical part of the FFF printing process. Discussion will focus on both the data-driven and model-based approaches.

Data-driven product surface roughness quality monitoring

Data-driven approaches to monitoring product surface roughness errors have been developed. In the development of these purely data-driven methods, most of the research studies use cameras to get real-time images or video and then apply advanced algorithms to process the images and identify the product surface roughness condition.

ML algorithms are commonly utilized by researchers to detect product surface roughness. Jin et al. mounted a camera on the FFF printer [139], as shown in Fig. 2.12a. Different product qualities for under-extrusion, good-quality, and over-extrusion conditions are shown in Fig. 2.12b. These images are used to build a pre-trained data set, which is trained by a ResNet 50 architecture CNN classification algorithm. After that, real-time monitoring and refining is implemented, by feeding real-time images into the pre-trained classification mode, the current condition can be obtained. Liu et al. designed a system for defect recognition that uses the texture analysis-based image diagnosis (TA-ID) algorithm to process surface images taken by a digital microscope [173]. The system can recognize the effects when in conjunction with a supervised classification algorithm. The system could even adjust the related printing parameters (i.e., cooling fan on/off and material flow rate) to mitigate printing defects online through closed-loop feedback control. Wang et al. developed a vision-based online monitoring system to identify surface defects [311]. In this system, a mobile camera system is designed to monitor the product's exterior surface from different angles. By utilizing a CNN algorithm, this system can detect the blob, void, thick line, crack, and misalignment defects. Moreover, Banadaki et al. built a reliable real-time quality monitoring system with a DCNN algorithm [17]. After training the model on images taken at each layer, the proposed online model can classify five different quality grades resulting from overfilled or underfilled with an average accuracy of 94%.

Other data-driven approaches utilize various advanced algorithms that could also

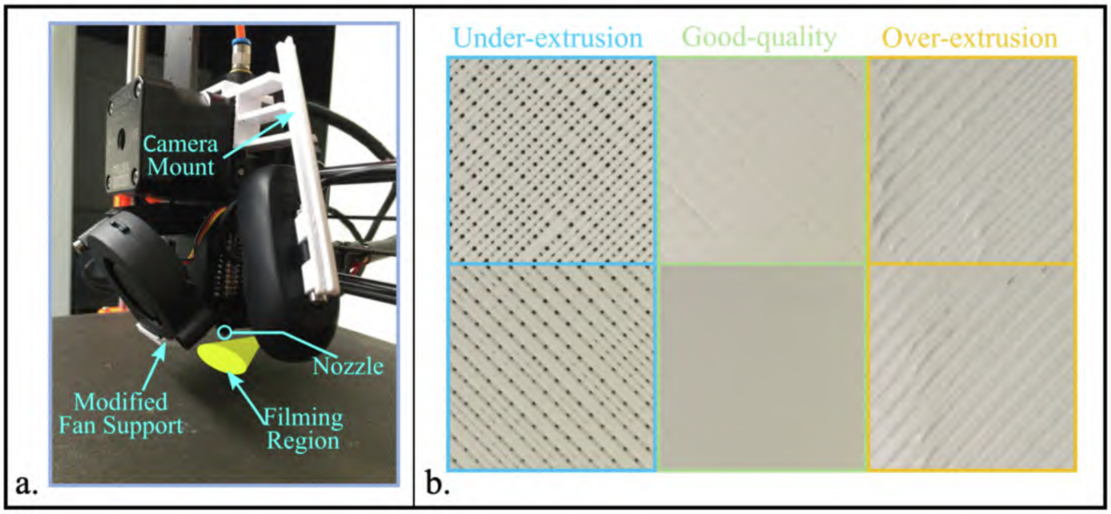

Figure 2.12: The experimental setup and product quality under different extrusion conditions: (a) camera setup on the 3D printer; and (b) representative images for 3D-printed blocks under different printing qualities categories of under-extrusion, good-quality, and over-extrusion.

achieve product surface quality assessment by analyzing the product texture images. Huang et al. proposed a statistical process monitoring (SPM) approach based on image data from a optical camera [126]. As the images' mean intensity values change when the printing process is in different states (e.g., overfill and underfill), the monitoring for the manufacturing process can be achieved by calculating the generalized likelihood ratio (GLR) statistics for each image's determination of regions of interest (ROI). The case study shows the proposed approach is effective in monitoring surface roughness for the FFF printing process. A series of methods were proposed by Okarma and Fastowicz et al. for assessing printing product surface quality. For example, an approach that investigates the product quality based on a gray-level co-occurrence matrix (GLCM) and Haralick features to analyze printing images for texture analysis was proposed by them [213]. This quality assessment method could lead to satisfactory surface distortion detecting and promising future deployment into the 3D printing process. A full reference image quality metric for assessing 3D printed product surface roughness was designed by Okarma et al.. It uses printing process images captured by a camera and could achieve a classification accuracy of 96.8% [215]. Another method is color independent product surface quality assessment based on image entropy, which can be applied for 3D printing monitoring as well as the product quality inspection without considering the filament color [214]. An improved color surface quality assessment approach based on image entropy was proposed recently by Okarma and Fastowicz [217]. With less than 100 samples, this color-independent quality assessment approach can identify the high or low surface quality. Beyond the approaches mentioned above, other quality assessment approaches were introduced by Fastowicz et al.. The first approach uses Hough transform and histogram equalization, which has a very promising classification accuracy, especially for the scanned images [156]. The second approach is an automatic product surface quality assessment based on the entropy of depth map images [74]. By equipping the fringe pattern-based 3D scanner on the 3D printer to get the most helpful product surface images, this approach's classification accuracy can exceed 90%, with the F1-score over 92%. As the authors have published a series of related papers about surface quality assessment, the remaining papers are listed here for the convenience of the reader [75–77, 216, 218, 219].

Model-based product surface roughness quality monitoring

In addition to the data-driven approaches discussed above, different model-based approaches have been developed, including the utilization of CAD model to detect and quantify the surface roughness, models to understand the printing parameters effect on surface roughness, and surface roughness prediction models.

Precision CAD model is a common component that is used to assess the surface roughness quality. These CAD model-based methods often utilize advanced algorithms in combination with models to detect and quantify product surface roughness. Sohnius et al., proposed a method which can predict the location of surface defects [265]. For this method, two 2D laser profilometers are mounted on the 3D printer's extruder to record the point cloud's surface profile. By comparing the target model path (taken from the CAD model) with the actual path (produce from the point cloud), the surface defects can be detected. In this research, three ML algorithms (random forest, decision tree, and artificial neural network) are utilized. The result shows that the classification accuracy for the gap defect is over 95%. Cheng et al. built a closed-loop online process control model for increasing the product surface quality [53]. In their system, 3D surface reconstruction is achieved by intensity images obtained from a CCD camera based on the CAD model. By applying a 3D reconstruction surface with a machine vision (MV) process feedback system, the filament flow rate could be adjusted to reduce the over-extrusion or under-extrusion defect. Hurd et al. developed a quality assurance for the AM process using mobile computing [128]. The system has four parts: (1) 3D printer, (2) a host computer that controls the 3D printer with G-code, (3) a mobile device to achieve the visual quality assurance, and (4) a server that supplies communication between the mobile device and host computer. Two image processing algorithms (i.e., image subtraction and image searching) are used to detect filament misprint by comparing the printing layer's image with the reference image from the 3D model. The proposed system can detect printing errors and determine whether the printing should be stopped.

Printing process parameters have been studied to quantify their effects on surface roughness. Jin et al. developed a new surface profile quantitative analysis model for the FFF product [10]. The printing process parameters (i.e., layer thickness and stratification angle) and fabrication process parameters (ratio between flow rate and feed rate) are the two groups used for examining their effects on the surface quality. The experimental result verifies the proposed model is feasible and effective on surface quality enhancement by optimizing the printing process parameters. Chaidas et al. investigated the temperature changing effect on the surface roughness [49]. The result shows that with the temperature increasing, the product surface roughness decreased. Akande et al. tried to find the optimum printing parameters to manufacture products with good surface quality and dimension accuracy [5]. Layer thickness, printing speed, and fill density are chosen as printing factors. The experimental results show that layer thickness is the most significant printing parameter for both surface roughness and dimension accuracy. Galantucci et al. carried out the research about the tip size, raster width, and slice height, three parameters that have an effect on product surface roughness [94]. The results verify that the slice height and raster width has a more significant impact on surface roughness.

The surface roughness prediction model based on printing parameters is also a research hotspot, as various prediction models have been developed. Ahn et al. put forward an approach to formulate the FFF product surface roughness, which is based on the main printing parameters (surface angle, layer thickness, cross-sectional shape of the filament, and overlap interval). The research investigates the effects of printing parameters on the product surface quality [4]. Comparing the result between the real surface data with the estimated values shows the proposed model can get elaborate surface roughness predictions. Kaji et al. proposed a methodology to predict the final product surface roughness quality, which is based on the layer thickness and local surface slope [140]. A data-driven product surface quality predictive model for the FFF process was proposed by Li et al. [171, 319]. The author chooses extruder temperature, layer thickness, and print speed ratio to extrusion rate as input data to understand the effect of the printing parameters on surface roughness. After the feature extraction process, a random forest algorithm is utilized to predict the surface roughness. Dambatta et al. built an adaptive neuro-fuzzy inference system (ANFIS) model to predict the surface roughness for FFF parts [62]. Layer thickness, deposition orientation, and geometry are chosen as the input data, the surface roughness is the output. The ANFIS model shows a 93.34% surface roughness prediction accuracy for a fabricated prototype. Vahabli et al. built a hybrid model to estimate surface roughness distribution [299]. The layer thickness and deposition are chosen as input, while the response variable is arithmetic mean surface roughness. The proposed model is demonstrated on various printing samples, and the result shows that the hybrid model can enhance prediction accuracy compared to the previous models. Vahabli et al. raised a new methodology for estimating the surface roughness based on radial basis function neural networks (RBFNNs) [298]. In this research, the input data is layer thickness and build angle; output is the arithmetic mean surface roughness. After optimizing the RBFNNs model by the imperialist competitive algorithm (ICA), the optimized model can get a surface roughness estimation of 3.64% mean absolute percentage error and 2.27% mean squared error.

To reduce surface roughness predictive errors in FFF, Taufik et al. proposed an innovative method to analyze and estimate the randomness in the build edge profiles' geometry [283]. The research result shows that the proposed method can reduce the surface roughness prediction error compared to the existed methods. Barrios et al. compared three different algorithms' (i.e., J48 algorithm, random tree, and random forest) surface roughness prediction performance in FFF [18]. The training data is the samples printed with five different factors (layer height, temperature, print speed, print acceleration, and flow rate) and three distinct levels. The results show that the random tree performs best in this case of study, with an 80% accuracy for the surface roughness in the direction parallel to the direction of extrusion and 86.67% for the surface roughness direction perpendicular to the direction of extrusion. Angelo et al. verified that to get an accurate surface error, it is not adequate to use just the roughness parameters [66]. Angelo et al. proposed a better index, the parameter P_a (ISO 4287 [1]), for evaluating product surface quality and a new model to predict P_a for the product surface roughness manufactured by the FFF process. The new model prediction result shows a good match with the empirical values.

A series of surface roughness prediction methods were put forward by Boschetto et al. for the FFF process. A feed-forward neural network surface roughness predictive approach was proposed [31]. The layer thickness and deposition angle are chosen as input data. After being fully trained, this method can predict the average roughness, matching the actual data very well. A methodology that integrates the surface quality prediction and process design was built by Boschetto et al. [35]. Using layer thickness and printing angle as input, this method can estimate the product dimension accuracy and average roughness. Since the product average roughness is not able to reflect the real surface roughness, Boschetto et al. proposed a novel 3D roughness profile model to extend the characterization to all roughness related parameters by a profilometric analysis [36]. The method can predict a whole surface characterization for the FFF process. Another model was built by Boschetto et al. to examine the coupled operations of barrel finishing and FFF [33]. The built model can predict the surface roughness by using printing process parameters (layer thickness and deposition angle) with the material quantity removed by barrel finishing.

In summary, these model-based approaches emphasize the important roles of artificial intelligence-based image processing and use of data-driven process parameters in the determination of surface roughness of FFF processes.

Product infill printing defect detection

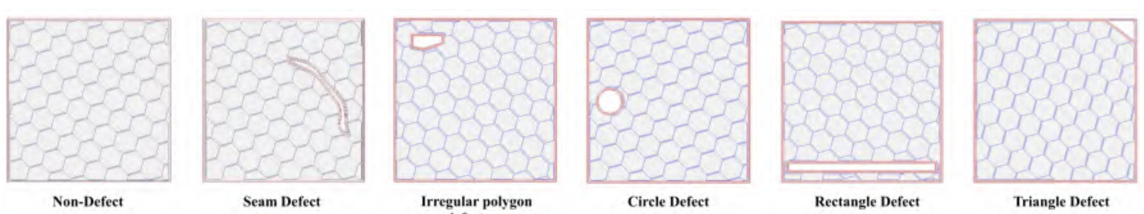

Figure 2.13: Types of designed infill defect patterns.

In contrast to the surface faults and defects discussed in the prior sections, the detection of defects in product infill is especially hard as once the part is completed, detecting internal faults is challenging. Due to the thin nature of the build-up on infill, often only one track wide, improper parameters setting (e.g., extruder temperature and printing speed) can cause faults in the infill affecting the exterior where more tracks help to compensate for small variations in printing. Furthermore, infill is especially vulnerable to cyber-attacks as an attacker could alter the infill in such a way to compromise the part that would be nearly impossible to detect once the part is fully completed [182, 276].

Monitoring of infill quality has been highlighted as a potential method for defending FFF from cyber-attacks [182]. Wu et al. presented an approach to detect malicious infill defects of the 3D printing process using a classification-based ML method to analyze the printing process images from the camera [324, 325]. With a naive Bayes classifier, an infill defects prediction accuracy of 85.26% could be achieved. Furthermore, if J48 decision trees are utilized instead of naive Bayes classifiers, 95.51% prediction accuracy could be achieved. Fig. 2.13 shows five different designed infill defect patterns: seam, irregular polygon, circle, rectangle, and triangle defects with 10% honeycomb infill, which almost cover the common infill defect patterns. Gao et al. built an online 3D printer monitoring system for defending against cyber-physical attacks where infill path attack is one of the monitored aspects [95]. The extruder's movement is tracked by using magnetometers to detect magnetic field intensity. The system matches the expected infill path with the measured infill. The average Hausdorff distance (Hausdorff distance describes the largest distance between sample points in two curves [202]) is obtained to quantify error. For 50 layers with different infill rates (from 5% to 20%), the measured error is less than 1 mm. Results show that this system could be used for monitoring the printing infill path error by comparing the extruder's movement path with the designed infill path. By integrating multiple heterogeneous sensors, a smart FFF 3D printer was built by Moretti et al., capable of monitoring the printing process and product quality [200]. With the camera's help, the poor cohesion defect between the infill and the outer wall can be detected from the layer image.

As different infill level impacts the printing product's structural integrity, the product's infill level condition was also studied. Straud et al. used visible light imagery to assess the quality and sufficiency of 3D printing product infill quantity [272]. According to the assessment result by the visible light imagery-based assessment method, this technique can determine if the product's current infill level is appropriate or not.

Product geometry error detection

A product's geometry accuracy is easily affected by printing process errors such as nozzle clogging, improper build platform temperature setting, or bad leveling calibration. These unanticipated anomalies influence either directly or indirectly product quality, which may produce defects such as product geometry deviation or deformation [43, 242]. For these reasons, an in situ monitoring and detection system for product geometry defaults is worthwhile to avoid wasted energy, time, and material [26, 244]. Discussion about the product geometry error detection also divides into data-driven and model-based approaches.

Data-driven product geometry error detection

To detect product geometry error, different data-driven approaches have been studied. In these studies, ML algorithms are often used to analyze real-time images or videos to monitor product geometry conditions. An automated geometry defect detection approach was developed by Narayanan et al. [204]. This approach could classify good and defective polymer products manufactured using the FFF process by analyzing the printing product geometry images. Two independent ML methods are implemented to detect geometry defects. First, the principal component analysis

(PCA) is utilized to reduce the images' dimension, and then an SVM algorithm is used to classify the product as good or defective, thus achieving an overall accuracy of 98.2%. Next, a deep-learning approach that utilizes a convolutional neural network (CNN) is investigated. This method achieves a 99.5% classification accuracy. Delli et al. built a real-time 3D printing monitoring system by integrating image processing with supervised machine learning [65]. The system is capable of detecting unintended failures such as filament runout or abnormal stop and can detect product geometrical defects during the printing process. Zhang et al. utilized a CNN algorithm to detect unfinished printing failure for the material extrusion 3D printing process [357]. The overall average accuracy in this approach is 70%.

Other non-ML approaches also show good potential for detecting product geometry errors. Yi et al. proposed a defect detection method for the FFF process, which combines the statistical process control (SPC) with machine vision [326, 343]. Utilizing SPC to analyze the extracted contour data from the printing process images captured by the camera, the contour profile defect can be detected. The result shows that the monitoring accuracy is 0.5 mm. A zero-defect AM process was developed by Faes et al. [71]. By implementing a closed feedback loop with sensors and a laser triangulation system, the developed algorithm can detect the deposited tracks and identify the z-direction geometrical error when the track is lower than one layer thickness. Li et al. applied a coherent gradient sensing (CGS) system to achieve the in situ deformation monitoring [163]. CGS technology is a full-field, lateral shearing interferometric method, with some advantages, such as real-time, full-field, intuitive, and vibration resistant. Moreover, this method can potentially be utilized in stress analysis for the 3D printing product. The study evaluates the deformation in three different cooling processes (i.e., fast cooling stage, transitional cooling stage, and slow cooling stage). Li et al. proposed a distortion defect diagnosis by utilizing the AE sensor to get the build platform's vibration data [162]. After filtering out the noise from the feature extraction process, this method shows promise for distortion detection; however, determining accuracy needs further study.

Model-based product geometry error detection

Model-based approaches are a common method for product geometry error detection. Significant research has been done to build models for evaluating product geometry conditions. Methods used include those based on CAD and point cloud models. These methods seek to improve and to predict product geometry conditions.

CAD models are one of the most widely used methods to detect product geometrical defects. By comparing the in-process product image with the final product image (e.g., 3-D reconstruction product images and CAD model image), the variations in geometry can be detected. Using cameras for "profile monitoring" has been investigated. For example, Nuchitprasitchai et al. developed a camera-based error detection system [209, 211]. In this research, two different systems with either a single-camera and double-camera setup were built. For the single-camera error detection system, the error is detected by comparing the STL image (2D profile extracted from the 3D STL file) with the camera image (2D image captured from the 3D printing product). After rescaling, a printing error is considered as a defect if the differences calculated by subtracting the STL image from the camera image are over 5%. To detect the printing error with the two-camera system, a 3D reconstruction product image was built. For this setup, the printed error between the 3D reconstruction product image and the 3D printed product is detected. If the difference is more than 5%, in which case, an action is taken to stop the printing. A profile monitoring based FFF product quality control approach was proposed by He et al. [118]. The product dimension deviation is obtained by comparing the built profile with the CAD profile layer by layer. The designed system could detect defects, such as dislocation, staircase, and gradual change.

More complex multi-camera systems have also been developed. Straub et al. built a multi-camera system that could achieve full image coverage of the product [271]. An imaging processing technique is applied in this research that compared images taken during the product printing progress to expected product images (e.g., CAD files) to assess the printing process. The system demonstrated a capability for detecting product defects. Fig. 2.14 shows the images of different printing stages and the pixel difference between the in-process image and the final product. Fig. 2.14a shows the finished object image. The partial object displays in Fig. 2.14b. Fig. 2.14c and Fig. 2.14d depicts the partial-complete difference comparison and threshold-exceeding pixels identification. By calculating the aggregate level of the pixel difference between the in-process image with the final product one, just as shown in Fig. 2.14, the system could detect the incomplete failure. Another error detection system was also developed by Nuchitprasitchai et al., which uses six web cameras to achieve 360 degrees coverage for printed product monitoring [210]. The printing error can be calculated in both the horizontal and vertical magnitude directions by comparing the current printing layer's 3-D reconstruction images with STL image models. Moreover, printing errors such as extruder clogging, filament runout, and incomplete product print can be automatically detected by checking the difference between 3-D reconstruction images and STL model images. The non-rescale and rectification image pre-processing technique in this study has a faster computation speed, and the detection accuracy can achieve 100%.

Laser scanning technology, which has high measurement accuracy and resolution, has also been implemented into 3D printing process monitoring. Comparing the point cloud from the laser scanning with the CAD model, the product geometry error can be recognized. Lin et al. proposed an online quality monitoring system for the material extrusion AM process, based on laser scanning technology [172]. There are three steps to detect the defects: (1) point cloud processing; (2) the comparison process

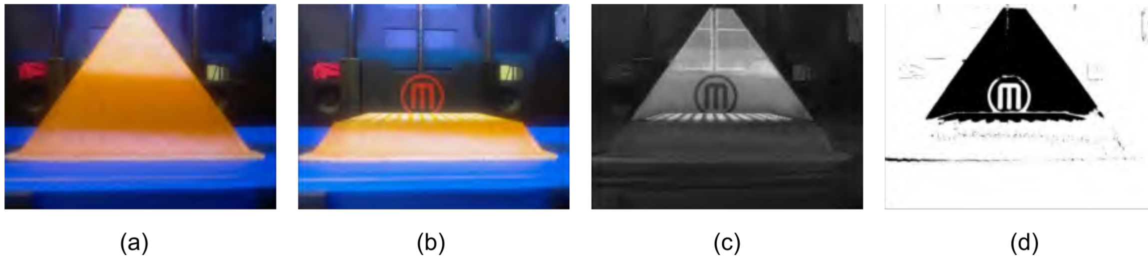

Figure 2.14: Images from one angle: (a) the completed product image; (b) the partial product; (c) and (d) the partial-complete difference comparison and threshold-exceeding pixels identification.

between the CAD data with the scan data; and (3) defect reconstruction process. The proposed system not only effectually detects the underfill defect (e.g., gaps in the thin wall, under-extrusion, and unfinished product) and overfill defect (e.g., over-extrusion and scars) but also can reconstruct the 3D defect model. By combining the laser-scanned coordinates dataset with the self-organizing map (SOM) unsupervised machine learning approach, Khanzadeh et al. presented an approach to quantify the FFF printed product geometric deviation [146]. The geometric deviation is gotten from a comparison between the laser-scanned data with the original CAD data. With the SOM cluster, the geometric deviation on magnitude (severity) and direction can be recognized. Rao et al. proposed a spectral graph theory (SGT) method to quantify the complex AM product dimension deviation [237]. An SGT method achieves the dimensional integrity monitoring by calculating the deviation value based on the normal distance from a point on the 3D point cloud to the CAD model's closest surface. Tootooni et al. built a ML dimensional variation classification method [293]. The spectral graph Laplacian eigenvalues are utilized to extract the features from the laser-scanned 3D point cloud data. Afterward, six ML algorithms (sparse representation, k-nearest neighbors, neural network, naive Bayes, support vector machine, and decision tree) are applied to evaluate the classification accuracy. The result shows that the sparse representation algorithm obtains the best classification accuracy (F1-score $> 97\%$).

In addition to the point cloud generated from the laser scanning, the point cloud from G-code has also been investigated. Holzmond et al. presented a quality assurance system named "certify-as-you-build" [121]. This system obtains product geometry using a three-dimensional digital image correlation (3D-DIC) and then compares the obtained product geometry with the point cloud generated from the G-code command to detect the defects. The local defect (e.g., filament blob) and global defect (e.g., low flow) can be detected and located. Lastly, Kopsacheilis et al. presented an in situ vision-based monitoring system for FFF [152]. By comparing the theoretical point cloud 3D model generated from the G-code command with the reconstructed printed model obtained from the camera, the system is able to evaluate the product accuracy in real-time.

Augmented vision is an emerging technology that has recently been applied to monitoring 3D printer product geometry quality. Ceruti et al. developed an error detection system by applying augmented vision [47]. In this system, a virtual product to be printed is matched to the actual one, so the product geometry in different printing stages can be monitored. If manufacturing errors are detected, the printing process could then be stopped automatically or by the operator. Malik et al. proposed a novel scan-based real-time layer-by-layer monitoring system for the AM process [187]. To reconstruct the 3D model, the printing process image is converted to a 3D file OBJ format. By viewing the generated OBJ file through augmented reality, defects can be clearly visualized.

To improve the product dimension accuracy in FFF, various models related to the printing parameters have been built. Boschetto et al. defined a mathematical formulation, which is related to layer thickness and deposition angle, and can be used as a guide for compensating dimensional derivations [34]. With this method, it is possible to get a product with good dimension accuracy without removing the physical error source. In three study cases, the model has a marked dimension deviation reduction. Kaveh et al. investigated how printing parameters affect the product precision and internal cavity for the material without knowing the printing parameters [142]. Then, the presented method tries to find the optimum extruder temperature and raster width settings. The demonstration shows that with the optimized printing parameters, the printed product internal cavity is negligible. Alizadeh et al. built a data-driven model that combines energy consumption with product geometric accuracy [8]. In this research, thickness deviation, product's out-of-tolerance percentage, build time, and energy consumption are selected as the variables. The proposed method is demonstrated and validated its effectiveness by printing on FFF product.

The prediction of a final product's dimension accuracy has also been performed using monitored or known printing parameters. To predict the product dimension deviation for the FFF process, Boschetto et al. developed a model [32] that uses the layer thickness and build orientation as inputs. The prediction value from this model shows a good fit with the experimental data. Noriega et al. proposed a model for predicting the actual printing product dimensions based on the designed characteristics [208]. An artificial neural network is used to predict the actual dimension values; then, an optimization algorithm applies to decide the CAD model's optimal dimensional values. Based on the optimal dimensional values, the CAD model is redesigned. The experimental result shows this methodology could reduce the external dimension error around 50% and internal dimension error around 30%. Yang et al. established a FFF product precision prediction model, which is based on printing process parameters [333]. The input process parameters data is the cable width offset, layer thickness, filling speed, extrusion speed, and the fallback speed. A cyber-model based on printing settings was developed by Miao et al. to predict the deformation during the FFF process [190]. The prediction experiment shows that the linear regression (LN) model performs best compared with the SVM and artificial neural network model. Based on this LN model, a cyber-physical system (CPS) was developed, which could adjust the nozzle temperature automatically. The evaluation experiment indicates this CPS could reduce the distortion remarkably. Song et al. addressed a shape deviation model for the FFF process [266]. In this research, the author attributes the extruder position error and processing error, including phase change and other occurred variation as two error sources, which impact the product consecutively. The Kriging method is applied to predict the extruder position error pattern. Both the experimental and case study shows that the proposed model could successfully grasp the deviation trend. Hebda et al. presented a method to predict the product geometric characteristics for the FFF process [119]. Base on the proposed equations, the product height, width, and cross-section area can be predicted for the given printing parameters.

2.5 Conclusion and future trends

FFF is the most commonly used 3D printing method due to its advantages of low production cost and the ability to create complicated geometries and shapes. However, its lower reliability means that significant work has been undertaken for the in situ monitoring of FFF as a first step to enabling robust closed-loop control of the FFF process. This paper summarized recent research focused on the in situ monitoring system for the FFF process. In situ monitoring is a driver for the next-generation of systems and polymers in AM. Their implementation will continue to increase the quality, efficiency, and sustainability of polymer components manufactured using the FFF process.

Sensing systems play a key role in the success of in situ monitoring systems. While indirect sensing methods (e.g., vibration and acoustic emissions) are useful for detecting whether a system is experiencing a fault, their ability to link a signal to a specific fault source is limited by the "uniqueness" in their signal. The "uniqueness" issue for indirect sensing methods represents a significant knowledge gap that limits the use of indirect sensing methods for in situ monitoring. Thermal and optical cameras are the most common method for in situ monitoring and have demonstrated success in fault detection, localization, and quantification.

A multitude of data-driven methods for detection printer fault scenarios have been developed, however, the issue with "uniqueness" in their signal remains. Model-based and data-driven approaches for the in situ monitoring of components during printing have also been developed. These methods have demonstrated success in detecting geometry and surface roughness errors.

Great progress has been made in the field of in situ monitoring for the FFF process. However, a knowledge gap still exists in how to use the information gained for the advancement of the FFF process in industrial manufacturing settings. Most importantly, the use of in situ monitoring for structural fault detection in components is yet to be realized. Structural faults are those that limit the structural functionality of a component, rather than just detecting faults related to component geometry or surface roughness that presently obtains. If structural fault detection can be widely achieved, the results could be integrated into part validation methodologies to empower in situ validation, with limited destructive or non-destructive testing of components post-print. Component validation using in situ monitoring would provide a significant forward leap to the FFF process. While data-driven approaches for structural fault detection may be relatively easy to develop, their "black box" approach makes them challenging to integrate into validation procedures for critical components. Model-based and or physics-based approaches that directly consider the thermodynamic and kinetic states of the component during the manufacturing process will allow for direct fault detection, localization, and quantification. Therefore, it is envisioned that the model-based approaches offer the most realistic approach to achieving in situ component validation.

A unified framework that can flexibly integrate available sensors, algorithms and computational models will truly accelerate technology advancement. In the foreseeable future, the FFF system will be realized without a human-in-the-loop [270, 293, 298]. Moreover, the ability to detect and prevent cyber attacks in real-time offers a path forward for future research [96, 263, 325].

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Chapter 3

Real-time Structural Validation for Material Extrusion Additive Manufacturing[1]

3.1 Abstract

Fused filament fabrication (FFF) is a material extrusion additive manufacturing (MEXAM) technology that produces a part through thermoplastic filament's controlled melting and extrusion. While popular for small-scale 3D printing, its wide-scale adoption for industrial applications still faces significant challenges due to the process's intrinsic uncertainties. A trivial defect in a specific position can significantly impact the whole product's structural quality and mechanical properties. Different defect detection systems have been built and have already achieved good detection accuracy by the researchers. However, the ultimate goal should focus on the defect's effect on product quality rather than detecting the defect. This research developed a real-time defect detection and product structural validation system on a common consumer-grade 3D printer with an optical camera. The layer-wise image information will be output from the fully trained CNN (convolutional neural network). The developed defect detection system links the output information with the product structural quality decision boundary to evaluate the defect impact and then make the proper decision. Experimental results show that the CNN model has a high training and validation accuracy, which means the CNN model can output the correct layer-wise

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information. Meanwhile, the decision boundary obtained from the support vector machine with radial basis function kernel also performs well on both training and validation data, which means the decision boundary can provide a solid reference based on layer-wise product information. The proposed real-time defect detection multi-dimensional accumulation-threshold-based decision-making system has an excellent accuracy based on different defect information to alert the users to continue or cancel printing.

3.2 Introduction

Material extrusion additive manufacturing (MEXAM), also known as fused filament fabrication or fused deposition modeling, is one of the fastest-growing, most promising, and widely-used additive manufacturing technologies [112]. The unique affordability of MEXAM has made it a popular technique for small-scale 3D printing. However, its wide-scale adoption for industrial applications still faces significant challenges, especially for Industry 4.0, due to the process's intrinsic uncertainties [68,158]. MEXAM is a repetitive process: the molten material squeezes from the nozzle to form a thin, tacky bead of plastic that adheres and bonds with the previous layer until the part is finished. During the printing process, defects, such as bonding gaps and/or material voids, can dramatically decrease the final product's structural quality and mechanical properties [72, 258]. Therefore, a defect detection system for MEXAM that reduces part-to-part variations is essential to ensure the product's mechanical properties.

Machine learning (ML) is an effective way to detect defects or guarantee product quality for MEXAM due to its ability to link implicit information with the final prediction or classification outcome [67]. Extensive ML systems for MEXAM have been built to detect various faults, including bonding quality problems and infill defects [86]. For example, Jin et al. used an optical camera mounted on a material extrusion printer to capture different product quality images for under-extrusion, good-quality, and over-extrusion conditions [139]. These images were used to build a dataset, which was used to train a ResNet 50 architecture convolutional neural network (CNN) classification algorithm. Online monitoring was implemented by feeding the captured images into the pre-trained classification model to get the current printing condition. Banadaki et al. built a reliable online quality monitoring system with a deep convolutional neural network (DCNN) algorithm [17]. After training the model with images captured at each layer, the proposed online model can classify five different quality grades resulting from overfilled or underfilled with an average accuracy of 94%. Wu et al. presented an approach to detect malicious infill defects injected into the 3D printing process using a classification-based ML method to analyze images from a camera [325]. The achieved infill defect prediction accuracies are 85.26% and 95.51% with a naive Bayes classifier and a J48 decision tree, respectively. Holzmond et al. presented a MEXAM product quality assurance system that uses a three-dimensional digital image correlation system to capture the geometry and then compare the obtained geometry with the computer model to achieve in situ printing error detection [121]. A validation experiment was done on a MEXAM printer, which demonstrates the in situ detection of localized and global defects. An in situ infrared temperature sensing for real-time fused deposition modeling defect detection was studied by Hossain et al. [160]. A statistical method for defect detection is exploited and applied to detect temperature deviations, which are caused by deliberately rooted defects. Khan et al. utilized a CNN algorithm to model to detect real-time malicious defects in the printing process, which is based on feature extraction of geometrical anomalies occurring in infill patterns [143]. Charalampous et al. presented a vision-based real-time monitoring of MEXAM for automatic manufacturing error detection [51]. The proposed method compared the high-resolution point cloud data of the printed part with the corresponding digital 3D model in various phases of the printing process to automatically monitor and detect the defect. Cases validate the efficacy of the proposed methodology.

The ultimate goal of defect detection is to guarantee product quality rather than just find defects. Degradation in functional qualities can be caused by the accumulation of smaller defects that, on their own, do not constitute a critical flaw in the component. Therefore, an effective online quality validation system must consider the accumulation of smaller defects when attempting to guarantee product quality. To expand, the proposed system should not only detect visible defects but also diagnose their direct and compound effects on product functional qualities (e.g., structural performance) caused by variations in the printing process (e.g., abnormal extrusion).

In this paper, a real-time ML-based product structural quality validation approach is proposed. The novel method is capable of inferring structural quality rather than just detecting surface defects and uses an accumulation-threshold-based decision-making system to account for the accumulation of smaller defects. If a single defect or the accumulation of smaller defects has a limited effect on product quality, this means the defect is ignorable, and the printing can continue. Otherwise, the printing process needs to stop once the defect is impactful. In addition, a novel component health index that links the size of defects across layers is developed in this work. The component health index accumulates defect information throughout the whole sample printing process. The ML-based product defect detection approach uses the CNN algorithm to classify the defect type. Tensile tests are done on the collected samples with different defects to get the effect-of-defect on product quality. The decision boundary is built in the component health index domain using a support vector machine (SVM) with a radial basis function kernel trained against the ultimate tensile load obtained from mechanical testing. Layer-wise images are fed into the trained CNN model to classify defects during the printing process. Then the accumulated component health index is mapped to the decision boundary model to evaluate the effect-of-defect on product quality. Based on the assumption that all un-printed layers of the part are of good quality, the multi-dimensional accumulation-threshold-based decision-making system will make the proper decision (i.e., continue or cancel printing) by referring to the evaluation result before the next layer is finished. The real-time evaluation and decision-making continue for every layer until the whole part is done or the impactful defect occurs.

The experimental setup and methodology are discussed in Section 3.3, which has five sections: experiment methodology, experiment setup, data-driven model for defect detection, component health index, and defect impact boundary. Section 3.4 discusses the experiment results and provides a discussion, which includes training results for the data-driven defect detection, results for the defect impact decision boundary process, and results of the real-time defect detection and decision-making process. Section 3.5 presents the conclusion and future work.

3.3 Experimental setup and methodology

Experiment methodology

The real-time structural quality validation system for MEXAM is diagrammed in Fig. 3.1. The system has three parts:

1. Offline defect detection training.

2. Offline decision boundary training.

3. Online real-time decision-making.

In the offline defect detection training process, a dataset is built that consists of images taken at each layer during manufacturing with an optical camera. Images are then labeled manually with the defect information (i.e., defect length and location). The dataset is used to train the CNN model. The trained CNN model is deployed to infer the corresponding defect length and location during the normal printing process in real-time. The data-driven ML model is discussed in greater detail in Section 4.3.

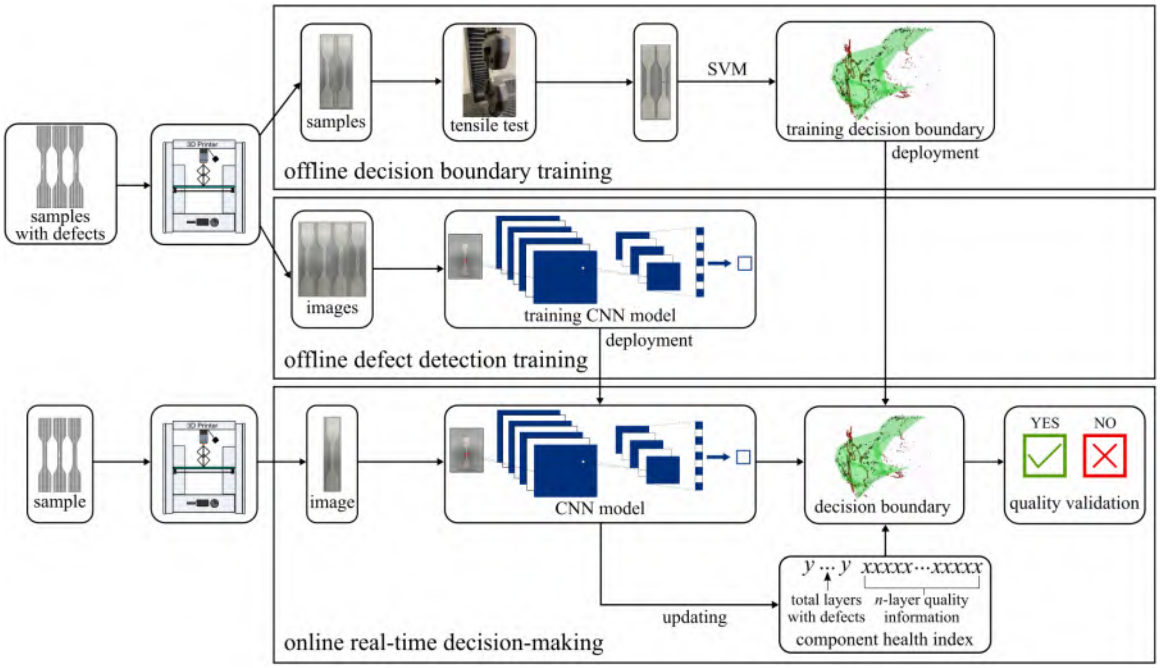


Figure 3.1: Diagram for the real-time structural quality validation approach in the MEXAM printing process.

The decision boundary is trained offline on data collected from the tensile tests for the samples with defects to evaluate the effect-of-defect [284, 285]. The defect is considered impactful when it reduces the structure's tensile strength by 3% [188], an arbitrary threshold based on the author's experience; and breaks at the seeded defect position. Otherwise, the defect is ignorable, which means the defect has a limited effect on product quality. The defect impact decision boundary is built using an SVM with the radial basis function kernel based on the tensile test results from the collected samples with various defects. The decision boundary lays the foundation for the accumulation-threshold-based effect-of-defect inference. The trained defect impact decision boundary is deployed for online real-time decision-making. A decision on the structural quality of the part is made for every layer in the build. At every layer, the inference from the CNN model will be accumulated with the previous information and then mapped to the defect impact decision boundary to evaluate the impact on product quality. Based on the evaluation result, the accumulation-threshold-based decision-making system will make the proper decision [262].

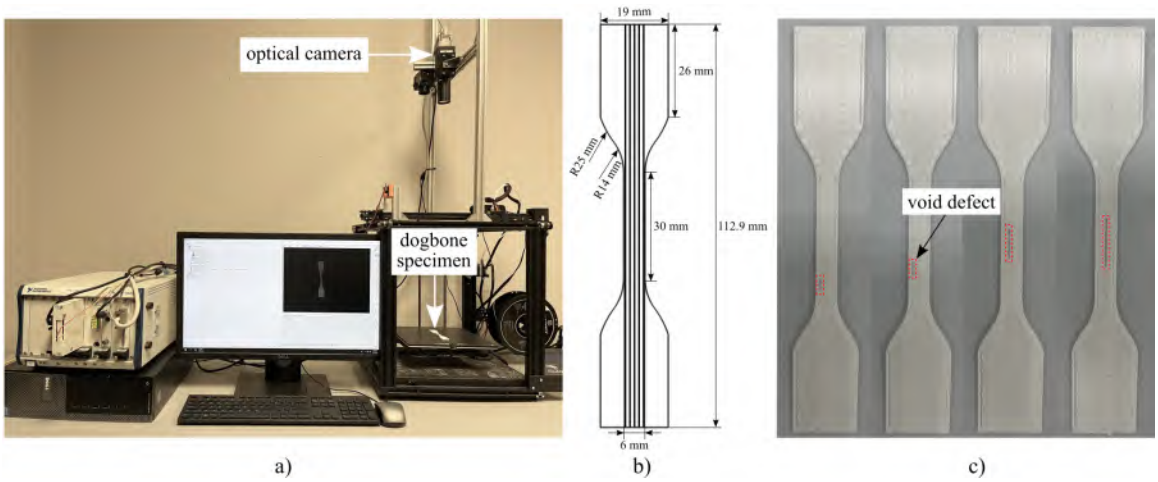
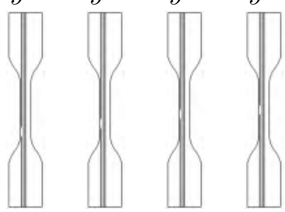
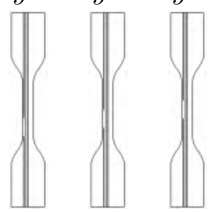
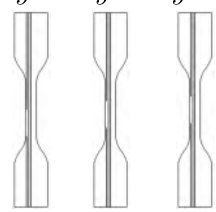
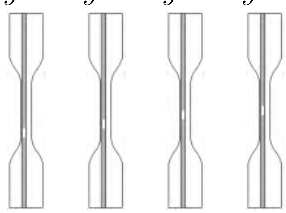
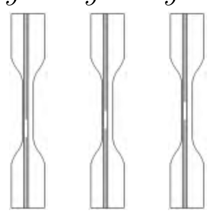
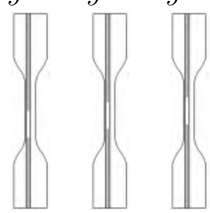
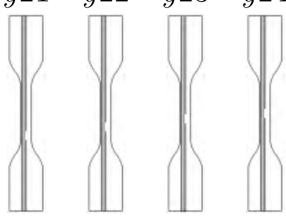
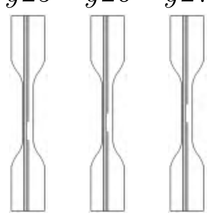
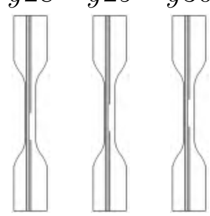


Figure 3.2: The experimental platform and designed sample: a) experimental platform for defect detection and product structural quality validation; b) the designed sample dimension; and c) various designed defects on real printed samples.

Experimental setup

Samples are printed with a printer (manufacturer: Creality, model: Ender 5, city: Shenzhen, country: China) with Polylactic Acid filament (manufacturer: Hatchbox, thickness:1.75 mm, city: Los Angeles, country: United States). Fig. 3.2a) presents the setup of the experimental platform. An optical camera (manufacturer: JAI, model: CV-M4+ CL, city: Los Angeles, country: United States) with a resolution of 1368 $\times$ 1020 is fixed to the printer frame above the extruder to collect the printing images. The designed sample's G-code was modified before printing, so when one layer is completed, the printing head moves to a fixed position, which relocates out of the camera's field for the duration of the picture. At this time, the camera is triggered through a LabVIEW code to capture the layer image of the sample. With the setup, the relative position between the camera and specimens is fixed, providing consistent

Table 3.1: The designed defect with different lengths and locations within the dog-bone specimen.

length / location	5 mm	10 mm	15 mm
$x1$	$y1$ $y2$ $y3$ $y4$ 	$y5$ $y6$ $y7$ 	$y8$ $y9$ $y10$ 
$x2$	$y11$ $y12$ $y13$ $y14$ 	$y15$ $y16$ $y17$ 	$y18$ $y19$ $y20$ 
$x3$	$y21$ $y22$ $y23$ $y24$ 	$y25$ $y26$ $y27$ 	$y28$ $y29$ $y30$ 

images with an identical scale across layers.

As MEXAM manufactures the part by depositing the plastic filament, each layer is bonded to the adjacent layers. The overlap regions (inter-layer necking) between the adjacent layers bond together through filament interface diffusion to produce a semi-solid part [58, 257]. The voids between tracks introduced during the printing may be ignorable, which means they have a tolerable impact on the part's mechanical property. However, large gaps in the extruded polymer undermine the product's quality [78].

The defects in this research are produced by turning the extrusion off, which simulates the abnormality of the failure-to-feed during the printing process. The dimensions of the designed dog-bone are shown in Fig. 3.2b), which is based on the American Society for Testing Materials (ASTM) D638 Type IV dog-bone test specimen [55]. The sample thickness is 3 mm with 100% infill. The diameter of the printing nozzle is 0.8 mm to accelerate printing speed and reduce the sample collection time. There are nine layers for the sample: the bottom and top layers are printed in the horizontal direction, and no defects happen on these two layers; the internal layers are printed in the vertical direction, and all defects are introduced into these seven layers. As shown in Fig. 3.2b), only five extrusion traces cross through the whole dog-bone, which are the actual extrusions that the defect occurs on. To increase the size of the training dataset, data augmentation (i.e., horizontal and vertical flipping, Gaussian blur, and Gaussian noise) is implemented.

Based on finite element analysis of failures for the printing parts by Garg et al., the stress distribution on the sample printed by MEXAM varies in diverse locations in gauge length and layers [100]. Thus, the defect in different locations and layers may have varying impacts on product structural quality. To figure out the defect's effects on product structural quality, four defect variables are used in this study: defect length, defect location, defect occurring layer, and the number of layers with defects.

Table. 3.1 reports the 30 designed defects with three defect lengths (5, 10, and 15 mm) used to investigate their impact on product quality. The x, y position is located at the center of the defect (Note that only the 30 mm straight traces cross through the whole dog-bone), which is used for calculating the defect location based on equation $\sqrt{x^2 + y^2}$. For the 5 mm defect, there are four y locations in the lower-left quadrant and one that crosses into the upper-left quadrant at one x location (x location is the traces cross through the entire dog-bone, which are termed line 1, line 2, and line 3). The defect on one line is either 2.5, 7.5, 12.5, or 15 mm in one of four different positions. Therefore, based on the equation above, the 5 mm defect has four potential locations (2.7, 7.6, 12.5, 15) on line 1. This style of defect is repeated for the 5, 10, and 15 mm defect, as shown in Table. 3.1. Fig. 3.2c) shows the different defect lengths in various locations. Note that even though the sample infill density is 100%, due to the big diameter of the nozzle, the slicing resolution is low, which produces the long void along the edge (the long voids shown in Fig. 4.4b) and c)).

Since the accumulated defects' impact on product structural quality is important to consider [37, 295], two variables, the layer of defect occurrence and the number of layers with defects, are defined. The details are presented in Section 3.3.

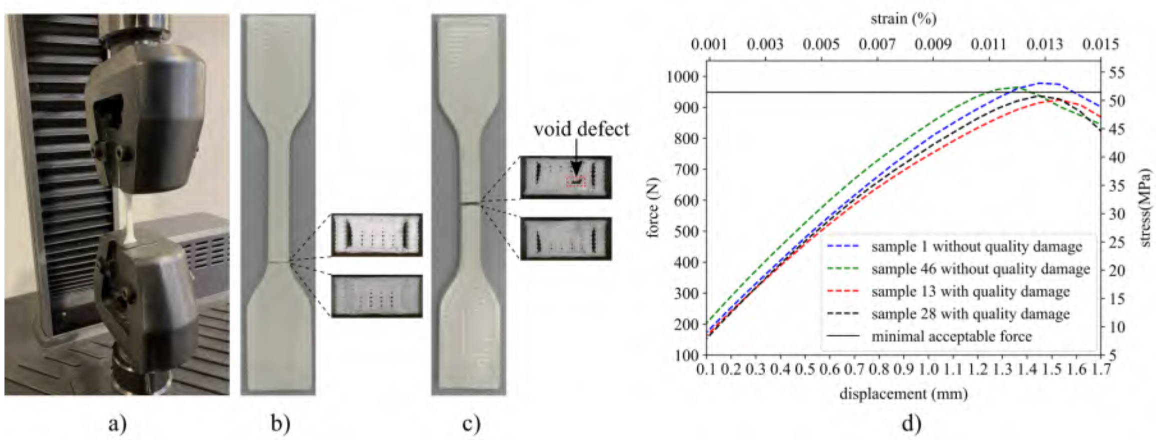


Figure 3.3: Printed sample tensile test: a) tensile test setup; b) the broken good quality sample with its cross-section; c) the broken sample with defect and its cross-section; and d) the tensile force results.

To train the product structural quality defect impact decision boundary, tensile tests are done for the collected samples. Mechanical validation is performed using ASTM D638 on an electromechanical load frame (manufacturer: MTS, model: Exceed E43, city: Minneapolis, country: United States), as shown in Fig. 3.3a). To have a thorough and accurate evaluation of defect impact, tensile force reduction with the rooted defect is considered [73]. The average tensile force for ten samples without seeded void defects is 977.9 N, with a standard deviation of 9.92 N. In addition, the average tensile force for ten good samples where images are not collected and therefore the layers are not allowed to cool during image acquisition is 982.4 N, with a standard deviation of 8.43 N. The tensile force for the good samples with and without image acquisition interval are within one standard deviation and therefore no significant difference is found. In this work, a 3% reduction in the ultimate force of the good sample with image capture is chosen as the minimal acceptable structural threshold [38, 305]; this relates to a 29.34 N reduction in the tensile force at failure. For example, even if a sample breaks at the location of a defect but the tensile force is larger than 948.6 N, the defect's effect on the product's mechanical properties is still treated as ignorable due to the maximum tensile force having no meaningful difference. Therefore, the hypothesis in this research is that the defect is considered impactful if and only if the specimen breaks with a tensile force of less than 948.6 N at the location of a detected defect. Fig. 3.3b) and c) show the broken sample's cross-section after the mechanical testing. Fig. 3.3b) displays the sample's broken cross-section without defect. Fig. 3.3c) is the sample with a 10 mm defect in the center of the $x2$ location; the break occurred at the defect (a red rectangle marks the void defect), which means this defect impacts the product's structural quality. Fig. 3.3d) shows the tensile forces of the samples with and without quality damage. Note that all the tensile tests in the dataset are recorded with displacement and force. The dataset developed for this work is made available in a public repository [Dataset-in-situ-validation-for-polymer-components], where it is Dataset-1 [87].

Data-driven model for defect detection

The CNN model in this research is developed with three convolutional layers, three max-pooling layers, two fully connected hidden layers, and one output layer. A network similar to this was previously demonstrated by the authors to have excellent defect detection capabilities [85]. The designed model architecture is shown in Fig. 3.4 and is based on lenet5 [157]. The convolutional kernel size adopted in this model is 3×3, and the filters are 32 with one stride. Except for a softmax activation function for the last output layer, all the rest are Rectified Linear Units (ReLUs). The raw image dimension of the sample is 1368 $\times$ 1020. To increase the training speed, the image resolution is reduced by resizing to 684 $\times$ 684 pixels before being input into the model. After the first convolution operation on the resized input image, the output feature map size is 682×682. A max-pooling layer with a 2×2 kernel size is performed after the first convolutional layer. The feature map reduces from 682×682 to 341×341 after downsampling. By performing the subsequent convolutional and max-pooling layers twice, the output feature map size changes to 83 $\times$ 83. After flattening all the feature maps, two fully connected hidden layers are used to depict the whole image's feature map by a one-dimensional vector, which dramatically reduces the size from 881792 to 128. The last output layer is also a fully connected layer with a total node of 31 (30 defect types with defect length and location, good quality). The output nodes represent the prediction categories from the algorithm, where each node maps to one defect type or good quality. The CNN model is trained offline using Keras and Tensorflow using the Adam optimizer [151]. After data augmentation, the final dataset contains 1550 images. For the labeled sample images, 80% of them are randomly chosen as training data, and the remaining 20% are used as test data. Training is done over 200 epochs. The defect length and defect location are output by the CNN model; however, with just an unlabeled image, the CNN can not tell which layer has the defect and the number of layers with defects. This is the job of the component health index, which is discussed next.

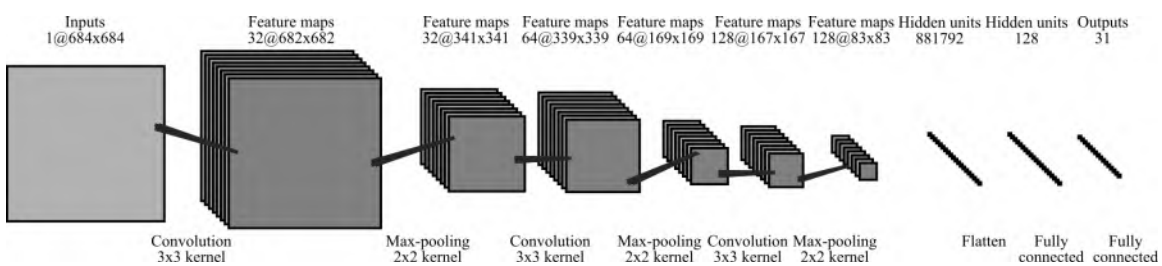


Figure 3.4: Designed data-driven CNN model structure with three convolutional layers, three max-pooling layers, two fully connected hidden layers and one output layer.

Component health index

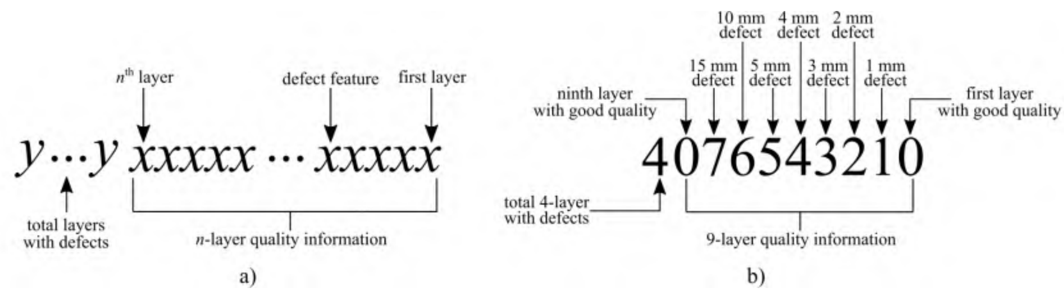


Figure 3.5: The proposed component health index for recording the product layer quality: a) the generalized component health index; and b) a 9-layer component health index for the dog-bone specimen.

The proposed component health index is a standardized way to accumulate the effects of defects in the printed component while assuming that all future layers of the part will be of perfect quality. The developed component health index is applied to record product layer information, which contains two parts: which layer has a defect and the number of layers with defects. A graphical representation of the generalized component health index is presented in Fig. 3.5(a). Starting from the right, for a printed component with n layers, the first n digits are associated with an individual layer. Each layer quality written here as x, is quantified on a scale, which can be assigned to specific quality features. On the left, a single number represents the total number of layers with defects. This means that the number of y digits is defined as the total number of digits required to count the total number of layers. Mathematically, this is written as $\log_{10}n + 1$. The initial values of the component health index are all set to zero, which by default assumes all successive new layers are without defects.

Fig. 3.5(b) shows an example of the proposed component health index for the dog-bone specimen considered in this work. The dog-bone is a 9-layer print with seven defects. In this preliminary work, the first and last layers, corresponding to the bottom and top layers, are always considered to be without defects. Therefore, all defects occur in the seven infill layers. The component health index has seven damage quantization levels, which were designed for this research. 0 represents a good quality layer, while 1, 2, 3, 4, 5, 6, and 7 represent defects with lengths of 1, 2, 3, 4, 5, 10, and 15 mm, respectively. The generic component health index can be expanded to apply to any new cases.

Defect impact decision boundary

The total number of defect impact results from the tensile test is 19503; 80% are split as training data, and 20% are chosen as test data to plot the decision boundary. Of note is that assumptions based on the real defect impact result are used to expand the data obtained from mechanical testing. To be more specific, suppose a 10 mm defect on one layer of the sample damages the product's structural quality and is broken at the rooted defect position. Therefore, a sample with more defects than just the 10 mm defect would also impair the product's structural quality by rooted defects. Three variables, defect length, defect location, and component health index, are chosen to draw the defect impact decision boundary. The SVM model with radial basis function kernel is trained offline using scikit-learn [227]. As the dataset is somewhat imbalanced, the weights of the SVM's classes are balanced to avoid overfitting. The defect impact decision boundary here is only for a quadrant of the dog-bone specimen, and the symmetry of the part is leveraged to expand the decision boundary for the entire part.

Real-time defect detection and decision-making

Real-time defect detection has two steps. First, an image is captured of the recently extruded layer and fed into the trained ML model. The model outputs the layer's quality information (i.e., defect length and defect location or good quality). Second, the health index is updated to combine the layer-related information (good quality or defect with length, location, and layer information) together. The updated health

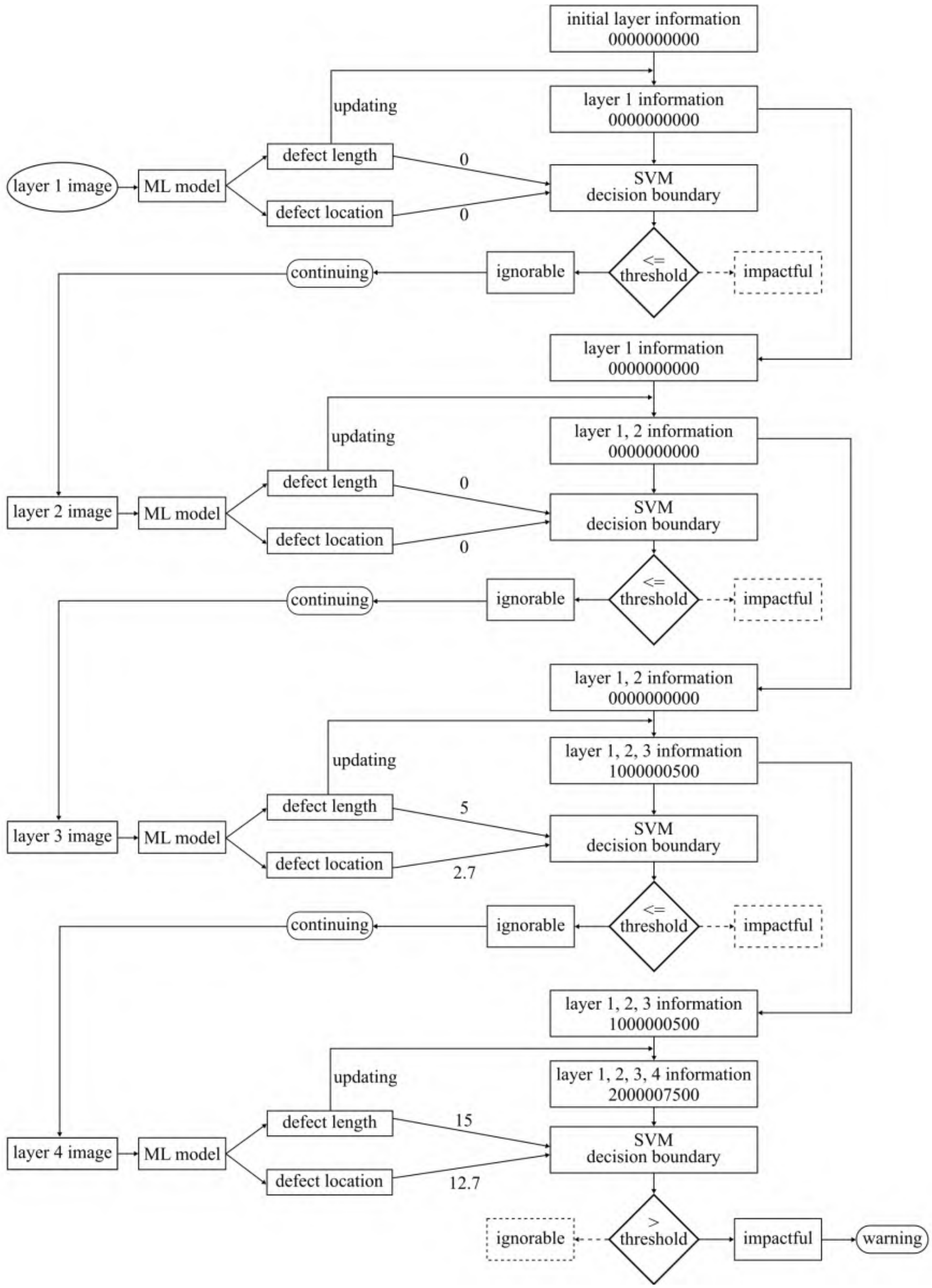


Figure 3.6: Visualization of the structural validation and decision-making process.

index is then mapped to the decision boundary for evaluation. If the accumulation of defects has a limited impact on structural quality impact and can be ignored, the printing and evaluation process will continue. However, suppose a combination of defects occurs such that the defect accumulation impact on the decision boundary shows that it is severe enough to undermine the product's structural quality. In that case, it may be prudent to cancel the print as the subsequent deposition of layers will not improve the quality of the part.

The SVM-based decision-making process uses a pre-computed boundary that is trained offline before printing is initiated. The current position of the part within the part's health performance space is computed using an SVM computation that uses input defect length, defect location, and the component health index. If the current position of the part's health marker is below the decision boundary, it is implied that the defect has a limited impact on structural quality and can be ignored. Otherwise, the product quality is undermined by the detected defects.

Fig. **??** shows a visualization of the structural validation and decision-making process. In the printing process, the first layer image is captured, and as it is the bottom layer and assumed to be of good quality in this introductory work, the component health index is not changed. Next, the second layer image is fed into the ML model. After obtaining the prediction result from ML, the quality of this layer is also good, therefore, the component health index remains unchanged. The printing continues to the third layer, with a 5 mm defect at position 2.7. The component health index is updated from 0000000000 to 1000000500. After evaluation against the decision boundary, the output result is below the threshold. Then the fourth layer image feeds into the ML model, and a 15 mm defect is detected at position 12.7 on this layer. After updating, the layer information is refreshed from 1000000500 to 2000007500, which means two layers have defects, a 5 mm defect on the third layer and a 15 mm defect on the fourth layer. By evaluating against the decision boundary function, it is determined that the value of the part's health marker is above the threshold value. This means the accumulated defects have an impactful effect on the product's structural quality. At this point, a warning for the part is provided. If deemed appropriate, the print may be canceled.

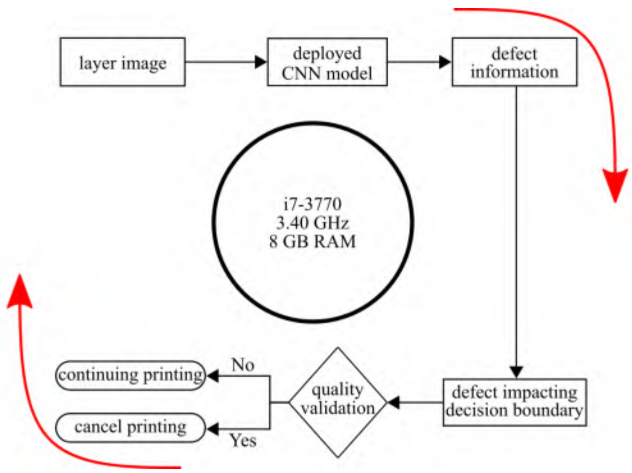


Figure 3.7: The diagram of the time range calculation for the real-time product structural quality validation.

To ensure the proposed algorithm can satisfy the timing constraints for real-time product structural quality validation, an investigation of the algorithm's timing is undertaken. Fig. 3.7 reports a diagram of the computational cycle used for the real-time product structural quality validation. For this investigation, the total cycle time is studied; this includes: capturing the image; processing data through the CNN model; and performing the multi-dimensional accumulation-threshold-based decision-making processes. In this work, a personal computer with an i7-3770 CPU @ 3.40 GHz with 8 GB RAM was utilized.

Training results for the data-driven defect detection

The CNN for defect classification is the first step of the process, which largely determines whether the following structural quality validation and decision-making are successful. Therefore, the accuracy of the data-driven defect detection from CNN is an important evaluation factor. The ultimate result for training data achieves 100% accuracy, recall, and precision with a 100% macro-averaged F1-score (the average of each class's F1-score), as the acquisition conditions of the images and geometry of the samples are always almost the same. In addition, the unbalanced nature of the dataset plays a role in these high values. The algorithm also performs well on test data, with a 99.78% average accuracy, 99.57% average recall, and a 99.57% average precision. The macro-averaged F1-score for test data is 99.57%. The training and validation results verify that the designed CNN model is precise and can be used for MEXAM defect prediction. Note that these results are excellent due to the somewhat large errors considered here and the maturity of CNNs for fault detection on hand-labeled data. Moreover, CNN has strong robustness for image classification.

Result of defect impact decision boundary process

Fig. 3.8 reports a 3D view of the part's health performance space where the volume of the green shape is considered to be a non-impactful defect (most of the impactful space is removed to provide a clear view of the decision boundary). The green is the prediction decision boundary for the ignorable defects, which means if the defect is located in this area, it has a limited (i.e., ignorable) effect on product quality. All area outside of the green decision boundary is treated as impactful to product structural performance. Moreover, location 0, 0, 0 is a special location in the health performance space as it represents a part without detectable defects. Note that all

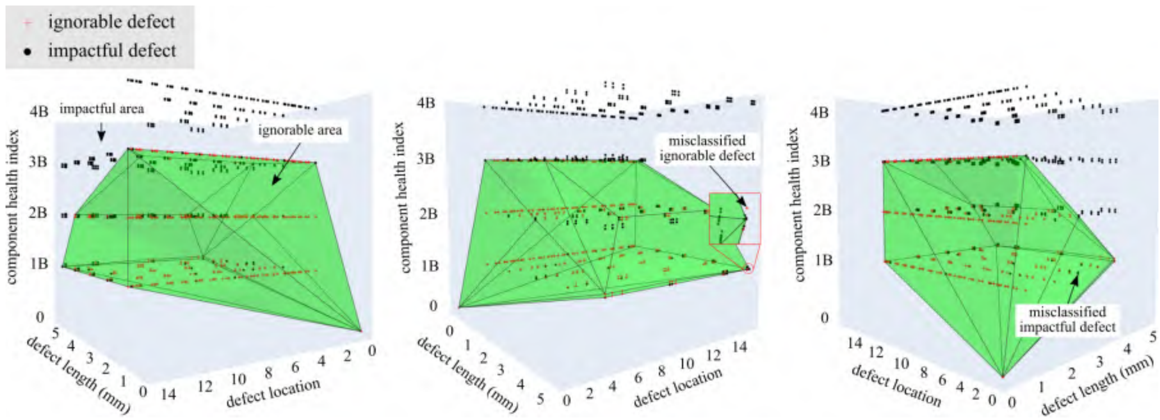


Figure 3.8: Defect impact decision boundary from different views (the green is the prediction decision boundary for the defects, red crosses are observations of ignorable defects, and black dots are observations of impactful defects.). An interactive 3D version of the decision boundary is available in this manuscript's supplemental documents. In the interactive version, blue diamonds are correctly classified test data on the decision boundary and the pink squares are incorrectly classified test data on the decision boundary.

three plots are different views of the same health performance space. In this 3D space, the x and y axis are defect length and defect location, while the z axis is the component health index. In the decision boundary plot, data denoted with a red cross means the defect is ignorable and the black dot means the defect is impactful to the product's structural performance.

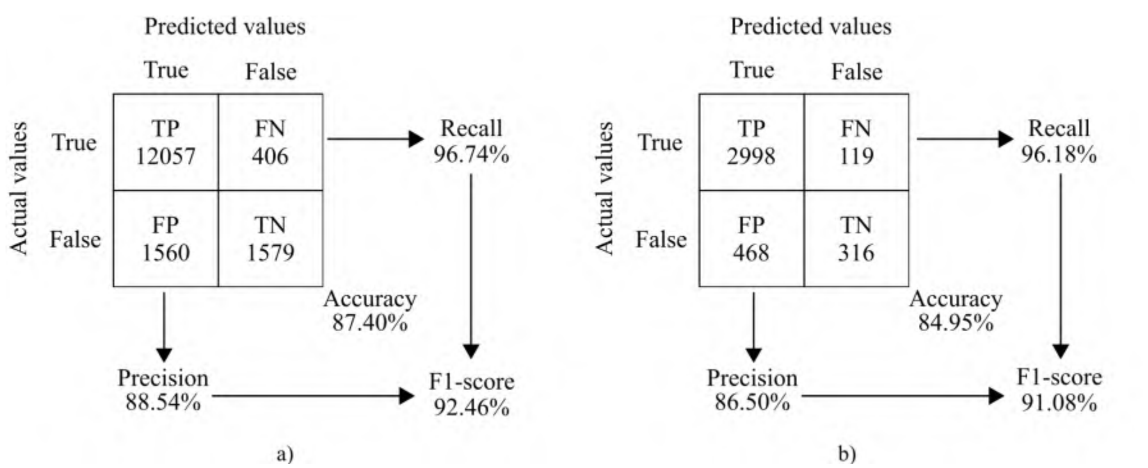


Figure 3.9: The confusion matrices for the decision boundary, showing: a) the confusion matrix for the training data; and b) the confusion matrix for the test data.

Note that in Fig. 3.8, plenty of impactful defects are deep within the area defined by the decision boundary. However, two ignorable defects are located outside of the decision boundary. The reason is that the training of the algorithm done by the team emphasizes recall over precision, as the miss classification of a faulty component as functional is more detrimental than a functional component as faulty [232]. Recall that the project's hypothesis is that the defect is considered impactful if and only if the specimen breaks with a tensile force of less than 948.6 N at the location of a detected defect. The decision boundary confusion matrix on the training and test data can be constructed, as shown in Fig. 3.9 a) and b), respectively. The F1-score for the test data shows the decision boundary accurately predicts the defect's effect on product structural quality. Again, note that the algorithm is biased towards recall, with 96.18% for the test data.

For completeness, a true positive is a specimen that is predicted to have impactful defects and fails under the 948.6 N threshold at the location of a detected defect. A true negative is a specimen that is predicted to have ignorable defects and fails above the 948.6 N threshold at any location. A false positive is a specimen predicted to have impactful defects yet fails above the 948.6 N threshold at any location. Conversely, a false negative specimen is predicted to have ignorable defects yet fails below the 948.6 N threshold at any location.

The test data with 30 samples is used to validate the trained decision boundary (please refer to the online interactive 3D decision boundary). Of these thirty samples, twenty-five samples are predicted correctly. The interactive 3D decision boundary plot shows the correctly predicted test data as blue squares, while the five incorrectly predicted samples are plotted with pink squares. Table 3.2 details the five incorrectly predicted test samples in an effort to investigate the reasons for the algorithm failures.

In these five incorrectly predicted samples, four miss classified ignorable cases were predicted to have impactful damage, and one impactful sample was incorrectly

Table 3.2: Incorrect predictions for the 30 sample test data.

sample number	defect length	defect location	layers with defect	tensile force achieved	defect predicted as	broken at seeded defect	meet tensile force requirement (948.6 N)	error type
0384	0.5 mm	10.4	2	953 N	impactful	No	Yes	false positive
0387	2.5 mm	6.08	3	950 N	impactful	Yes	Yes	false positive
0388	4.5 mm	5.10	1	941 N	ignorable	Yes	No	false negative
0396	1.8 mm	15.3	3	949 N	impactful	Yes	Yes	false positive
0409	5.5 mm	5.00	1	938 N	impactful	No	No	false positive

predicted as non-impactful. Considering impactful defects (the area outside the green decision-making boundary) as the positive class, samples 0384, 0387, 0396, and 409 are classified as false positives as the decision-making boundary incorrectly predicted these samples to have impactful damage cases. Sample 0388 was predicted to meet the tensile force requirement, but it did not, so it is a false negative error. Of the five incorrectly predicted samples, only two samples (0388 and 0409) did not meet the required tensile force and therefore are considered to be the more detrimental errors.

An important consideration is that samples 0384 and 0409 did not break at the seeded fault locations, suggesting that the detected defect was not the defect that limited their structural loading capability. It is hypothesized that sample 0384 was miss classified as it had the shortest defect length at 0.5, and the dataset does not have sufficient data at this defect size to accurately train a decision boundary. All five missed classifications are located near the decision boundary. As expected, samples located on or near the boundary are the most likely to be misclassified; this should be accounted for using the domain expertise of the practitioner when setting the load-reduction threshold used in building the decision-making boundary.

Result of real-time defect detection and decision-making process

Results for online defect detection are shown in Fig. 3.10, which reports the recorded times for 1,000 cycles tested. The longest process time is 688.1 ms, and the mean processing time is 461.8 ms. As one layer printing time requires 75 s, this means the

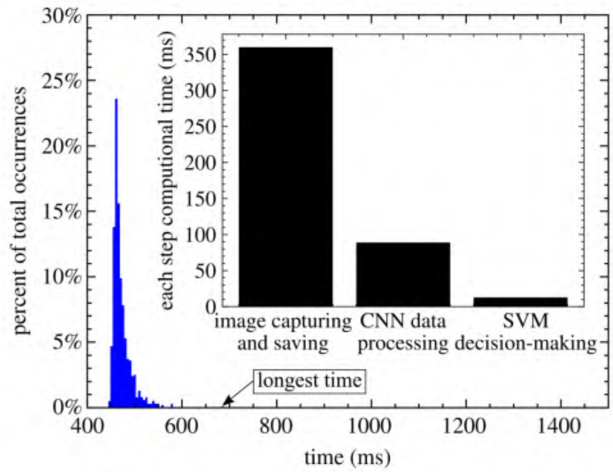


Figure 3.10: Time range calculation for the real-time product structural quality validation.

proposed product structural validation algorithm is feasible for real-time decision-making. Moreover, with a 100 times disparity between the required and available time, there is potential to deploy the proposed method on edge computing hardware [45].

To demonstrate the real-time defect detection and decision-making process, a specimen was designed with two defects during the printing process that is made up of four layers. The first and second layers are of good quality. Then a 5 mm defect at location 2.7 occurs on layer three, and another 15 mm defect is seeded at location 12.7. During the printing, as the first defect on layer three has a limited impact on product structural quality, the algorithm makes the decision to continue after evaluating the component health index. After finishing layer four, the new evaluation result shows that two defects stacked together undermine the product quality. Therefore, the algorithm displays a warning window to alert the user who may choose to cancel the print. Fig. 3.11 shows a screenshot of a video developed to demonstrate the defect detection and decision-making process in real-time. This video is included in the manuscript's supplemental material. This example demonstrates that the proposed real-time defect detection and decision-making system can make the right decision within a short time after a layer is finished.

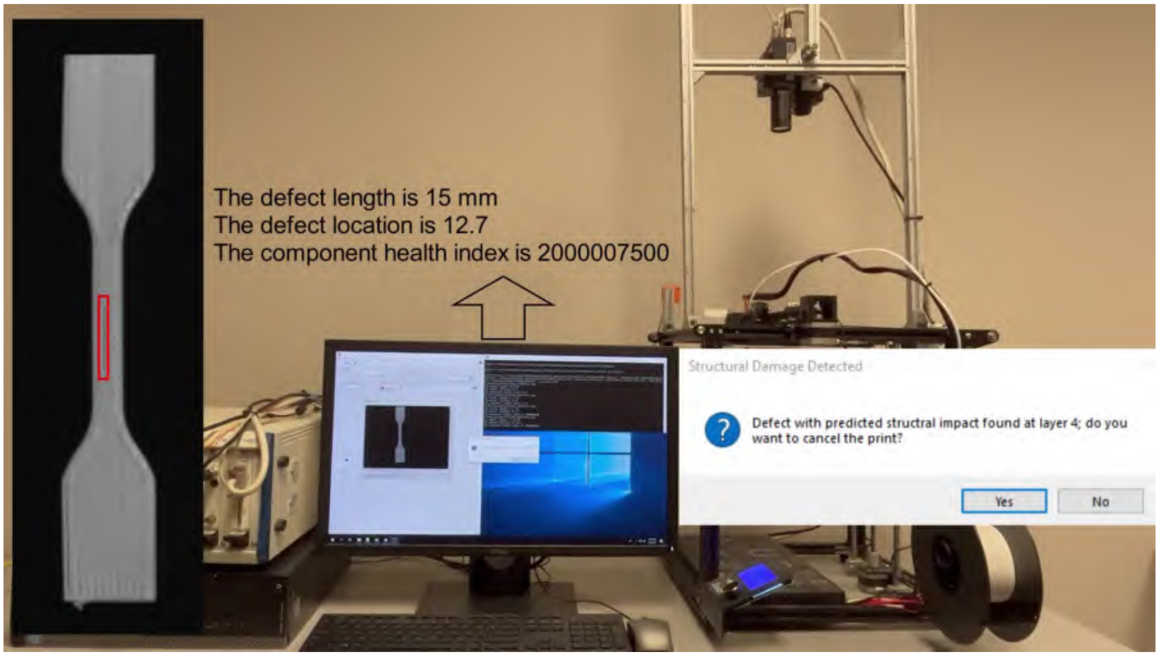


Figure 3.11: Screenshot of a video demonstrating the defect detection and decision-making process in real-time, where the warning window appears when an impactful defect occurs. The video is available in the paper's supplemental documents.

3.5 Conclusions

This paper presented a real-time methodology for product structural validation for components manufactured by the material extrusion additive manufacturing (MEXAM) method. This work links real-time defect detection with product structural quality validation rather than just focusing on defect detection. The specially designed material extrusion printer integrates an optical camera to obtain in situ images for training the convolutional neural network (CNN) model. The machine learning (ML) model detects damage and estimates every layer's quality. The output prediction from ML can be evaluated on the decision boundary made by the support vector machine (SVM) with the radial basis function kernel. Based on the evaluation results, the multi-dimensional accumulation-threshold-based decision-making system can decide whether to continue or cancel printing. The novel contribution of this method is its ability to link real-time defect detection with product structural quality validation.

The defect accumulation impact on different layers is considered, which makes the evaluation and validation process more comprehensive and precise. Results have shown that the developed product structural quality validation algorithm has good recall and F1 scores of 97.71% and 90.42% in a laboratory setting with the same image acquisition system used for training and validation. Further validation of the work requires that the algorithm be re-trained for more complex geometries and with different acquisition systems used for training and validation and the accuracy needs to re-evaluated. Meanwhile, the longest and mean process times are 688.1 ms and 461.8 ms, which means the proposed algorithm satisfies the timing constraints for real-time product structural quality validation and decision-making. While the proposed real-time structural validation method achieved good results, one key drawback is that it is time and effort-consuming to obtain the decision boundary from the tensile test result. Future work will focus on automating and expanding the component health index to multiple dimensions and supplementing the training process with numerical simulations, then transfer and extend the approach to other additive manufacturing technologies. The dataset developed for this work is made available through a public repository [87].

Acknowledgments

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Chapter 4

Real-time Finite Element Analysis for Structural Validation[1]

4.1 Abstract

End-use quality guarantee is critical for every batch production part, especially the load-bearing or fastening piece. Defects in sensitive positions can substantially impact the product's structural quality and mechanical properties. Various numerical simulations with finite element analysis (FEA) have been studied to evaluate and validate the product's qualities and properties. However, relatively few works have been focused on specimens with intrinsic or extrinsic defects. Generally, specimens with defects depend on pre-designed or post-defect manual updates within the FEA model. Obtaining stochastic defects in the specimen is challenging, let alone updating the relevant defect information in the FEA model. In addition, the product structural validation system should determine whether a part with a single defect or accumulated defect will meet the designed structural requirements. To address the mentioned issues, this paper proposes an advanced image segmentation-based real-time automatic FEA product structural quality validation system. A case study on a consumer-grade 3D printer validates the proposed system. The defect information is segmented from the layer image with a fully trained U-net. Two different defect update approaches are utilized in the Abaqus Python script: 1) the obtained poly-

[1]Yanzhou Fu, Austin Downey, Lang Yuan, and Hung-Tien Huang, 2023, To be submitted to Additive Manufacturing.

gon information for the large defect (i.e., seeded void) is used to form the defect by extrude-cut operation, which emphasizes the position sensitivity and structural quality impact, 2) the generic and intrinsic defect information is applied to update the layer void-related Young's modulus. The simulation process runs automatically after the Abaqus Python script updating is completed. A warning window appears if the maximum stress after fracture reduces to more than 4% compared with a good specimen, which means the product's structural quality is damaged. The online validation of the proposed real-time structural quality validation system is performed on a test specimen with defects. The result indicates that the proposed system is reliable and contented for real-time online FEA structural validation.

4.2 Introduction

End-use product quality guarantee with structural validation refers to the assurance provided by manufacturers or sellers that a product not only meets specific quality standards but also undergoes a thorough validation process to ensure its structural integrity and performance [334]. This type of guarantee is essential for products that require robust structural properties, such as buildings [358], bridges [327], automotive components [177], and aerospace parts [150]. Additive manufacturing (AM) products are commonly employed in aerospace due to their lightweight nature and great structural integrity. However, AM is an iterative process that inherently contains uncertainties. Defects in sensitive positions can dramatically impact the product's overall structural quality, which may cause serious consequences [86, 89]. Structural validation presents an effective solution to this challenge by subjecting a product to diverse tests, simulations, and analyses. This process evaluates the product's capacity to endure loads, stresses, vibrations, and other environmental factors it might face in its intended operational context. The goal is to confirm that the product's design, materials, and construction align safety and performance standards. [335].

Finite Element Analysis (FEA) is a computer-based simulation technique that is commonly used to analyze the structural behavior of the AM product under different conditions [279]. It helps identify potential weaknesses, stress concentrations, or areas with defects that may require reinforcement, allowing manufacturers to optimize the product's performance and mitigate potential risks associated with structural failures. Substantial research focused on the numerical simulation with the FEA method to evaluate the AM product structural quality. For example, Rahman et al. proposed a FEA method to examine the behavior of continuous carbon fiber-reinforced AM [236]. Based on four different volumes of fiber reinforcement configurations, it was found that the simulation method is reliably applicable for predicting the behavior of end-use components when dealing with fiber volume fractions of approximately 11%. Cao et al. developed an advanced FEA model that integrates the equivalent boundary condition method and dynamic mesh method to enhance the speed of thermo-mechanical simulations in laser powder bed fusion printing [44]. The developed model accurately predicts the thermo-mechanical behavior of multiple powder bed fusion deposits across various process parameters. The result demonstrated excellent agreement with experimental data regarding in-situ temperature and residual stress. A three-dimensional thermo-elastic-plastic model is introduced by Yang et al. for accurately forecasting the thermo-mechanical response during the laser-engineered net shaping process of Ti-6Al-4V by employing the FEA method [336]. The model successfully reproduces the computed thermal history and mechanical deformations, exhibiting strong concurrence with experimental measurements. To enhance the ability of simulation with FEA, Wang et al. introduced a novel Gaussian process-constrained general path model for estimating high-fidelity FEA simulation results by utilizing low-fidelity results voxel-by-voxel [308]. This model effectively measures the heterogeneous discrepancies between the low and high-fidelity FEA simulation results by integrating product design information, such as Cartesian coordinates of the deposition sequence, and process design information obtained from FEA simulation inputs. As a result, this approach enables the efficient validation of new product and process designs by providing timely access to simulation results with the desired accuracy. The existing research studies in FEA simulation focus on good quality specimen quality evaluation but lack the consideration of defects in the actual printing process. However, the manufacturing process is inherently complex and often prone to printing deficiencies. Obtaining stochastic defects in the samples and updating the relevant defect information in the FEA model poses significant challenges. Additionally, it is important to note that individual small defects may not cause critical quality damage alone. Nevertheless, when these ignorable defects accumulate, they can collectively result in significant degradation of functional qualities [88]. To address these limitations, a comprehensive and innovative approach is required for product structural quality validation. Such an approach should automatically identify observable defects, incorporate them into the FEA model, and assess their individual and combined effects on the product's functional properties to ensure end-use quality.

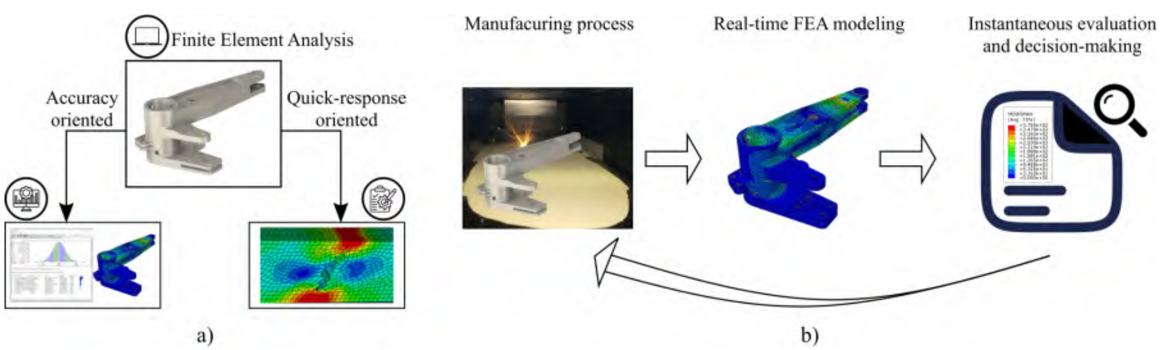

Figure 4.1: Orientation and workflow for FEA structural validation: a) orientation for real-time FEA structural validation; b) workflow for real-time FEA structural validation in AM.

This paper proposes a novel real-time image segmentation-based product structural quality validation approach. Unlike the traditional methods, this system can automatically validate product structural quality by updating the related defect information in the Abaqus Python script. If the simulation result indicates that a single defect or the accumulation of defects does not minimize product quality, the defects are deemed negligible. However, the printing process must stop if the structural property fails to meet the desired value. The U-net image segmentation algorithm is used to identify the defect, and the extracted defect polygon information is utilized to update the extrude-cut sketch or the void-related layer Young's modulus in the Abaqus Python script. After the update, the simulation automatically runs using pre-developed code. To demonstrate the proposed system is feasible and reliable, a case study is investigated on a 3D printer. The unprinted layers of the specimen utilize the averaged void-related Young's modulus derived from the printed layers. Based on that assumption, the product structural quality validation system makes the appropriate decision (continue or cancel manufacturing) by referring to the maximum tensile stress result from FEA before the next layer begins. Every layer's real-time evaluation and decision-making process continues until the entire part is completed or harmful defects occur. The result indicates that the proposed system is trustworthy and contented for real-time FEA online structural validation.

The experimental setup and methodology are discussed in Section 4.3, which has five sections: experiment methodology, experiment setup, data-driven model for defect segmentation, Abaqus Python script update, and real-time structural validation and decision-making. Section 4.4 discusses the experiment results and provides a discussion, which includes the training results for the data-driven defect segmentation and results for the Abaqus Python script update and real-time structural validation. Section 4.5 presents the conclusion and future work.

Experiment methodology

There are two orientations for FEA: accuracy oriented and quick-response oriented, which is shown in Fig. 4.1a). Accuracy oriented FEA suggests that the primary focus of the analysis is on obtaining highly accurate and reliable results, even if achieving such precision might take more time or resources. However, quick-response oriented FEA emphases on achieving rapid or quick responses in the analysis or modeling process. In this work, the goal is to achieve real-time structural validation during manufacturing by updating the live corresponding layer information in the FEA model, which belongs to quick-response oriented FEA. The workflow for the real-time FEA structural validation is shown in Fig. 4.1b). Specific result in the output database is chosen as validation metrics, which is further used for instantaneous decision-making.

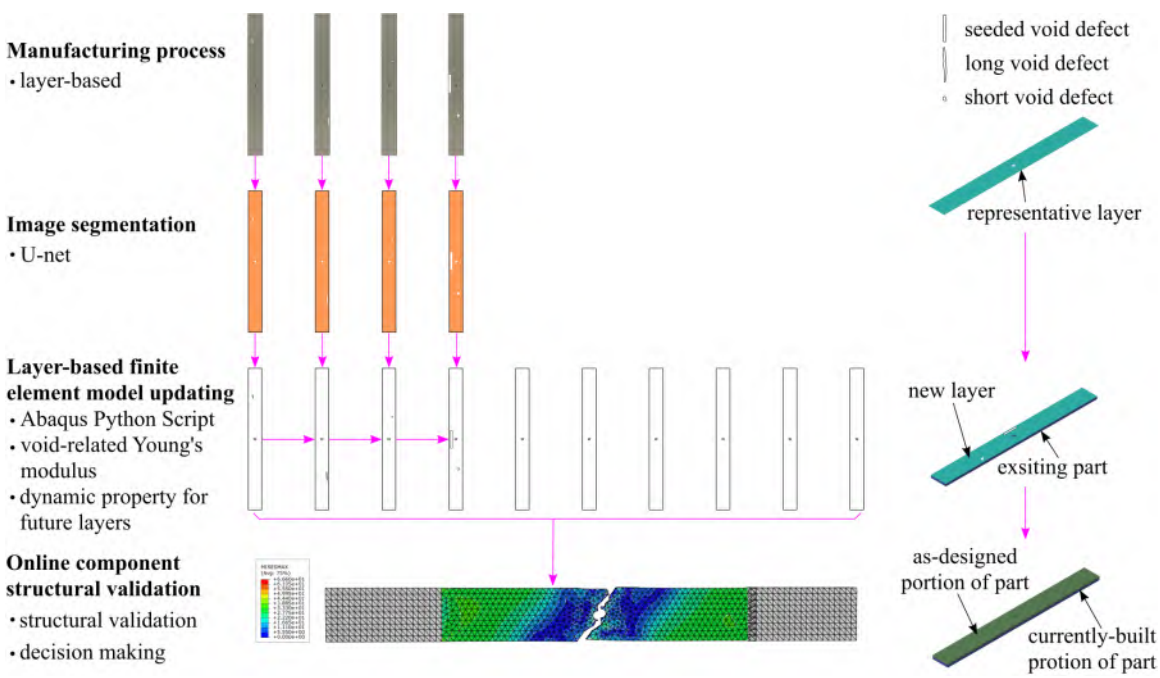

Figure 4.2: The architecture of real-time automatic FEA structural quality validation.

Fig. 4.2 illustrates the real-time automatic validation system for structural quality in detail. The system comprises three processes: offline defect segmentation training, tensile test pre-simulation (Python script generation), and online real-time structural validation and decision-making. For the offline defect detection training process, a dataset of images taken at each layer during printing with defects is created using an optical camera. Images are annotated manually with the seeded void defect, short void, long void, and specimen by Labelbox. The dataset is used to train the image segmentation model U-net. The fully trained U-net is deployed to output the defect information. After coordinate transformation, the defect information is used for layer-based FEA model updating in Abaqus Python script. When the update is completed, the pre-developed code runs the simulation automatically. The online component structural validation and decision-making are based on the FEA result. The data-driven U-net model is discussed in greater detail in Section 4.3. The defect information update details are presented in Section 4.3.

Samples with tensile tests have been done to get the real tensile stress and strain. The related material properties setting in Abaqus are obtained from the stress and strain data. Through FEA, the structural quality is validated. The defect is considered impactful when the structure's maximum tensile stress decreases larger than 4%, compared with the tensile stress from good specimens. If the defect has a limited impact on product quality, it can be considered ignorable. Given a uniform averaged empirical void value on all unprinted layers, the structural quality validation system can decide whether to proceed or stop the printing process before completing the current layer. This real-time evaluation and decision-making process continue for every layer until the entire part is completed or harmful defects occur.

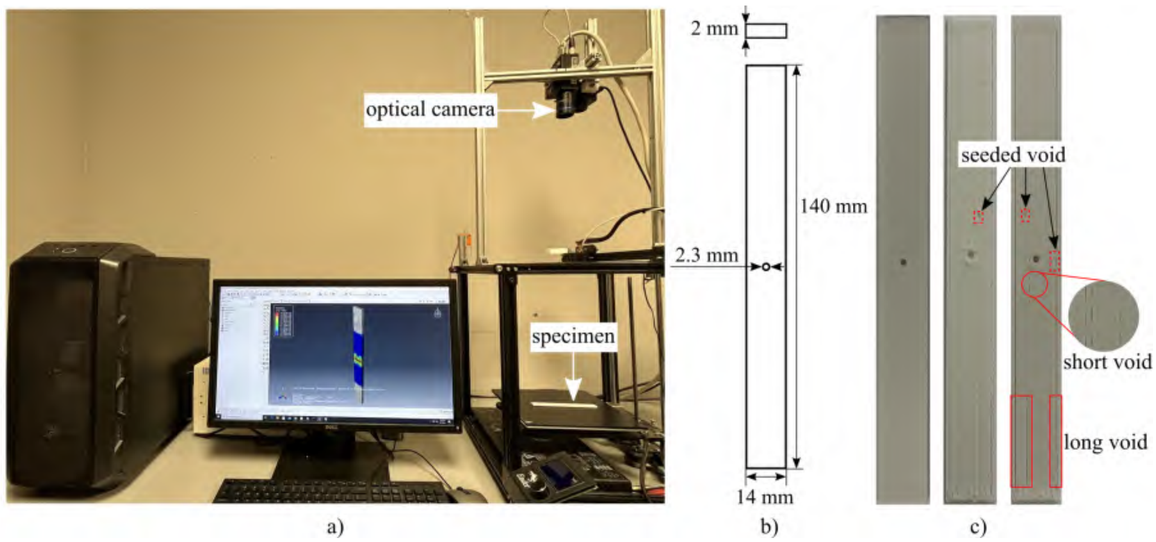

Figure 4.3: The experimental platform and designed specimen: a) experimental platform for product structural quality validation; b) the designed specimen dimension; and c) a good quality layer of the specimen and two layers with various defects on different positions.

Experimental setup

The experimental platform for this research involved printing specimens using a Creality Ender 5 printer with Polylactic Acid (PLA) filament, as shown in Fig. 4.3a). To collect printing images, an optical camera (JAI CV-M4+ CL) with a resolution of 1368 $\times$ 1020 is set above the extruder on the printer frame. The G-code for the designed specimen is modified before printing, and a LabVIEW code activates the camera to capture the layer image of the specimen after each layer is completed. The unique characteristic of Ender 5 is the relative position between the camera and specimens is fixed, which provides consistent images with an identical scale across layers.

The designed specimen is based on the American Society for Testing Materials (ASTM) D5766, which generates data where the product's final application may require fastener holes or to simulate flaws in a material component [55]. The hole allows for stress concentration and reduced net section, while the test method calculates ultimate strength based on the gross cross-sectional area, disregarding the hole. As

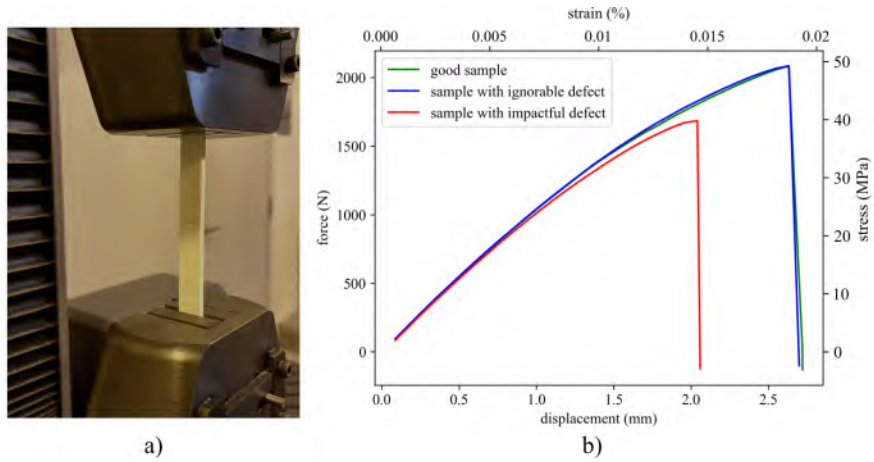

Figure 4.4: Printed specimen tensile test: a) tensile test setup; b) the specimen tensile force results.

the original specimen dimension is too big to print with Ender 5. All the dimensions are re-scaled to 0.56. The dimensions of the specimen are shown in Fig. 4.3b). The specimen has a thickness of 2 mm with a 100% infill density. The specimen comprises ten layers. The top and bottom layers print in a horizontal direction, and the middle layers print in a vertical direction.

To obtain the right simulation result on Abaqus, tensile tests are done for the collected specimens as references. Mechanical validation is performed on an MTS Exceed E43 electromechanical load frame, as shown in Fig. 4.4a). Fig. 4.4b) shows the tensile forces of the specimens with and without quality damage. The material properties used in FEA simulation are derived from the actual tensile test data.

Data-driven model for defect segmentation

Image segmentation is an approach to breaking a digital image into various subgroups called image segments. This process helps reduce the image's complexity to make further processing or analysis of the image simpler. Similarly, segmentation assigns labels to pixels. All picture elements/pixels in the same category have a commonly assigned label. The image segmentation algorithm used in this research is U-net, the most famous image segmentation model. The U-net architecture is composed of a contracting path and an expansive path [245]. In the contracting path, which follows the typical architecture of a convolutional network, two 3 × 3 convolutions (unpadded convolutions) are repeatedly applied, each followed by a rectified linear unit (ReLU) and a 2 × 2 max pooling operation with stride 2 for downsampling. At each downsampling step, the number of feature channels is doubled. On the other hand, each step in the expansive path involves upsampling the feature map, followed by a 2 × 2 convolution ("up-convolution") that reduces the number of feature channels by half. Then, the corresponding cropped feature map from the contracting path and two 3 × 3 convolutions, each followed by a ReLU, are concatenated and processed. Cropping is necessary due to the loss of border pixels in each convolution. At the final layer, a 1 × 1 convolution is used to map each 64-component feature vector to the desired number of classes. Overall, the U-net architecture has 23 convolutional layers. With the help of image segmentation, the defect information can be obtained, which is outputted as polygon points. This information is used in the Abaqus Python script model updating step. To increase the size of the training dataset, data augmentation (i.e., flip, random rotation, Gaussian blur, and Gaussian noise) is implemented. After data augmentation, the final dataset contains 1000 images. For the labeled specimen images and masks, 80% of them are randomly chosen as training data, and the remaining 20% are used as test data. The raw image dimension of the specimen is 1368 × 1020. To accelerate the training speed, the image resolution is reduced by resizing to 512 × 512 pixels before input into the model. Training is done with 100 epochs.

Abaqus Python script update and real-time structural validation

The first crucial step for the Abaqus model update is generating the standard Python script file, which is the base for the following update. Abaqus provides a Python API that allows users to interact with Abaqus/CAE and Abaqus/Viewer using Python scripting [278]. Abaqus/CAE can generate a Python script that records every operation during a session. This script can be modified to replicate the model development process. Therefore, the FEA simulation, assuming all the product layers are of good quality, is generated on Abaqus. The generated Python script is used as the base file for updating. The script containing all layers' information is executed through the command line interface (CLI). The image segmentation results from the U-net model are used for updating the defect or the porosity-related Young's modulus in the FEA model. Specifically, the polygon information from the big void defect is used to cut extrusion to form the defect inside the specimen. The defect information from the un-rooted defects (normal printing gap and porosity) is chosen to update the porosity-related Young's modulus. It is also worth noting that the cut extrusion method emphasizes the significance of the sensitive defect position, and the porosity-related Young's modulus focuses on the generic layer property. The following unprinted layers of Young's modulus are based on averaging the previously printed layers to simulate precise mechanical properties. Based on the research from Dewey et al., the basic form of the equation for Young's modulus is:

$$E_p = E_0(1 - aP)$$

where E_p is Young's modulus of the porous body, E_0 is the modulus of a non-porous body of the same material, a is a constant dependent on Poisson's ratio of the matrix material, and P is the volume porosity [54, 246]. To simplify the calculation and simulation process, the constant dependent a is equal to one, which means Young's modulus negatively correlates with the volume porosity.

The FEA is simulated with the explicit scheme and modeled the tensile with a fixed loading displacement. The boundary condition where one end of the specimen

Table 4.1: Material property from tensile stress-strain for FEA.

Property	Density (g/mm^3)	Young's modulus (MPa)	Poisson's ratio	Yield stress (MPa)	Yield strain (%)
Value	1.36	3600	0.4	49.56	1.62

is encased by constraining all six degrees of freedom, U1 = U2 = U3 = UR1 = UR2 = UR3 = 0. The displacement along the longitudinal axis is imposed on the other side, and displacement/rotation from all other sides is fixed, U1 = U3 = UR1 = UR2 = UR3 = 0, U2 = 3. The material properties used for model simulation are assumed to be homogeneous and isotropic for each layer. To have a thorough and accurate analysis of defect impact, material properties are derived from the real tensile test, which is shown in Table 4.1. The other FEA simulation setting is the approximate global size of 1 mm, and the element shape is a 4-node linear tetrahedron (C3D4). To obtain the right product structural quality, the specimen is partitioned into several regions. The gauge range region is treated as a separate set from where the maximum stress is obtained.

Fig. 4.5 shows the flowchart of the real-time automatic product structural quality validation and decision-making process for a real case: the specimen with a 3 mm rooted defect on layer two, 3mm and 5 mm rooted defects on layer four, and with short/long voids on each layer. The FEA structural validation is skipped as the first layers have no defect. Then, a 3 mm rooted defect shows on layer two. The defect information outputted from the image segmentation model is used to update the defect sketch and void-related Young's modulus in the Abaqus Python script. The designed code automatically activates the FEA simulation. After comparing with the stress threshold, the maximum stress is larger than the set threshold, which means the defects on this layer are ignorable. The printing continues to layer 3 only with short and long void, and the FEA validation is triggered and processed and, therefore, goes into layer four. Layer 4 has two seeded defects(a 3 mm defect and a 5 mm defect), and voids. All the defect information from layers 2, 3, and 4 is accumulated to update the

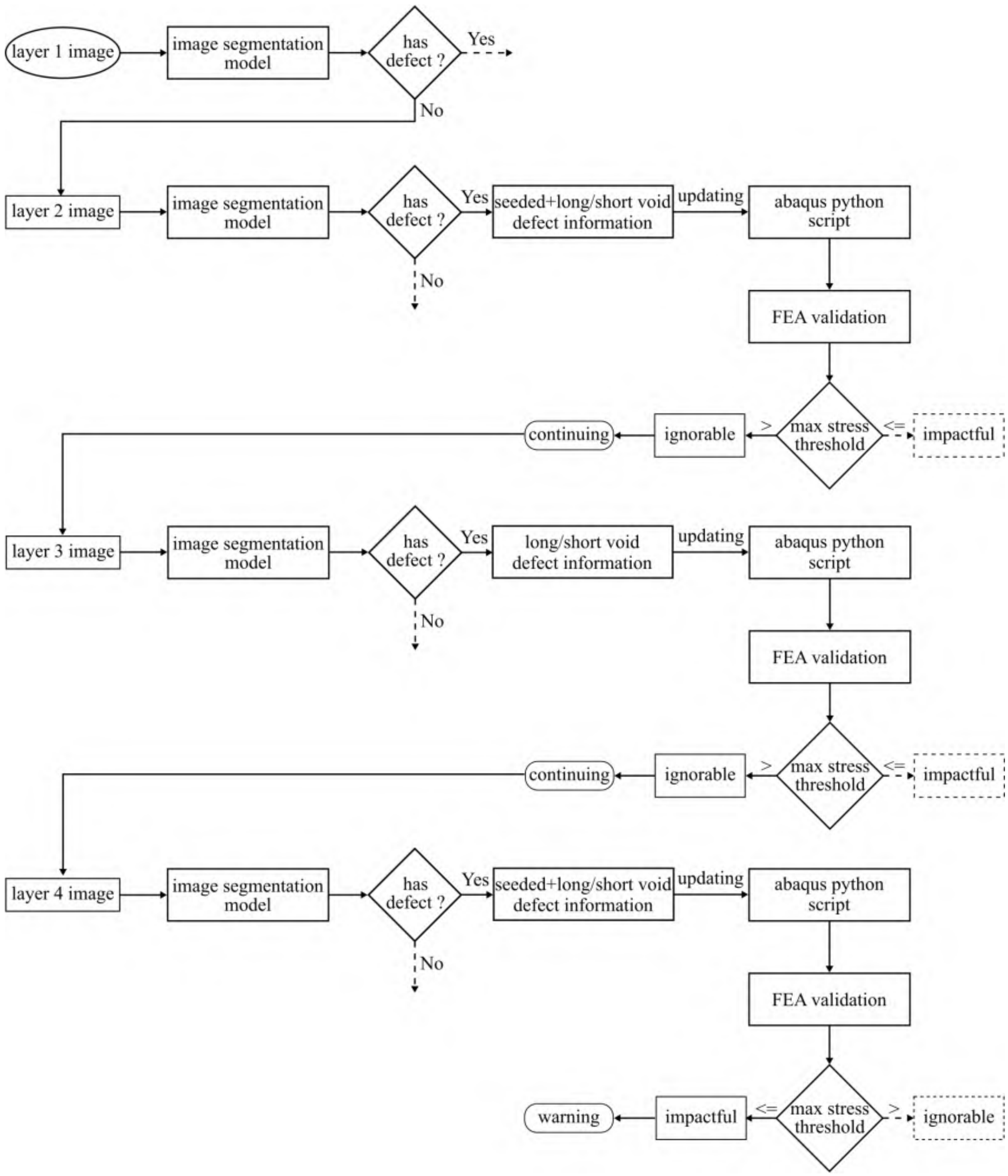

Figure 4.5: The flowchart of the real-time automatic product structural quality validation and decision-making process.

corresponding layer in the Abaqus Python script. The final maximum stress from the simulation is lower than the stress threshold, which means the accumulated defects damage the product's structural quality. Then, the warning window pops up on the screen to alert the cancellation.

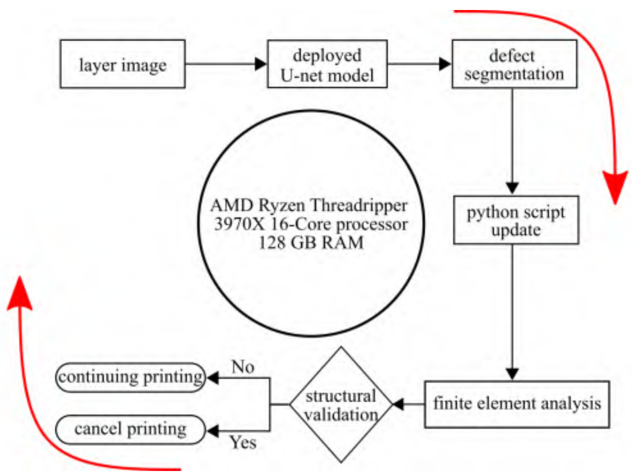

Figure 4.6: The diagram of the computational time calculation for the real-time automatic product structural quality validation.

Real-time structural validation has three steps. First, an image of the currently printed layer is captured and fed into the trained U-net model. The model outputs the seeded defect information as polygon lines, which means after a transformation, the real defect size in Abaqus can be re-plotted by the polygon lines and the short/long void volume is recorded. After the defect sketch is updated, extrusion-cut is applied to form the rooted defect, and the void volume in this layer calculates the void-related Young's modulus. Second, the Abaqus Python script automatically runs in the command prompt by pre-designed code after updating. Third, analyzing the maximum tensile stress and making the final decision based on the analysis. By evaluation, it is determined whether the requirement of the part's structural quality is satisfied or not. If not, this means the accumulated defects have an impactful effect on the product's structural quality. At this point, a warning for the part quality damage is provided. If deemed appropriate, the print may be continued. Note that the Abaqus Python file has ten layers of information; the defect information only updates on the corresponding layer. After updating, all the layers merge to form a part.

Real-time means the response time for various operations in computing or other processes must be within a specified range (generally a very short time). In other words, a real-time FEA validation process occurs in defined time steps of maximum duration [212]. An investigation of the computational timing is undertaken to check the timing constraints for real-time product structural quality validation. Fig. 4.6 shows a computational cycle diagram for real-time product structural quality validation. For this investigation, the total cycle time is studied; this includes defect segmentation, Python script update, and tensile test simulation and decision-making. This work uses three personal desktops with different system configurations to measure the structural validation computational time. The system configurations are shown in Table 4.2.

Table 4.2: Three different system configurations for real-time constraint test.

Hardware	System 1	System 2	System 3
Processor	Intel Core i7-3770 CPU	AMD Ryzen Threadripper 3970X	Two Intel Xeon Gold 6250 CPU
Cores x GHz	4 Cores, 8 Threads @ 3.4 GHz	32 Cores, 64 Threads @ 3.7 GHz	16 Cores, 32 Threads @ 3.9 GHz
GPU	None	NVIDIA Quadro P4000	NVIDIA Tesla V100S
RAM	8 G	128 G	96 G
Opertaing system	Windows 10, 64 Bit	Windows 10, 64 Bit	Windows 10, 64 Bit

4.4 Results and discussions

Training results for the defect segmentation

The accuracy of the data-driven defect segmentation using U-net is crucial as it determines the success of the subsequent steps in the process, including structural quality validation and decision-making. Therefore, the accuracy of the data-driven defect segmentation from U-net is an important evaluation factor. To evaluate the image segmentation accuracy, Intersection-Over-Union (IoU) is utilized as the evaluation metric [241]. The IoU is calculated by dividing the area of overlap between the predicted segmentation and the ground truth by the area of union between the predicted

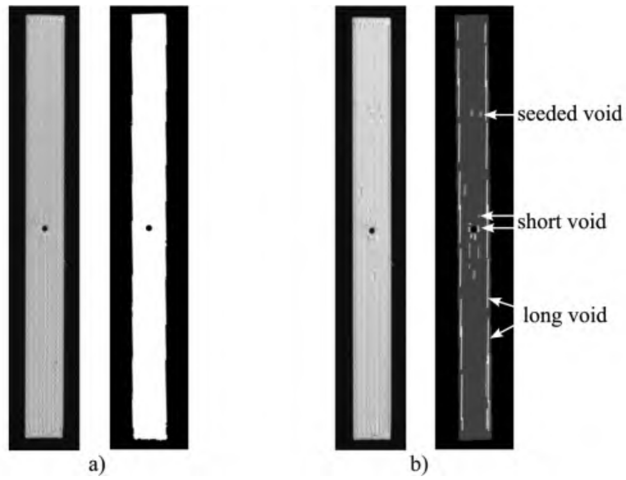

Figure 4.7: The U-net segmentation results: a) the segmentation result for a good layer and b) the segmentation result for the layer with defects.

segmentation and the ground truth. The metric ranges from 0 – 1 (0 – 100%), with 0 indicating no overlap and 1 meaning perfectly overlapping segmentation. The ultimate IoU for training data achieves 95.32%. The algorithm also performs well on test data, with a 92.79% IoU accuracy. The training and validation results verify that the applied U-net model is accurate and can be used for defect segmentation. The segmentation results for good layer and layer with defects are shown in Fig. 4.7.

Result of Abaqus Python script update and real-time structural validation process

In Fig. 4.8, the generated defects and defect polygon information from sketches are displayed, which are transformed from the defect segmentation result. The defect polygon sketch is formed from point to point. Fig. 4.8a) and b)show the defect update results on layer two and layer four with sketches. The porosity and gap volume for layers two, three, and four are 1.35%, 1.16%, and 1.42%, respectively. Therefore, based on the porosity-related Young's modulus description, Young's modulus for the current three layers and following layers is 98.65%, 98.84%, 98.58%, and 98.69% of the basic solid material.

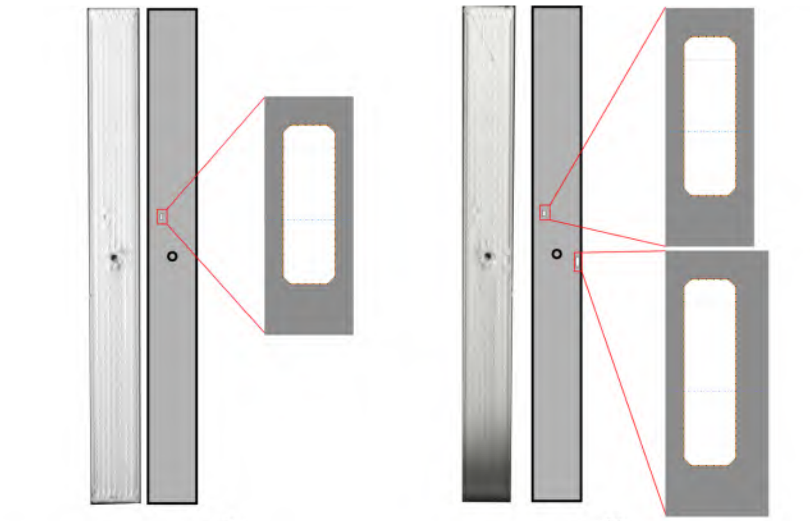

Figure 4.8: The defect updating results on specific layers: a) a 3 mm rooted defect with a sketch on layer two and b) the rooted 3 and 5 mm defects sketch updating result on layer four.

Fig. 4.9 displays the FEA simulation result for the specimens with and without defects. From Fig. 4.9a), it can be concluded that the maximum stress for the good specimen after fracture is 47.15 MPa. As shown in Fig. 4.9b), the specimen with the rooted defect in an ignorable position (the defect in Fig. 4.3b)), the maximum stress is 47.09 MPa, which is not much different from the good specimen. The specimen with a rooted defect near the fastener hole has a 45.22 MPa maximum stress (accumulates all the defects Fig. 4.3b) and Fig. 4.3c)), which is shown in Fig. 4.9c). Therefore, in this work, a 4% maximum stress decrease is picked as the structural quality evaluation threshold; this relates to a 1.89 MPa decrease. This research hypothesizes that the defect is considered impactful if the simulated specimen breaks with a maximum tensile stress lower than 45.26 MPa.

The computational time for each step in real-time product structural quality validation is shown in Fig. 4.10, which is tested with the system 2 in Table 4.2. Note that the computation time for image segmentation is always the same, which is 0.4 s. Whereas the time for the Abaqus Python script updating and FEA quality validation

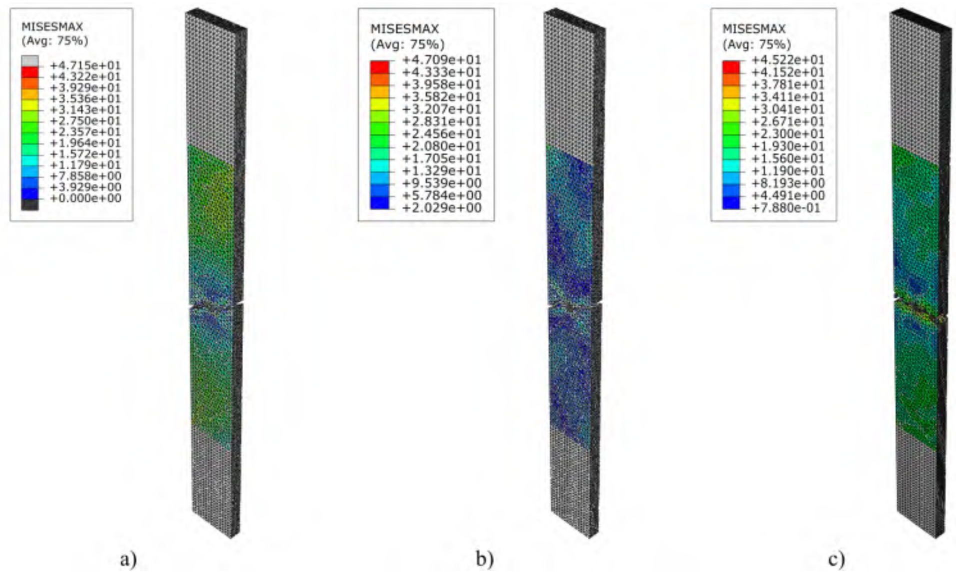

Figure 4.9: The FEA simulation result: a) the simulation result for the specimen without defects inside, b) the simulation result for the specimen with a 3 mm defect on layer two, and c) the simulation result for the specimen with a 3 mm defect on layer two and 3 mm, 5 mm defects on layer four.

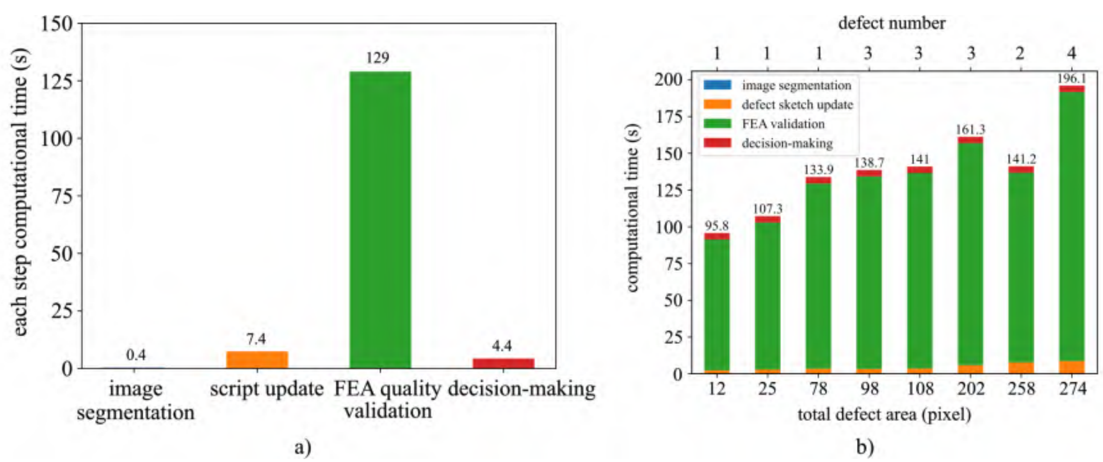

Figure 4.10: The computational time for each step in real-time product structural quality validation: a) the computational time for each step and b) the computational time for FEA validation with various defect sizes.

relates to the defects. To calculate the computational time for each step and the relationship between computational time with used tokens, the layer with two defects (total defect pixel is 258) is chosen, which is shown in Fig. 4.3c). The results are shown in Fig. 4.10a) and Fig. 4.10c). For a 3 mm and 5 mm defect, the script updating time is 7.4 s. FEA structural validation takes most of the computation time, which is 129 s, with 16 tokens in use. The decision-making time is 4.4 s to traverse all the frames to obtain the maximum stress. The total computation time is 141.2 s. As shown in Fig. 4.10c), the more tokens in use, the faster the simulation runs. From 4 to 12 tokens in use, the computation time saves around 20 s. However, changing the token numbers from 12 to 16, the acceleration is not noticeable. The computational time for FEA validation with various defect sizes is displayed in Fig. 4.10b). Eight layers with different defect sizes are used. It can be concluded that the computational time is not only related to the defect size but also influenced by the number of defects. Typically, more and larger defects take a longer time to update the defect sketch and do the FEA simulation. To ensure the proposed algorithm can meet the timing constraints for real-time product structural quality validation, logistic regression decision boundary is plotted for achievable and unachievable defects, as shown in Fig. 4.11. As one layer printing time needs 140 s, which real-time structural validation is not achievable when the defect's computational time over this constraint. The threshold for achievable and unachievable defects is 140 s, and the decision boundary is plotted based on the computational time with three systems shown in Table 4.2. In Fig. 4.11a), all the defects over the real-time limitation. The computational time for the minimum defect needs 459 s. From Fig. 4.10b) and c), it can concluded that more powerful computer can accelerate the entire process. It is worth highlighting that the minimal detectable defect size is 12 pixels. The defect, more minor than that, is undetectable or meaningless.

To demonstrate the real-time defect detection and decision-making process, a specimen is designed with one defect on layer two and two defects on layer four (the same process shown in Fig. 4.5). The first layer is of good quality, so the FEA structural validation is skipped. Then, a 3 mm void and some gaps occur on layer two, which is in the insensitive position and has a limited impact on the structural

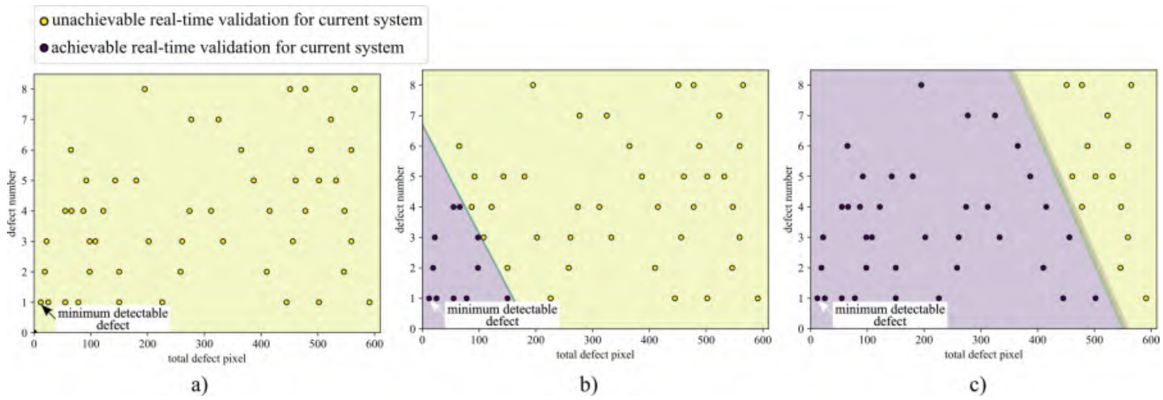

Figure 4.11: The decision boundary of the achievable and unachievable real-time structural validation defect for different computer system configurations.

quality. After evaluation, the simulation system is automatically exited. Layer 3 has some voids, which are negligible. On layer four, 3 mm and 5 mm rooted defects with voids occur at different locations on this layer. After getting the defects' polygon information, the defect is updated in the generated Python script file. As one of the defects is located near the fastener hole. From the simulation result, the maximum tensile stress is 45.22 MPa lower than the set threshold, implying that the presenting defects damage the product's structural quality. Therefore, the algorithm displays a warning window to alert the user who may choose to cancel the print. Fig. 4.12 shows a screenshot of the warning window during the real-time FEA validation and decision-making process.

4.5 Conclusions

This paper presented an automatic real-time methodology for product structural validation for components. The defect information is extracted by the image segmentation algorithm U-net. Then, update the defect information in the pre-generated Abaqus Python script. The updated defect information is used for the extrusion cutting or the dynamic void-related Young's modulus to simulate the printing defect. Then, the script is run by the pre-designed code in the command prompt to achieve

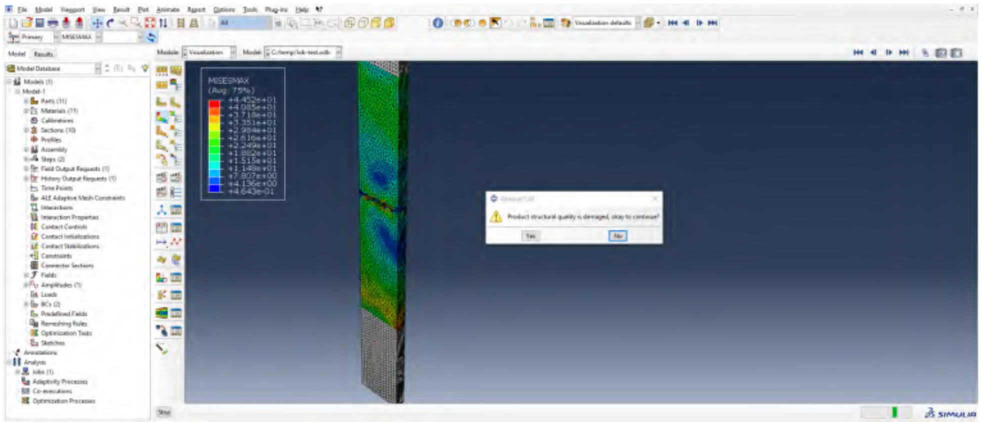

Figure 4.12: Screenshot of a video proving the real-time automatic defect detection and product structural quality validation, where the warning window appears when impactful defects occur.

the structural validation. The maximum tensile stress is compared with the presetting threshold to validate whether the product's structural quality is still good. In the verification case study, an optical camera is integrated into the specially designed printer to obtain in situ images for training the U-net. The automatic structural quality validation is processed when layers with defects are detected. The structural quality validation and decision-making system can decide whether to continue or cancel manufacturing based on the evaluation results. The novel contribution of this method is its ability to achieve automatic real-time defect detection with product structural quality validation. Furthermore, the dynamic void-related Young's modulus derived from the printed layers for the unprinted layers is taken into consideration for a more accurate product property evaluation.

Results have shown that the applied image segmentation model U-net has good Intersection-Over-Union (IoU) on training and validation data, with 95.32% and 92.79% accuracy on defect segmentation, respectively. Meanwhile, the demonstration case verifies the proposed algorithm satisfying the current automatic real-time product structural quality validation and decision-making. While the proposed real-time structural validation method achieved good results, some potential key factors still need further investigation. For example, the mesh size impacts on computational time, real-time simulation process optimization [124, 201], and a more flexible and affordable simulation ecosystem [56]. In addition, the proposed automatic structural quality validation system's transfer application to other additive manufacturing technologies [175], such as wire arc additive manufacturing, selective laser sintering, etc., should make it more practical and broader for widespread industrial adoption.

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Chapter 5

Conclusion

5.1 Summary of contributions

This work has made various contributions to the field of structural validation for fused filament fabrication. The main contributions of this dissertation are described as follows:

1. Systematic and comprehensive review and summarisation of different methods, devices, and achievements in a range of monitoring systems for 3D printing. Sensor types and devices used in the existing researches for printer health-state monitoring and printing process product quality monitoring. This work is presented in Chapter 2.

2. Proposed and experimentally verified an algorithm that fuses the defect detection and structural validation on defect impact decision boundary. This work is presented in Chapter 3. A novel component health index that links the size of defects across layers is developed in this work. The component health index accumulates defect information throughout the whole sample printing process. In addition, a multi-dimensional accumulation-threshold-based decision-making system is developed to make the proper decision (i.e., continue or cancel printing) by referring to the evaluation result before the next layer is finished.

3. Developed an image segmentation-based real-time automatic finite element analysis product structural quality validation system. This work is presented in Chapter 4. Proposed the dynamic layer porosity-related Young's modulus in

the Abaqus Python script and an automatic decision-making system by referring to the maximum tensile stress.

5.2 Outlook and future work

Significant progress has been achieved in the domain of in situ monitoring for the Fused Filament Fabrication (FFF) process. Nevertheless, the integration of in situ monitoring with structural validation has limited exploration. A notable knowledge gap persists regarding how to leverage the acquired information to advance FFF structural quality validation or delve deeper into correlation with industrial manufacturing contexts. The future in situ monitoring outcomes should be seamlessly integrated into part validation methodologies, enabling in situ validation with minimal reliance on destructive or non-destructive post-print testing of components. This would mark a significant leap forward for the FFF process.

The future development of FFF in situ monitoring and structural validation is expected to revolve around three key areas:

1. Data Fusion: The signals generated during the FFF process exhibit significant temporal and spatial resolution variations. Synchronizing sensors with different levels of accuracy and resolution is essential to comprehensively understand the printing process. Moreover, the data integration from various perspectives will be helpful, such as thermal-couple, vibration sensor, optical camera, etc., which can provide systematic and comprehensive information for product structural validation.

2. Advanced Algorithms: This encompasses not only in-process signal processing and defect detection but also real-time correction and decision-making. Machine learning and computer vision are anticipated to play increasingly vital roles in this field.

3. Real-Time Feedback Control: Although active work has been undertaken in this domain, existing control systems are often tailored to specific defects and limited to particular printers. A unified framework capable of flexibly integrating available sensors, algorithms, and computational models would greatly accelerate technological advancements.

For the real-time finite element analysis structural validation, several crucial factors warrant further exploration and refinement. These include:

1. Mesh Size Optimization: The impact of mesh size on computational time should be thoroughly investigated and optimized. Balancing computational efficiency with accuracy is essential to ensure the method's practicality and effectiveness.

2. Real-Time Simulation Process Enhancement: Continuous efforts should be made to enhance the real-time simulation process, making it more efficient and responsive. This optimization is vital for seamless integration into industrial workflows.

3. Affordable Simulation Ecosystem: Developing a simulation ecosystem that is both flexible and cost-effective is essential. This would help lower the barriers to entry for smaller manufacturers and encourage broader adoption of the technology.

4. Cross-Technology Application: The proposed automatic structural quality validation system should be adaptable to various additive manufacturing technologies beyond FFF, such as wire arc additive manufacturing and selective laser sintering. Ensuring its compatibility with multiple technologies will make it more practical and applicable across a wide range of industries.

In the foreseeable future, the FFF structural validation system may evolve to operate without human intervention but with well-rounded, useful information. Moreover, the advanced structural validation capability makes FFF a more practical and versatile solution for widespread industrial adoption and for Industry 4.0.

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Appendix A

Machine learning algorithms for defect detection in metal laser-based additive manufacturing: A review[1]

A.1 Abstract

Laser-based additive manufacturing (LBAM), a series of additive manufacturing technologies, has unrivaled advantages due to its design freedom to manufacture complex parts with a wide range of applications. Although advancements in LBAM processes and materials have led to increased manufacturing capabilities, the printing process's repeatability, durability, and reliability still face significant challenges. Therefore, a defect detection system for the LBAM processes is essential, as it promises to guarantee product quality and increase the efficiency of the printing process. As a practical and widely applied technology, machine learning methods have been providing novel insights into the manufacturing process, which has proven advantages for defect detection in LBAM. This paper summarizes the machine learning algorithms for defect detection in the metal LBAM processes. To have a comprehensive and systematic summary, machine learning algorithm, material type, defect type, dataset type, and algorithm accuracy for various LBAM technologies are described.

[1]Yanzhou Fu, Austin Downey, Lang Yuan, Tianyu Zhang, Avery Pratt, and Yunusa Balogun, 2022, Journal of Manufacturing Processes, vol. 75, p. 693-710. Reprinted here with permission of the publisher.

A.2 Introduction

Additive manufacturing (AM) is a rapidly growing technology that has proven advantages for unlocking design freedom for lightweight components with complex geometries and as a disruptive technology, it offers exciting new manufacturing capabilities [132, 134]. AM has been actively utilized in various industries, and it shows enormous economic potential [102, 263]. According to the Wohlers 2021 Report on AM, in 2020, the AM industry grew 7.5%, or nearly $ 12.8 billion, despite the global COVID-19 pandemic [317]. Among various AM techniques, laser-based additive manufacturing (LBAM) displays considerable potential for industrial adoption and has already changed the manufacturing process by enabling complex design and innovative application development [64]. The principle of LBAM technologies lie in using a laser beam to yield thermal energy for sintering/melting and consolidating additive materials or emitting light quanta of a particular wavelength to induce a chemical curing response in vat polymerization. The materials utilized in the LBAM processes can be in the form of powders (metals, ceramics, and polymers), solids (paper, plastics, and metals), or liquids (resin). In LBAM, according to the difference in material feeding approach, there are three major systems: powder blown system, powder bed system, and wire-feed system. In the powder blown system, the material is carried out through single or multiple nozzles and melted by a laser beam. The powder bed system generally has two chambers, a build chamber for printing and a powder chamber with a coating roller to spread the powder material across the build chamber. In the wire-feed system, the wire is fed and melted to manufacture the metal parts.

Due to the convenience of fabrication and the practicability of mechanical properties enhancement, LBAM is regarded as a highly favored manufacturing technology for Industry 4.0 over conventional technologies [194]. However, its full potential is held back by variabilities inherent to the process. The LBAM processes are dominated by complicated physics processes, including laser energy absorption and transmission, material evaporation, remelting and solidification, melt pool fluid dynamics, and microstructure evolution via epitaxial growth and nucleation. Defects, such as cracks, pores, distortion, and insufficient melt frequently occur in the printing process, impacting the manufactured parts' mechanical and functional properties. The severity and density of these defects can be seen as a semi-stochastic function of material parameters and printing parameters related to laser energy density and its associated parameters, such as power, speed, and spot size [296]. In addition to the above factors, the LBAM process consists of different print technologies. Even though they share the same basic principle, individual characteristic are exhibited by different LBAM processes, which makes defect detection harder. Also, part geometry and the specific powder utilized for fabrication influence defect formation dramatically, making defect detection in LBAM complicated and challenging. To accelerate the industrialization of the LBAM processes, a novel and effective defect detection system to detect and eliminate defects and guarantee product quality is essential.

Machine learning (ML) has progressed from a laboratory curiosity to a fundamental process used in various industries and fields, including smart manufacturing [259, 281], civil engineering [15], and biomedical science [153]. For the LBAM processes, ML has proven itself as a useful way to monitor product quality or detect defects [307]. Applying ML to a defect detection system offers new insights into LBAM processes due to its ability to discover implicit knowledge and build the relationship between printing parameters and product quality [189].

This paper reviews and summarizes the latest ML algorithms that have been utilized for LBAM defect detection systems, examines the achievement of these systems, and discusses the current and future research direction. It is important to point out that this paper only focuses on metal LBAM processes. In the author's search, no defect detection ML algorithms for laser-based wire-feed systems were found; all research focused on powder blown systems and powder bed fusion systems. Powder blown system is commonly known as directed energy deposition (DED), laser metal deposition (LMD), or laser engineered net shaping (LENS), while powder bed fusion system, is commonly known as laser powder bed fusion (LPBF), direct metal laser melting (DMLM), direct metal laser sintering (DMLS), or selective laser melting (SLM). As these technologies essentially are the same, DED will be used for all powder blown systems and LPBF for all powder bed fusion processes to unify the naming conventions. To have a systematic and comprehensive summary, ML algorithm, material type, defect type, input data type, and algorithm accuracy for various LBAM technologies are described. In this paper, the background is described in section A.2, and section A.3 presents different defect types in the LBAM processes. In section A.4, ML algorithms used in defect detection systems for the LBAM processes are discussed; a conclusion to the review and future trends are discussed in section A.6.

A.3 Defect types in LBAM

Although different LBAM processes have their own characteristics, they share similar manufacturing methodologies. The two most common technologies for LBAM are DED and LPBF and are shown in Fig. A.1. As shown in Fig. A.1(a), the center of a typical DED system is the nozzle head, which consists of the energy source (e.g., laser beam, electron beam) to melt additive material at the point of deposition and the material delivery nozzle for feeding powder or wire [133]. Typically, the nozzle head is fixed on either an articulated arm or a multi-axis computer numerical control (CNC) head. The laser beam melts the powder/wire and forms the melt pool on the substrate at the start spot along the build track. This process continues until the whole part is completed. The LPBF system is shown in Fig. A.1(b). In the LPBF printing process, the laser beam scans at a controlled speed on the powder bed, and the selected locations of the powder are fused to form a solid track. The powder bed is sunk by the predefined layer thickness, and a new powder layer is spread and leveled after the previous layer is finished. The process repeats until the part is completely built. As the LBAM processes are complicated; many printing parameters, such as laser scan speed, laser power, scanning pattern, material type and size, and chamber environment, are involved in this process. Any improper settings in printing will produce defects. These defects can be divided into four types: product geometric and dimensional defect, porosity, incomplete fusion, and cracks. This section describes defect types and is organized as follows: section A.3 shows the geometric and dimensional defect, section A.3 presents the porosity defect, incomplete fusion is described in section A.3, and section A.3 discusses the crack defect.

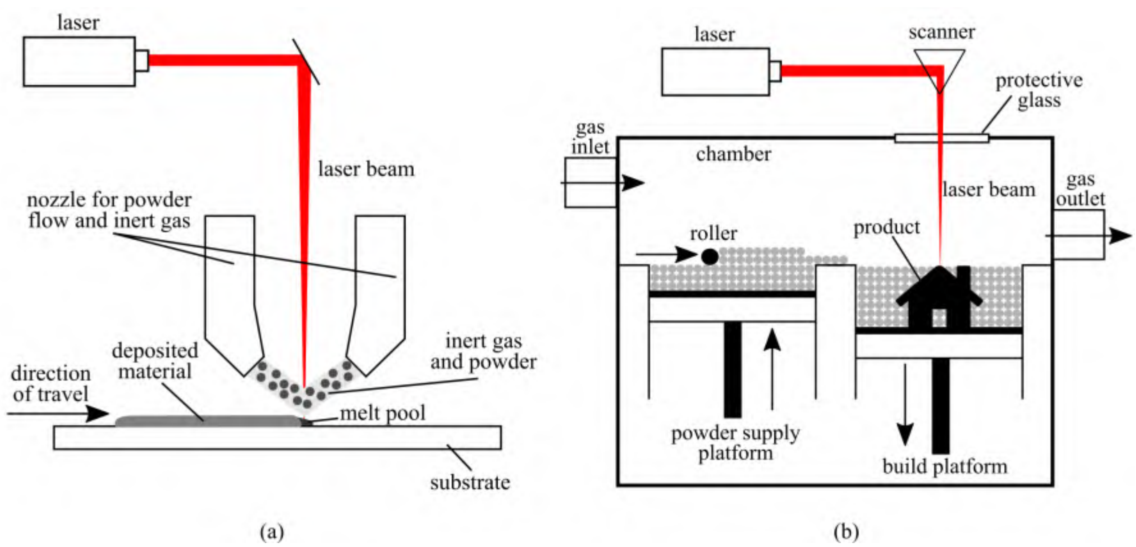


Figure A.1: Schematic illustration of LBAM processes with two different feeding systems: (a) the DED technology of the powder blown system; and (b) the LPBF technology of the powder bed fusion system.

Geometric and dimensional defects

Geometric deviation is one of the common defects in the LBAM processes. Geometric defects may be caused by machine errors and errors in the generation of machine

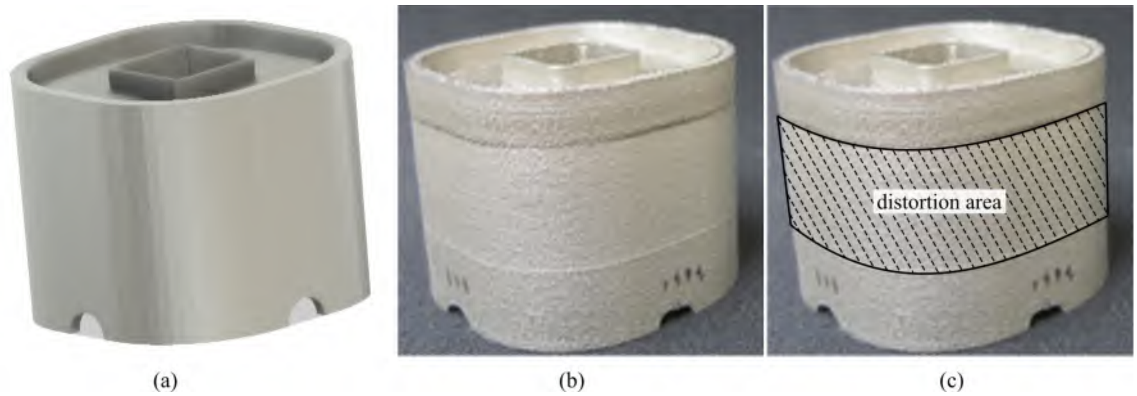


Figure A.2: An distorted product of LPBF process with stainless steel 316L: (a) CAD model of the designed product; (b) manufactured product; and (c) the distortion area on the manufactured product.

code that controls laser movement. Laser position error and platform movement error are two machine errors that lead to geometric defects [27]. Layer-based slicing can introduce step-wise marks on surfaces, where their contours may also introduce inaccuracy due to melt pool dimensions.

For dimensional inaccuracy, the main influence factor is shrinkage/distortion. In the LBAM processes, there are two kinds of shrinkage: sintering shrinkage and thermal shrinkage. Sintering shrinkage is mainly produced by densification, while thermal shrinkage is caused by cyclic heating and cooling, which leads to significant residual stress, thus, local plastic deformation [207,226,361]. An example of a distorted canonical geometry is shown in Fig. A.2, where shrinkage was observed near the top.

Another factor influencing product geometric and dimensional accuracy is surface finish accuracy, also known as surface roughness [205]. During the printing process, partially melted particles and spatters attach to the product surface, causing the final dimension to vary from that designed [294]. Poor surface roughness not only impacts the product's usefulness but also influences the material's properties (e.g., fatigue life) [50]. Fig. A.3 shows a typical surface morphology of the LPBF process in 316L stainless steel, where melt pool tracks and attached particles can be observed.

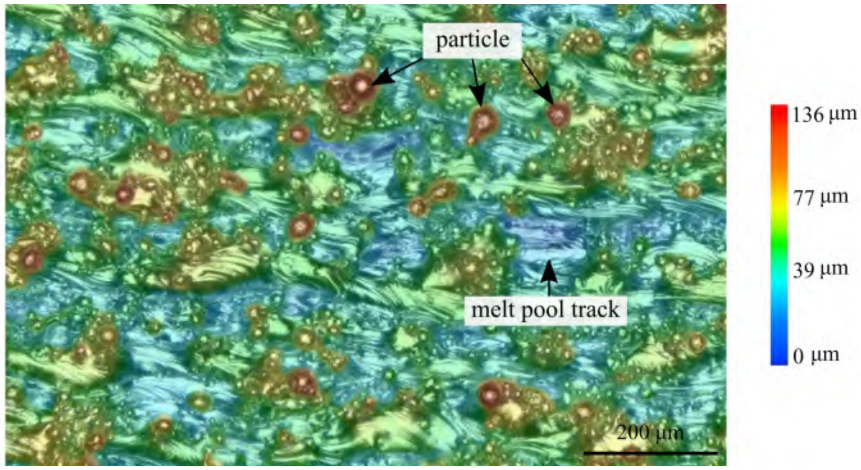


Figure A.3: Product's surface metrology of LPBF process with stainless steel 316L.

Porosity

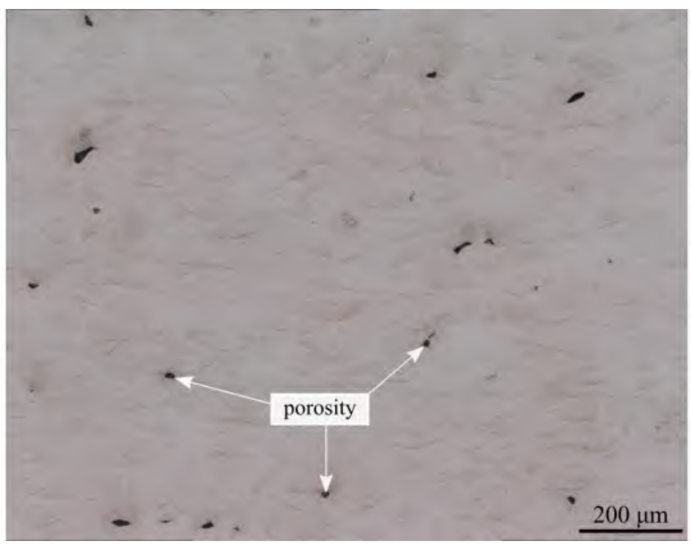


Figure A.4: Optical image of porosities in LBAM printing part with stainless steel 316L.

Porosity is a common defect in the LBAM processes and negatively impacts product density [3]. It is particularly deleterious to mechanical properties, such as fatigue for structural components. The porosity size varies with distinguishing shapes under different forming mechanisms [348]. Typically, pores form due to insufficient powder spreading, lack of fusion (as part of the "incomplete fusion defect" discussed next), keyholing, shrinkage, and gas involvement. The first three types are large in size and are due to improper process parameters. For example, low metal powder packing density (e.g., less than 50%) and spatters may lead to large voids in the powder bed, which cannot be filled during melting. High laser energy density leads to keyholing, and metal vapor pressure results in pores at the bottom of the deep melt pool. Through optimization of LBAM processes parameters, these defects can be fully eliminated [287]. The latter two types, in general, are smaller in size. Pores due to solidification shrinkage are normally located between grains and follow the shape of grain boundaries [301]. Gas-induced pores are spherical in shape and can possibly originate from the gas trapped in the powder feedstock particles due to the gas-atomized process, gas dissolved in metal due to different gas solubility at different temperatures, metal vapor generated during melting, or moisture from the powder surface. An example of porosity is shown in Fig. A.4. This type of pores cannot be completely avoided due to the nature of its forming mechanism.

Incomplete fusion

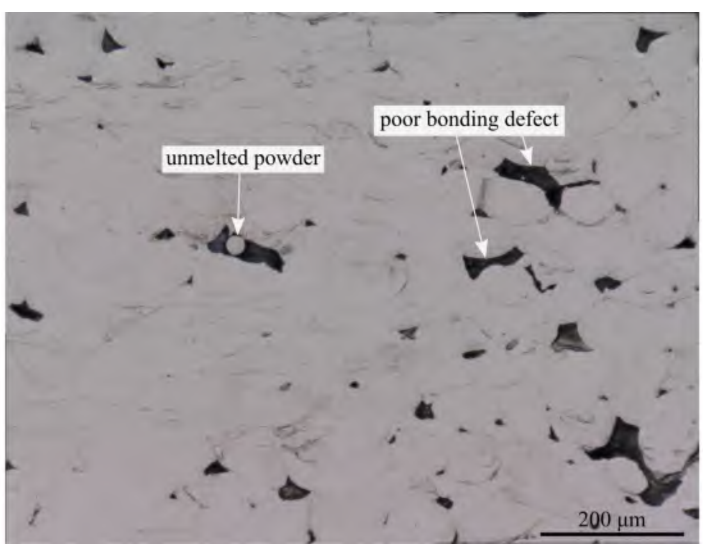


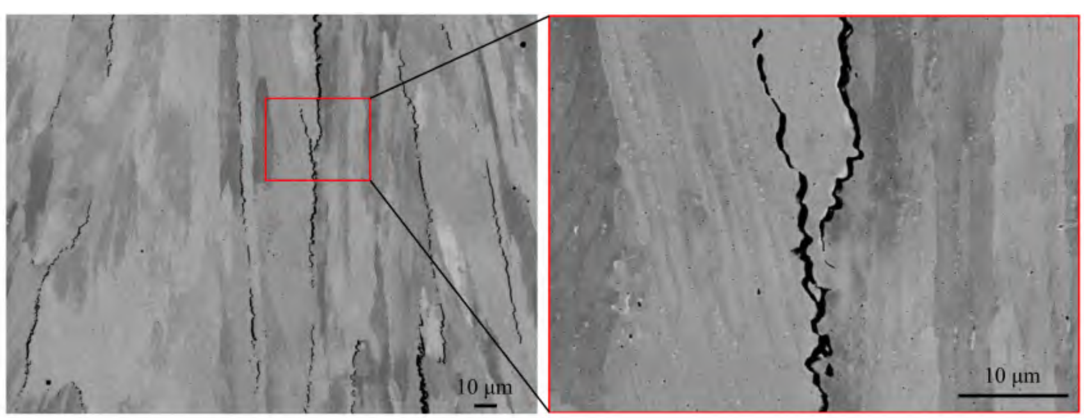
Figure A.5: Optical image of the lack of fusion defects in LBAM printing part with stainless steel 316L.

The incomplete fusion hole is mainly produced by a lack of energy input in the

LBAM printing processes, also known as lack of fusion defects, as shown in Fig. A.5. Lack of fusion defects are produced mainly because the metal powder is not fully melted to deposit a new layer on the previous layer with sufficient overlap between them [105, 302]. The lack of fusion defects can be categorized into two types: poor bonding defects formed by insufficient molten metal powder in a solidification process and defects due to unmelted metal powder. It is also one of the reasons for large pores in LBAM.

When the scan path of the laser density is low, the molten pool width is small, which produces an insufficient overlap between each track. The insufficient overlap then results in unmelted metal powder between the scan tracks as there is insufficient energy to fully melt the powder for a new layer [176]. As a result, incomplete fusions are produced and unmelted powder remains in the LBAM printed product, especially between the scan tracks and the deposited layers. Moreover, the defect formation area makes the product surface rough, directly impacting the molten metal's flow to form interlayer defects. Then the interlayer defects may gradually extend and propagate to form more significant multi-layer defects in the continuing printing process [359].

Cracks



(a) (b)

Figure A.6: Image of crack morphology from an LBAM fabricated part with stainless steel 316L: (a) crack morphology; and (b) enlarged crack morphology.

In the LBAM printing processes, with a high laser energy input, the metal powder undergoes fast-melting and rapid solidification. The cooling rate in the melting pool can reach more than 1.6×10^6 K/s following high-temperature gradients [247]. Cracks initiate during and after solidification based on the materials, processes, and part design. For example, solidification cracking forms due to insufficient liquid feed at the last stage of solidification. Strain-age cracking, a type of ductile cracking, forms due to the formation of precipitates during heat treatment processes. Also common is the tremendous amount of residual stress is developed in the printed product, which may exceed the ultimate strength of the materials and lead to cracks [111, 354]. In Fig. A.6(a), solidification cracking was developed along grain boundaries in the LPBF printing process fabricated part. Fig. A.6(b) shows the enlarged crack morphology of the red box in Fig. A.6(a).

A.4 Machine learning algorithms for defect detection

ML is a subset of artificial intelligence (AI) that provides systems the capacity to learn and improve from experience without being explicitly programmed. Lots of the ML models have been built and applied by researchers for defect detection in AM [189]. This section reports ML algorithms used explicitly for defect detection during the metal LBAM processes. The field of ML can generally be divided into four domains, as shown in Fig. A.7: 1) supervised machine learning; 2) unsupervised machine learning; 3) semi-supervised machine learning, and; 4) reinforcement machine learning. This section presents the state-of-the-art in defect detection and is organized by domain: section A.4 presents supervised machine learning algorithms; section A.4 describes unsupervised algorithms; section A.4 lists the semi-supervised machine learning algorithms; and section A.4 discusses reinforcement learning algorithms.

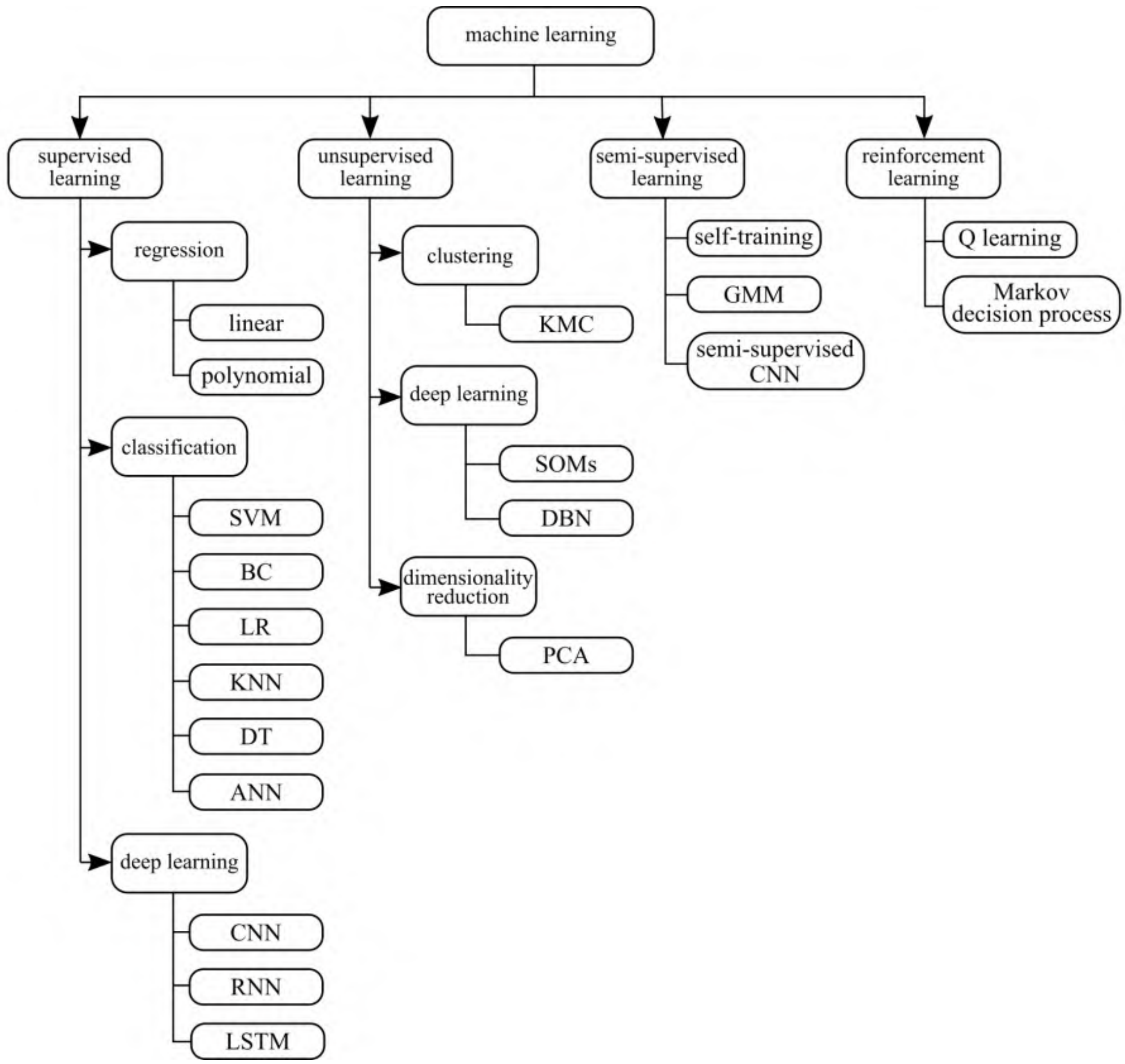


Figure A.7: Categories of different machine learning algorithms (all the abbreviations are listed in the nomenclature and can be found in the paper).

Table A.1: Supervised ML algorithms utilized in defect detection for LBAM processes.

ML category	ML algorithm	AM technology	Material type	Defect type	Dataset type	Accuracy	Reference
Supervised learning	Logistic regression	LPBF	17-4 precipitation stainless steel	Anomalous, good quality	Melt pool thermal image	99.6%	[185]
		DED	Mixing Ti-6Al-4V with H13 tool steel	Pore, crack	AE signal	1.72973 (MSE)	[93]
	Gaussian process regression	LPBF	17-4 stainless steel	Porosity	laser power, scanning speed	No specific accu-racy	[282]
		LPBF	Stainless steel 304L	Underheating, medium underheating, normal, medium overheating, overheating	Acoustic signal	89.13%	[342]

Table A.1: Supervised ML algorithms utilized in defect detection for LBAM processes.

ML category	ML algorithm	AM technology	Material type	Defect type	Dataset type	Accuracy	Reference
		DED	Ti-6Al-4V	Porosity	Melt pool thermal image	97.97%	[144]
		LPBF	17-4 precipitation stainless steel	Anomalous, good quality	Melt pool thermal image	99.5%	[185]
		DED	Ti-6AL-4V.	Healthy layer, unhealthy layer	Melt pool thermal image	91.65%	[255]
		LPBF	Inconel 718	Desirable, balling, severe keyholing, keyholing porosity, or under-melting	Melt pool morphology	85.1%	[253]
		LPBF	Stainless steel GP-1	Anomalous, good quality	Layerwise Optical imaging	80%	[229]

Table A.1: Supervised ML algorithms utilized in defect detection for LBAM processes.

ML category	ML algorithm	AM technology	Material type	Defect type	Dataset type	Accuracy	Reference
		LPBF	Ti-6Al-4V	Porosity	Layerwise Optical imaging	89.36%	[130]
		LPBF	Stainless steel 316L	Track continuity	Optical image	90.1%	[355]
		LPBF	Stainless steel GP-1	Anomalous, good quality	Optical image	85%	[104]
		DED	Stainless steel 316L	Track depositing height	Printing parameters	2.89E-08(MSE)	[181]
	Bayesian classifier	LPBF	Inconel 625	Porosity, quality of fusion	Visual image	89.5%	[9]
		LPBF	AlSi10Mg aluminum powder	Key hole, lack of fusion	Optical image	77%	[19]
		DED	Ti-6Al-4V	Porosity	Melt pool thermal image	98.44%	[144]

K-Nearest neighbors

Table A.1: Supervised ML algorithms utilized in defect detection for LBAM processes.

ML category	ML algorithm	AM technology	Material type	Defect type	Dataset type	Accuracy	Reference
		LPBF	Ti-6Al-4V	Porosity	Layerwise Optical imaging	78.60%	[130]
		LPBF	17-4 precipitation stainless steel	Anomalous, good quality	Melt pool thermal image	99.1%	[185]
		LPBF	Inconel 718	Porosity	Photodetector data	90%	[198]
		LPBF	Stainless steel 316L	No defect, bulge defect, dent defect, wavy defect	Point cloud	93.15%	[52]
	Random decision tree	DED	Al-5083 powder	Macropores, micropores, elongated pores	Optical microscope image	94.41%	[98]
		LPBF	Stainless steel GP-1	Anomalous, good quality	Layerwise Optical imaging	90%	[229]

Table A.1: Supervised ML algorithms utilized in defect detection for LBAM processes.

ML category	ML algorithm	AM technology	Material type	Defect type	Dataset type	Accuracy	Reference
		LPBF	Ti-6Al-4V	Porosity	Layerwise Optical imaging	84.40%	[130]
		DED	Mixing Ti-6Al-4V with H13 tool steel	Pore, crack	AE signal	1.702703 (MSE)	[93]
		LPBF	Inconel 625	Surface texture	Printing process parameters	Refer to paper	[221]
		LPBF	Stainless steel 316L	Balling, lack-of-fusion, conduction, key hole	Pyrometer data, video camera data	Refer to paper	[90]

Table A.1: Supervised ML algorithms utilized in defect detection for LBAM processes.

ML category	ML algorithm	AM technology	Material type	Defect type	Dataset type	Accuracy	Reference
		LPBF	Ti-6Al-4V	Geometry variation	Layerwise image	92.50 $\pm$ 1.03%	[129]
		LPBF	Titanium alloy, STM B348 Grade 23 Ti-6Al-4V	Printing condition variation	Powder bed image	97.14%	[339]
	Convolutional neural network	LPBF	Stainless steel 316L	Track continuity	Optical image	92.7%	[355]
		DED	Stainless steel 304, stainless steel 316, Ti-6Al-4V, AlCoCrFeNi alloys, Inconel 718	Good quality, crack, gas porosity, lack of fusion	Digital microscope image	92.1%	[60]

Table A.1: Supervised ML algorithms utilized in defect detection for LBAM processes.

ML category	ML algorithm	AM technology	Material type	Defect type	Dataset type	Accuracy	Reference
Supervised learning	Convolutional neural network	LPBF	Ti-6Al-4V	Thin wall's thickness, density, edge smoothness, discontinuity	Optical image	85%	[91]
		LPBF	CL 31	Good quality, average quality, bad quality	OM-meltpool 3D-generated image	98.9%	[154]
		LPBF	Stainless steel 316L	Track continuity	Printing process video	93.1%	[347]
		LPBF	ASTM F75 I CoCrMo	Under-melt, beautiful-weld, over-melt	Micrograph image	100%	[352]
		DED	Stainless steel 316L	Dilution	MWIR image	2.8% (RMSE)	[106]
		DED	Sponge Titanium powder	Porosity	Melt pool image	91.2%	[351]

Table A.1: Supervised ML algorithms utilized in defect detection for LBAM processes.

ML category	ML algorithm	AM technology	Material type	Defect type	Dataset type	Accuracy	Reference
		DED	Stainless steel 304, stainless steel 316, Ti-6Al-4V, AlCoCrFeNi	Gas porosity, crack, lack of fusion	Optical image	92.1%	[60]
		DED	Ti-6Al-4V	Porosity	Pyrometer image	100%	[288]
		DED	0Cr18Ni9 powder	Printing parameter variation	Thermal image	80%	[168]
		LBPF	AlSi10Mg, bronze, Inconel 625, Inconel 718, stainless steel 316L, Ti-6Al-4V, Fe-3Si, stainless steel 17-4 PH	Recoater hopping, recoater streaking, porosity, swelling, spatter, soot, debris, super-elevation, part damage, incomplete spreading	Powder bed image	Refer to paper	[254]

Table A.1: Supervised ML algorithms utilized in defect detection for LBAM processes.

ML category	ML algorithm	AM technology	Material type	Defect type	Dataset type	Accuracy	Reference
		LPBF	Alloys, Inconel 718 alloys	Short feed defect, warpage, part shifting defect	Powder layer optical image	94%, 96%, 94%	[328]
		LPBF	AlSi10Mg, bronze, Inconel 625, Inconel 718, stainless steel 316L, Ti-6Al-4V, Fe-3Si, stainless steel 17-4 PH	Recoater hopping, recoater streaking, debris, super-elevation, part failure, incomplete spreading	Powder bed image	97%	[252]
		LPBF	Inconel 718, stainless	Overheating, normal, irregularity, balling	Optical image	99.7%	[356]
		DED	Stainless steel 316L, Ti-6Al-4V, Fe-3Si,	Distortion	Thermal image	24 μm (RMSE)	[84]

Table A.1: Supervised ML algorithms utilized in defect detection for LBAM processes.

ML category	ML algorithm	AM technology	Material type	Defect type	Dataset type	Accuracy	Reference
		LPBF	Stainless steel 17-4 PH	Defects from standard energy, low energy, high energy, very low energy	Powder layer images, part slice images after laser scanning	99.4%	[40]
		LBAM	Not mentioned	Metal fracture, metallographic defects	Metal fracture microscope images, Metal metallographic images	82%, 87.5%	[114]
Supervised learning	Convolutional neural network	LPBF	Ti-6Al-4V, CM247-LC	Porosity, surface imperfections	Acoustic spectroscopy, optical image	No specific accuracy	[314]

Table A.1: Supervised ML algorithms utilized in defect detection for LBAM processes.

ML category	ML algorithm	AM technology	Material type	Defect type	Dataset type	Accuracy	Reference
		LBAM	Ti–6Al–4V	Laser additive manufacturing transvers and longitudinal, laser cladding	Microstructure	90.4%	[170]
		LPBF	H13 tool steel	Delamination, splatters, good quality	Thermographic off-axis imaging	96.80%	[23]
		LPBF	Stainless steel 316L	Poor, medium, high quality	AE signal	83% - 89%	[260]
	Long-term recurrent convolutional networks	DED	Ti–6Al–4V	Porosity	Thermal image	92.07%	[289]

Supervised learning

Supervised learning is the most widely used ML technique. The training dataset needs to be labeled with input values and the corresponding output values. During the training process, the ML algorithm uses this labeled dataset and learns the relationship between input data and output data. Supervised learning is appropriate for classification and regression; therefore, many defect detection systems use supervised learning approaches to detect and classify different defects in LBAM. For clarity, the supervised learning subsection of this paper is divided into three parts: traditional classifier, artificial neural network (without convolutional layers), and convolutional neural network. As most of the reviewed research utilizes several classification algorithms in one paper and compare their performance, this paper lists all traditional classification algorithms together. Traditional classification algorithms include support vector machine, Bayesian classifier, logistic classification, K-nearest neighbors, and decision tree. Artificial neural network (ANN) can also be formulated as a classifier and are discussed here in detail with one subsection. Furthermore, ANNs that utilizes convolutional layers in their architectures are separated into their own subsection due to their prevalence in the literature. Table A.1 compiles a list of supervised ML algorithms utilized for detecting defects in components manufactured using the LBAM processes. In Table A.1, key features of the reviewed research are listed, including ML domain, dataset type, material type, defect type, and algorithm performance.

Traditional regression

Regression is a method of modeling target values based on independent predictors. This method is mainly used to uncover cause and effect relationships between variables. Regression techniques differ primarily depending on the number of independent variables and the relationship between the dependent and independent variables. Regression algorithms may be linear as well as non-linear.

Different regression algorithms also have been applied to defect detection in the LBAM process. Mahmoudi et al. built an anomaly detection system for the LPBF printing process [185]. This system detects printing process deviations by using the thermal signals obtained from the thermal images of the melt pool. Logistic regression is utilized to decide the printing process quality (in control or out of control) through a step-by-step process. The result shows this framework had a very low error rate for cavity defect detection, which can detect the 750 μm diameter cylindrical cavity on a 5.5 $\times$ 8 $\times$ 9 mm rectangular prism printed by 17-4 precipitation hardening stainless steel powder. Gaja et al. used a logistic regression model to detect product defects simulated by mixing Ti-6Al-4V powder with H13 tool steel powder in the LMD printing process [93]. With the AE signal as the input data, this approach can detect two primary defects between crack and porosity. Moreover, Tapia et al. proposed a spatial Gaussian process regression model to learn and predict porosity for the SLM process [282]. Two printing parameters, laser power and scanning speed, which have the most significant effect on porosity formation, are fed into the Gaussian prediction model. The case study validated on 10 mm $\times$ 10 mm $\times$ 10 mm test coupons printed with 17-4 PH stainless steel powder. The result shows that the proposed model provided accurate porosity prediction under any printing parameters.

Traditional classifiers

Classification is a class of ML that focuses on predicting the label associated with a given piece of data. Sometimes classes are termed as labels, categories, or targets. Classification predictive modeling is the task of approximating a mapping function (f) from input variables (X) to discrete output variables (y). In a classification task, the output can be a binary or multi-class classification according to the chosen classifier, such as good or bad quality, defect types.

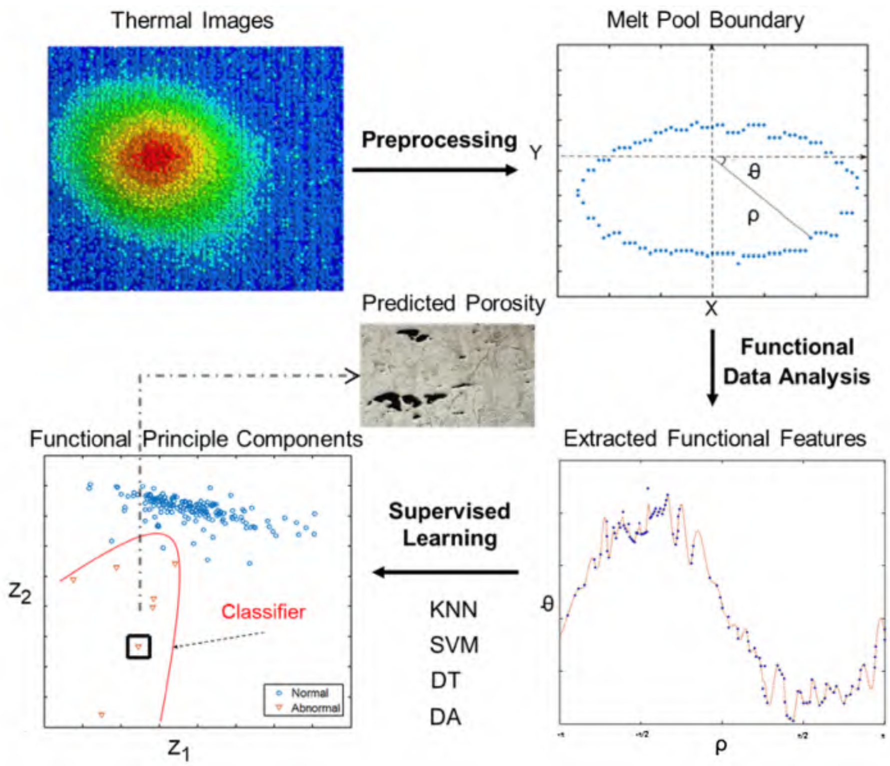


Figure A.8: Demonstration of porosity prediction procedure using supervised machine learning.

Support vector machine (SVM) is a popular ML tool that offers a solution for both classification and regression. Various researchers have used this algorithm for defect detection; for example, Khanzadeha et al. applied multiple supervised learning methods (i.e., decision tree (DT), K-nearest neighbor (KNN), SVM, linear discriminant analysis (LDA), and quadratic discriminant analysis (QDA)) to predict porosity on the single track thin wall specimens (Ti-6Al-4V) by using melt pool thermal images [144]. In the SVM method, when selecting the polynomial as the kernel function, the porosity prediction accuracy is 97.97%. Fig. A.8 shows the overall SVM methodology for porosity prediction. To classify the melt pool morphologies, Scime et al. developed a flaw formation identification method for the LPBF process with a multi-class SVM, which can detect five melt pool types on a 10 mm × 20 mm rectangle made with Inconel 718 powder: under-melting, balling, spatter, porosity, and desirable [253]. Gobert et al. applied SVM to detect defects on a staircase cylinder fabricated with stainless steel GP-1 powder in the LPBF process [104]. By labeling different flaws (i.e., porosity, incomplete fusion, crack, or inclusions) with computer tomography (CT), the properly trained model achieved an over 80% defect detection accuracy. The flaw with a diameter larger than 47 μm can be identified. Petrich et al. developed an approach based on the SVM and neural networks to detect defects during the LPBF process with high-resolution layer-wise images [229]. By utilizing a cross-validation methodology, the proposed algorithm in situ anomaly detection achieved a 90% accuracy for a staircase cylinder made with stainless steel GP-1 powder. Imani et al. investigated the printing process to figure out the effects different printing process conditions had on fusion porosity in LPBF [130]. By analyzing the in-process layer-by-layer optical images, the result shows the process conditions significantly impact the porosity generation. This approach satisfies statistical fidelity by linking the features extracted from the layer-by-layer images to the process condition with ML methods. Several ML classifiers (SVM, complex tree, LDA, KNN, bagged trees, and feed-forward neural network) are applied in this research. The SVM classifier achieved an 89.36% porosity defection accuracy with F-score for the cylinders of 10 mm diameter $\times$ 25 mm height manufactured with Ti–6Al–4V. The resolution for the pore detection is larger than 65 μm. Mahmoudi et al. built an anomaly detection system for the LPBF printing process [185]. This system detects printing process deviations by using the thermal signals obtained from the thermal images of the melt pool. Classifiers, including logistic regression (LR), KNN, SVM, and random forests (RF), are utilized to decide whether the printing process is in control or out of control through a step-by-step process. The result shows this framework had a very low error rate for cavity defect detection.

SVM is also utilized by researchers for product quality prediction and inspection. Seifi et al. proposed a method to detect product anomalies in real-time for LBAM and verified on the DED printing process [255]. After extracting the key layer-wise signature features from the melt pool images with multilinear principal component analysis (MPCA), an SVM classifier is used to extract key layer-wise signature features to predict the product's quality. The proposed methodology prediction accuracy obtained an F1-score of 91.65% and was validated on a thin wall fabricated with Ti-6AL-4V. Lu et al. utilized the least square support vector machine (LS-SVM) to predict the track's depositing height for the DED process [181]. With laser scanning speed, laser power, and powder feeding speed as inputs, this model provided knowledge on the whole printing part's precision. The experimental result on a thin wall printed with AISI316L powder shows that the LS-SVM algorithm prediction value had a high correlation coefficient with the experimental value of 0.971. Ye et al. proposed a defect-recognition approach for the LPBF process [342]. Using SVM and extracted features from recorded acoustic signals, this method achieved accurate diagnosis for five different melted states (i.e., overheating, medium overheating, underheating, medium underheating, and normal) for the single tracks printed by 304L stainless steel powder.

The Bayesian classifier is a probabilistic classifier that gives probability information about the examined product layer being defective, which could also be considered a product quality quantitative method. Therefore, the Bayesian classifier is also a useful tool for defect detection in the LBAM processes. A Bayesian classifier was built and trained by Aminzadeh et al. to classify the product quality fabricated by Inconel 625, which can detect the defective regions or abnormal layers in LPBF [9]. The training data is labeled printing process images with detailed layer defects and porosity. The result shows that with appropriate feature selection, the defect identification performance is 89.5%. To predict the product quality (stainless steel 316L powder) printed by LPBF, Hertleina et al. proposed a Bayesian network, which combines the printing process parameters with product quality characteristics [120]. After training, the forecasted mean of a printed product quality characteristic (hardness) is within 0.41 standard deviations of the true value. Bartlett et al. utilized a three-dimensional digital image correlation system to predict microstructural defects (i.e., lack of fusion and keyhole) from AlSi10Mg aluminum powder bed quality in direct metal laser sintering (DMLS) printing process [19]. By feeding the powder anomaly topology images into the Naïve-Bayes classifier, the algorithm can predict microstructural defect formation probability according to the in-process powder bed error. The defect and no defect prediction accuracy for three energy density printing conditions (i.e., low, standard, and high) are 66%, 77%, and 72%, respectively.

In addition, other classifiers are also utilized by researchers for detecting defects in the LBAM processes. For example, Montazeri et al. presented an in-process porosity monitoring approach using optical emission spectroscopy [198]. The experimental result shows that using the graph Fourier transform coefficients as the input feature within the KNN model had the best layer porosity-level prediction accuracy for two-level classification (90% F-score) on the disk manufactured with nickel alloy 718 powder. In addition, the computation time only needs less than 0.5 s. Chen et al. proposed a rapid surface defect detection method for the DED technology, which integrates the in situ point cloud processing with ML [52]. The algorithm used in this research combined unsupervised with supervised ML to identify and classify the surface defect. The unsupervised algorithm is used to segregate the potential surface defect region from the point cloud. This approach registered the defect's existence and typed recognition by inputting the clustering result from an unsupervised algorithm into the supervised algorithm. The surface defects are categorized into four classes: no defect, bulge defect, dent defect, and wavy defect. To compare the classification accuracy, eight algorithms are applied; the most accurate algorithm for classification is the KNN classifier, with an accuracy of 93.15%. Garcia-Moreno et al. presented an image-based porosity classification method by utilizing a random forest algorithm for the LMD printing process [98]. The random decision tree classifier achieved an accuracy of 96.45%, 94.56%, and 92.21% for the macropores, micropores, and elongated pores on the dogbone sample fabricated with Al-5083 powder, respectively. The proposed method can segment and classify pores between 5 and 250 μm.

Artificial neural network

Artificial neural networks (ANN) are a class of ML algorithms modeled loosely after the human brain and designed to recognize patterns. A typical ANN usually has three parts: the input layer (first layer), one or more hidden layers (middle layers), and the output layer (last layer). Each layer consists of a collection of neurons, which are nodes connected with the other nodes via links (also termed synapse). Each link has a weight, which decides one node's impacting strength to another node. During the training process, the ANN model forms probability-weighted associations between the input and target. The target output is given; thus, training is performed by reducing the difference between the network's processed output and the target output. Backpropagation is used to adjust the model's weights and biases, given the change of error at each step (also termed the gradient) and a predefined set of learning rules. After successive and sufficient adjustments, the network's output is increasingly similar to the target output, which means the network has been fully trained and can be used for future prediction.

When fed a labeled dataset for training, the ANN is helpful for clustering and classification. Petrich et al. developed an approach based on the ANN to detect defects during LPBF with high-resolution layer-wise images [229]. By utilizing cross-validation strategies for training and testing, close to 90% accuracy is achieved when using the ANN model to detect the defect for a staircase cylinder fabricated by stainless steel GP-1 powder. Gaja et al. used an ANN model to detect product defects in the LMD printing process [93]. With the AE signal as the input data, this approach can detect crack and porosity simulated by mixing Ti-6Al-4V powder with H13 tool steel powder. To predict the product surface texture in LPBF, Ozel et al. applied several ML algorithms to establish the relationship between printing process parameters (i.e., laser energy density and scan strategy) and measured surface texture parameters [221]. For the ANN algorithm, it is able to reflect the actual conditions when the measurement data is sufficient; for the genetic programming (GP), it is good at selecting the best LPBF process parameters for the cubes fabricated with nickel alloy 625 with the dimension of 16 mm $\times$ 16 mm $\times$ 15 mm. Gaikwad et al. developed and evaluated an ML-based quality assessment model for a single track printed with stainless steel 316L powder by LPBF process [90]. The study researched various printing parameters (i.e., laser power and laser velocity) effects on single-track's quality and analyzed the quality measurement with height-map by using the extracted mean and standard deviation of width and percent continuity. The experimental result shows that the proposed sequential decision analysis neural network (SeDANN), which uses the sensor data-derived features, performs better than the other ML models, such as long short-term memory recurrent neural network (LSTM-RNN), in speed and accuracy for balling, lack-of-fusion, conduction, and keyhole detection. Fig. A.9 shows the schematic of the SeDANN proposed by Gaikwad et al.

A deep neural network (DNN) is an ANN with multiple layers between the input and output layers. The DNN algorithm finds suitable mathematical manipulation to exploit the input information into the output, whether it is a non-linear or linear relationship. DNNs have also been applied to defect detection in the LBAM processes. For example, Imani et al. designed a DNN model for real-time incipient geometry defects detection from the spatial characterization images in the LPBF process [129]. The experimental result shows that the proposed DNN model effectively achieved geometry flaw detection on drag link joint object (23.7 mm $\times$ 13.3 mm $\times$ 27.3 mm). The accuracy achieved is 92.5% for the 750 μm cylinder and cube defect. As laser power has a huge impact on the pores and cracks formation, which directly determines the printing product quality, Kwon et al. applied a DNN in LPBF to find the link between melt-pool images and laser power [155]. The cuboid product with a dimension of 8.5 mm $\times$ 8.5 mm $\times$ 4 mm is fabricated with stainless steel 316L. The proposed DNN model's classification accuracy on this product is 98.9%, which is helpful for understanding the product microstructure formation by abnormal laser power. Mohammadi et al. presented a novel approach for defect detection in the LPBF process by utilizing the elastic waves from the AE sensor [195]. The proposed approach can detect three types of defect (i.e., parts with minimum defects, parts that had only intentional cracks, and parts that had both intentional cracks and porosities) on the cylindrical parts with a dimension of 20 mm diameter and 10 mm height fabricated with H13 tool steel powder. Three ML algorithms are utilized to analyze and interpret the data. Firstly, a K-means clustering is applied for data labeling, then followed by a DNN to match the AE signal with the correct defect type. Secondly, a PCA is employed to reduce the data's dimensionality. A Gaussian Mixture Model (GMM) is utilized to accelerate the defect detection speed. Thirdly, a variational auto-encoder method is applied to get a general feature of the AE signal that could be an input for the classifier. The unique contribution of this work is that the proposed approach can generalize the classifier to be used for different materials without the need for training.

Convolutional neural network

A convolutional neural network (CNN) is a class of ANN that incorporates one or more convolutional layers into the architecture of the neural network for feature extracting, one or more pooling layers for subsampling, and then with one or more fully connected layers for the output. CNN is the most popular deep learning (DL) algorithm for image recognition, image classification, and object detection due to its outstanding

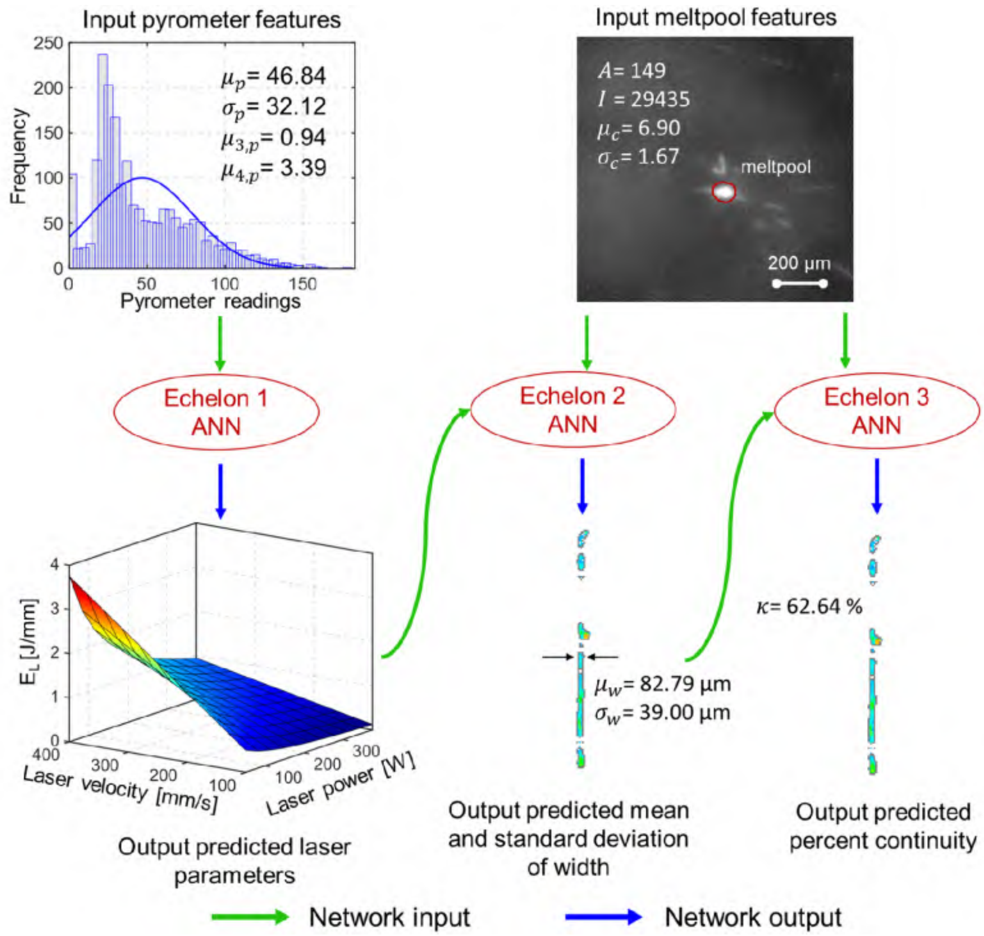


Figure A.9: A schematic of the sequential decision analysis neural network (SeDANN). The sensor data and height map shown above belong to a single-track deposited at linear energy density (EL) of 0.33 (i.e., balling regime). The statistical probability distribution features extracted from the pyrometer are used in the first echelon ANN to predict the laser process parameters (P and V) followed by melt pool features derived from the high-speed video camera to predict the mean width and standard deviation and single-track continuity at higher echelons.

image processing and pattern recognition performance.

The CNN algorithm has been used for defect detection in the LBAM process. For example, Scime et al. utilized a CNN algorithm for autonomous anomaly detection and classification in LPBF [252]. The proposed model achieved six types of flaws detection (i.e., recoater hopping, recoater streaking, debris, super-elevation, part failure, incomplete spreading) with an overall accuracy of 97%. A case study validated the proposed algorithm's ability by printing a heat exchanger model with Inconel 718 powder. Fig. A.10 shows the flowchart of the ML technology, which is utilized by Scime et al. for defect detection. Baumgartl et al. developed a printing defect detection system by combining thermographic off-axis images with a DL-based CNN [23]. By using the depthwise-separable convolutions to reduce the dimension of channels into the neural network architecture, this technique achieved an accuracy of 96.80% for delamination and splatter defects. Furthermore, the model's architecture is small and therefore has reduced computational cost, which shows great potential to apply the proposed algorithms for online defect detection. Cui et al. utilized a CNN algorithm to inspect the defect for the product produced by LMD [60]. The materials in this research include AISI 304 stainless steel, AISI 316 stainless steel, Ti-6Al-4V, AlCoCrFeNi alloys, and Inconel 718 alloys. The proposed algorithm had an accuracy of 92.1% for defects, such as gas porosity, crack, and lack of fusion. Zhang et al. described a porosity prediction CNN model for the LBAM process [351]. With the melt pool image from the single track manufactured by sponge titanium powder, the algorithm had a 91.2% porosity detection accuracy and is able to predict the micropores below 100 μm. Moreover, Guo et al. presented a physical-driven CNN model to predict porosity on thin-wall structure from Ti-6Al-4V powder by using metal pool thermal images obtained from a pyrometer for the DED process [288]. When combining the data-driven feature from the pyrometer image with the physical feature from finite element analysis, the proposed algorithm had a reported 100% porosity prediction accuracy. Zhang et al. proposed an online monitoring method for LBAM products, which relies on a CNN algorithm [352]. The experiment results show that when using a small local image block, the classification accuracy for bonding quality (i.e., under-melt, normal, and over-melt) is 82%; when utilizing the full images, the model accuracy was reported as 100%.

Product quality guarantee can also be achieved with CNN algorithms for the LBAM processes. Yuan et al. built a CNN model to monitor the LPBF printing

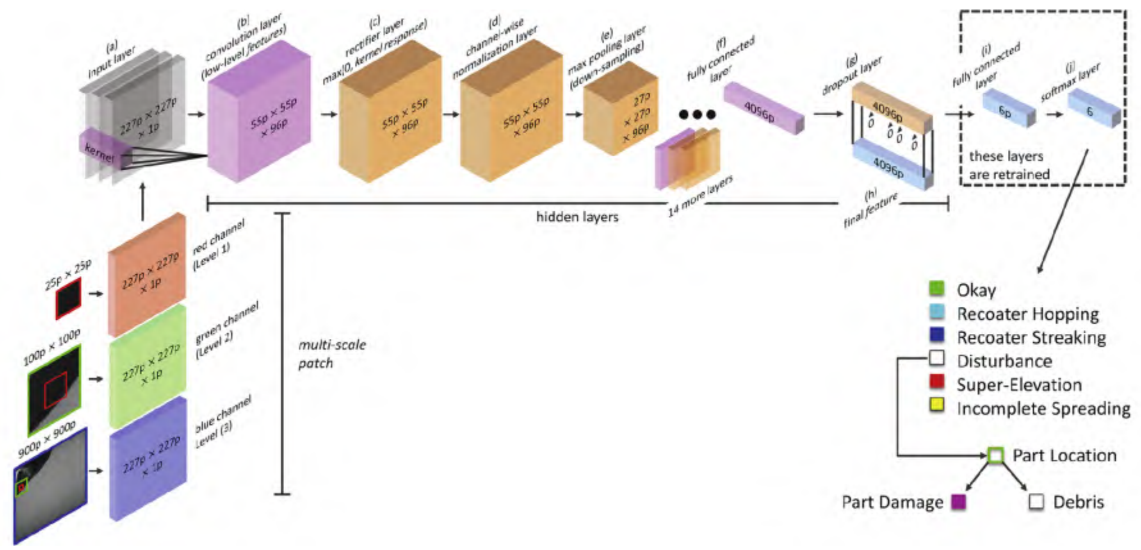


Figure A.10: Flowchart of the implementation of the multi-scale CNN ML technique.

process by analyzing the single track quality made with 316L stainless steel [347]. After training with the frames obtained from the melt pool video, this model is able to predict the track width, width standard deviations, and track continuity. Zhang et al. utilized CNN for product quality level identification in the LPBF process [355]. The input data is melt pool, plume, and spatter images obtained by a high-speed camera from the 8 mm single track printing process with stainless steel 316L powder. The proposed CNN model's quality level identification accuracy is 92.7%. Furthermore, one image identification only takes 8.01 milliseconds. A CNN-based approach was proposed by Gaikwad et al. to monitor and predict the thin-wall build quality with Ti-6Al-4V in LPBF [91]. After training with the layer-wise thin-wall optical images captured by an optical camera in the LBPF printer, the model can predict the X-ray computed tomography (XCT) derived statistical quality features, such as thickness and edge consistency, with an accuracy exceeding 85%. Kunkel et al. proposed a product quality assurance method for LBPF by utilizing a CNN-based image classification algorithm [154]. This CNN method had a promising quality identification accuracy of 98.9% for the product made with AlSi10Mg powder. Li et al. proposed a DL-based process monitoring for the DED technology [168]. The proposed CNN model used thermal images from the straight stick part manufactured with 0Cr18Ni9 powder as input, which had an accuracy of over 80% for different printing condition recognition (i.e., normal, lower power, low and high speed).

Deep convolutional neural network (DCNN) algorithm is a subclass of CNN, and therefore ANN, that utilizes more layers in the architecture. A DCNN is defined as one that utilizes at least 20 layers in its formulation in this work. There are several specific models for DCNNs with distinct structures, such as GoogleNet (also called InceptionNet), VGGNet, and ResNet. Generally, DCNN is better suited at detecting non-linear relationship and feature extraction than traditional CNN but require more data and longer training time than their more-traditional counterparts. Due to their capability to detect non-linear patterns, DCNN algorithms have been widely applied to detect defects in LBAM. Han et al. exploited defect detection in LBAM by utilizing a DCNN (Inception-v4) to analyze and classify images from the manufacturing process [114]. In this research, the DCNN model was applied to two datasets, metal fracture microscopic images and product parts of metal metallographic images. For the metal fracture microscopic image dataset, the model achieved an accuracy of 82%, recall of 93.8%, and precision of 96.8%; the model trained on the metal metallographic images, the accuracy, recall, and precision is 87.5%, 96.3%, and 97.6%. Li et al. built a novel CNN model structure based on a dense convolutional network to recognize the microstructure in LBAM [170]. The result shows the accuracy is 90.4% and one microstructure image processing time only needs 0.1 s. Gonzalez-Val et al. proposed a novel ConvLBM (CNN for laser-based manufacturing) method to estimate dilution in the LMD process [106]. Dilution is an important quality indicator: a low dilution may produce insufficient bound and generate warping, but a high value means a large heat-affected zone, which has a high defect probability because of the thermal expansion [184]. ConvLBM is a modified CNN model based on ResNet [116] to extract key features from the raw medium wavelength infrared coaxial images. The trained ConvLBM model estimated dilution with a root-mean-square error (RMSE) of 2.8%.

Ensembles of CNNs have also been developed to provide better fault detection capabilities than the traditional stand-alone CNNs. Shevchik et al. investigated combining spectral CNN with acoustic emission for quality monitoring in LBAM [260, 261]. The classification confidence varies between 83% and 89% on a cuboid with a dimension of 10 mm × 10 mm × 20 mm manufactured with stainless steel 316L. Xiao et al. developed a two-stage convolutional neural network (TS-CNN) to predict different kinds of defects in LBAM [328]. The research demonstrated that this approach had high accuracy and efficiency in coping with geometrical distortion manufactured with pure polyamide (PA) and PA/TiO2 composite. The respective prediction accuracy for the warpage, part shifting, and short feed defects is 94%, 96%, and 94%. Williams et al. developed a CNN model named Densely connected convolutional block architecture for multimodal image regression (DCB-MIR) to detect product defects in metal LBAM [314]. This proposed model utilized the printing product's SRAS(spatially resolved acoustic spectroscopy)-derived acoustic velocity maps as input data and decoded them to a resembling optical micrograph as output. The model had high defects detection accuracy (i.e., porosity and surface imperfections) for the titanium alloy and nickel alloy samples. The defects were not distinctly recognizable in the as-measured SRAS acoustic map and blurred in the optical image. Zhang et al. proposed a hybrid CNN to monitor the printing process in LPBF [356]. This hybrid CNN architecture consists of two CNN models; the first one is utilized to study the spatial features from a single printing process image and the second one is applied to product quality classification. The model's overall detection accuracy on the 10 mm single track manufactured with stainless steel 316L powder is up to 99.7% for the overheating, normal, irregularity, and balling defects. Scime et al. built a new dynamic segmentation CNN (DSCNN) model for real-time pixel-wise semantic segmentation with layer-wise powder bed images as input [254]. The proposed DSCNN model had a wide application range of powder-bed-based AM machines, spanning from LPBF, to electron beam PBF, to binder jetting. The model segmented and located the defect, include recoater hopping, recoater streaking, incomplete spreading, swelling, spatter, soot, debris, super-elevation, part damage, and porosity, with good accuracy. Yazdi et al. proposed a hybrid DL model to monitor the printing process parameters, which potentially affect the porosity defect production [339]. The input data is the statistical features extracted from powder bed images by wavelet transform and texture analysis. The proposed model had a 97.14% F-score porosity prediction accuracy on a cylinder (a height of 25 mm and a diameter of 20 mm) fabricated with titanium alloy and Ti-6Al-4V powder. By applying a bi-stream DCNN model, Caggiano et al. built an online defect recognition system for LBAM [40]. The pre-alloyed Inconel718 powder layer image and laser scanning layer image are input into an ML algorithm as training data to manifest the defect produced by improper process conditions. The study result shows that a mean defective condition-related image pattern recognition accuracy of 99.4% was achieved on the final disc with a 40 mm diameter and 20 mm height.

Novel ensembles of ML techniques integrate different algorithms together to enable the new one to take full advantage of the strengths from every integrated algorithm for detecting defects in LBAM, which researchers also utilize. For example, Francis et al. developed a novel DL approach named CAMP-BD (Convolutional and Artificial Neural Network for Additive Manufacturing Prediction using Big Data), which integrates CNN with an ANN algorithm for analyzing the thermal images. It used the relevant process/design parameters as input data to predict fabricated product distortion [84]. The whole CAMP-BD model is displayed in Fig. A.11. The distortion prediction result on a disk (5 mm thick and 45 mm diameter) made with Ti-6Al-4V powder, shows that most of the predictions are within the metal LBAM machines'

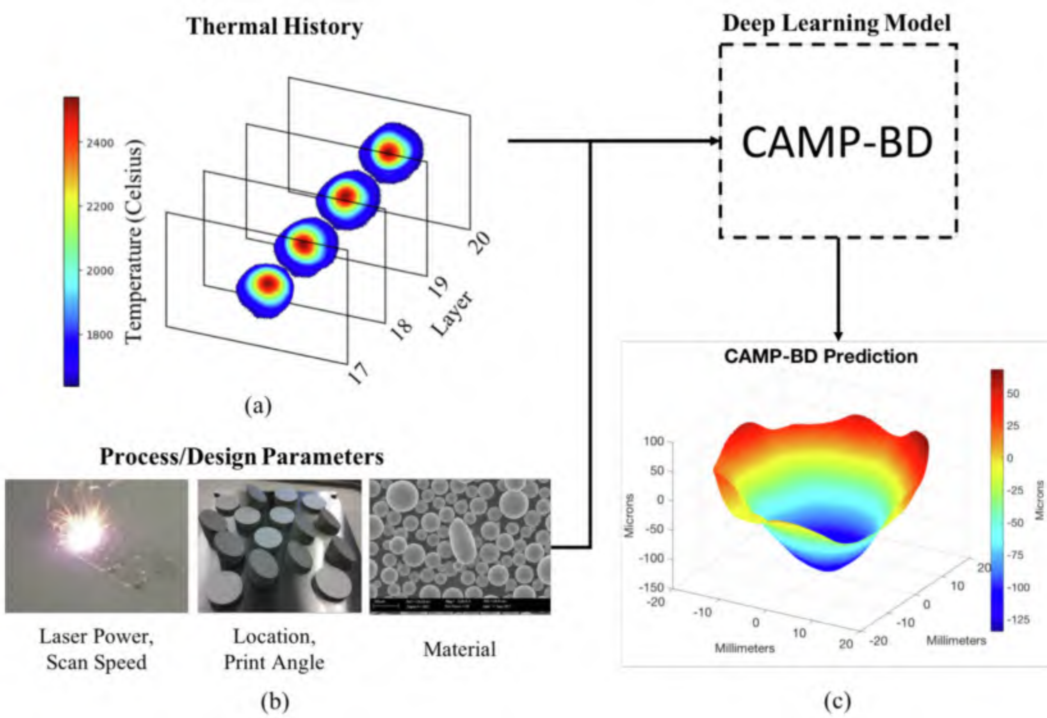


Figure A.11: Displaying the CAMP-BD model: (a) the tensor structure of thermal history is visualized; (b) various process/design parameters are listed as examples of additional inputs to CAMP-BD; and (c) an example prediction of CAMP-BD.

tolerance limits. The distortion prediction RMSE on training data is 24 μm. Tian et al. developed a DL-based in situ porosity detection method for LBAM, which uses the melt pool thermal images to monitor the melt pool and predict porosity [289]. A PyroNet based on CNN and an IRNet based on long-term RNN two different algorithms are utilized to correct layer-wise porosity. For the PyroNet, the input is in-process images obtained from pyrometry, and for the IRNet, the input data is sequential thermal images captured from a thermal camera. To have a higher porosity prediction accuracy, two algorithms are fused together when making the final decision. The experimental result on a Ti–6Al–4V thin-wall structure shows that the average porosity prediction accuracy on six-folds is 98.93%.

Table A.2: Unsupervised ML algorithms utilized in defect detection for LBAM processes.

ML category	ML algorithm	AM technology	Material type	Defect type	Dataset type	Accuracy	Reference
Unsupervised learning	K-means clustering	LPBF	AlSi10Mg, bronze, Inconel 625, Inconel 718, stainless steel 316L, Ti-6Al-4V, Fe-3Si, stainless steel 17-4 PH	Recoater hopping, recoater streaking, debris, super-elevation, part failure, incomplete spreading	Powder bed image	98%	[251]
		LPBF	Ti-6Al-4V, Inconel 718, Ti-5553, Haynes 282	Key hole, lack of fusion	X-ray computed 3D pore tomography, 2D pore micrograph	No specific accuracy	[264]

LPBF	Stainless steel 316L	Undesired overheating defect	Optical image	No specific accu-racy	[107]
LPBF	AlSi10Mg	Drift, no drift	Optical tomography image	No specific accu-racy	[331]
DED	Mixing Ti-6A1-4V with H13 tool steel	Crack, porosity	AE data	No specific accu-racy	[92]
DED	Ti-6Al-4V	Process condition variation	AE signal	87%	[280]

	DED	7075-Al alloy powder	Surface roughness porosity	Spectral	Refer to paper	[240]
Deep belief network	LPBF	Stainless steel 304	Balling, slight balling, normal, overheating, slight overheating	Acoustic signal	95.93%	[341]
	LPBF	Stainless steel 304L	Over-melted, middle over-melted, normal-melted, middle under-melted, under-melted	NIR image	83.40%	[340]
Self-organized maps	DED	Al-5083	Super-micropores, cracks, hybrid pores, inter-micropores defect	Optical microscope image	95%	[97]
	DED	Ti-6Al-4V	Porosity	Melt pool image	96%	[145]

Unsupervised learning

Unsupervised learning is a category of ML that looks for formerly undetected patterns in a dataset without pre-existing targets or labels and only involves minimal human supervision. Compared with supervised learning, which uses training data with labels, unsupervised learning, also termed self-organization, can model probability densities based on the input data. The advantage of unsupervised learning is that no labeled data is needed. Table A.4 shows a list of unsupervised ML algorithms utilized for detecting defects in components manufactured using the LBAM processes, which contains key features of the reviewed research, including dataset type, material type, defect type, and algorithm performance.

K-means clustering algorithm

The K-means clustering (KMC) algorithm is one of the simplest yet most powerful unsupervised learning algorithms for solving clustering problems. The procedure follows a simple method to classify a given dataset through a pre-defined number of clusters (assume k clusters) fixed apriori. The main idea is to define k centers, one for each cluster. Scime et al. presented an approach for in situ monitoring and analysis of powder bed images, which shows promising potential to be used as a real-time control system in the LPBF process [251]. A computer vision algorithm is used in this approach for automatic defect detection and classification during the powder spreading process. Defect detection and classification are implemented by using a standard k-means unsupervised clustering algorithm to process the printing image. The final algorithm achieved six types of flaw detection (i.e., recoater hopping, recoater streaking, debris, super-elevation, part failure, and incomplete spreading). Snell et al. trialed an unsupervised ML method (KMC) for rapid pore classification in metal additive manufacturing [264]. In this research, the pore data was collected by two different techniques: 3D pore data from the XCT and 2D pore data from microscopy, with multiple alloy (Ti-6Al-4V, Inconel 718, Ti-5553, and Haynes 282). The result shows that KMC is suitable for 3D pores, it performs well both on lack-of-fusion pores and keyholes pores, but it can not be used to analyze the 2D pore data. Grasso et al. developed an in-process defect spatial detection method by utilizing image data analysis for the LPBF process during the layer-wise printing process [107]. The proposed image KMC algorithm achieved defect detection automatically by detecting and positioning the potential defect in each layer. Taheri et al. developed an in situ process condition monitoring method for the DED process [280]. By utilizing the KMC to process the acoustic signatures from a single-layer (Ti-6Al-4V) printing process, the algorithm identified different process conditions (i.e., normal, low powder, and powder spray) with an accuracy of over 87%. In addition, Gaja et al. presented a novel real-time defect detection and classification system for the LMD process, which uses an acoustic emission sensor and a KMC unsupervised ML algorithm [92]. By analyzing the AE signal, the experimental result shows that the KMC is able to recognize crack and porosity simulated by mixing Ti-6Al-4V powder with H13 tool steel powder, two different defects effectively. Ren et al. built an unsupervised model for product quality recognition manufactured by DED [240]. The proposed model has an LSTM-based autoencoder for extracting encoded features which contain information to express the original spectra. Then the extracted features are utilized to K-means clustering for product quality classification. Four 40 mm tracks with Al7075 alloy are printed in the research, and the result shows that the classification matches with the real track quality.

Deep belief network

A deep belief network (DBN) is a generative graphical model, or alternatively a class of deep neural networks, which is composed of multiple latent variable layers, and each layer is connected, but the units within each layer are not related. DBN is another useful ML algorithm, which has been utilized for defect detection in LBAM. Ye et al. demonstrated that acoustic signals are feasible for product quality monitoring, and DBN algorithms achieved a high defect detection rate among five melted states on single tracks manufactured with 304 stainless steel powder (i.e., balling, slight balling, normal, slight overheating, and overheating) without signal preprocessing [341]. Fig. A.12 displays a generic classification-based DBN architecture that can be used for defect recognition with the acoustic signals from the LBAM printing process. As the plume and spatter signatures have a very close relationship with the melted state and laser energy density. Ye et al. applied DBN to the plume and spatter NIR images obtained from the LBAM printing process, and this approach achieved an 83.4% classification rate for over-melted, middle over-melted, normal, middle under-melted, and under-melted melted states on single tracks produced with 304 stainless steel powder [340].

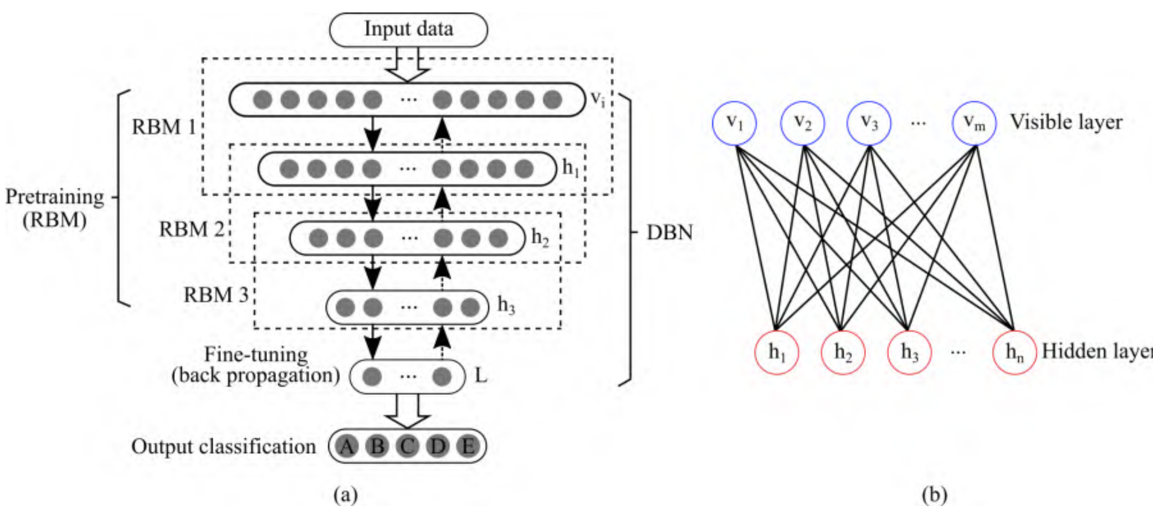


Figure A.12: An illustration of a generic classification-based deep belief network (DBN) with stacked RBMs (restricted Boltzmann machines): (a) deep belief network (DBN); and (b) restricted Boltzmann machine (RBMs).

Self-organized maps algorithm

Self-organized maps are a type of ANN trained by unsupervised learning to output a low-dimensional (generally two-dimensional), discretized representation of the input space of the training samples. As an effective dimensionality reduction method, self-organized maps are also applied to defect detection. Garcia-Moreno proposed an automatic porosity quantification method by utilizing an unsupervised ML classifier for the LMD printing process [97]. The final classifier based on self-organized maps obtained a 95% accuracy for the super-micropores, cracks, hybrid pores, and inter-micropores defect for the product manufactured with Al-5083 powders. The proposed algorithm is sensitive detecting the average size of 6.33 μm inter-micropores. Moreover, Khanzadeh et al. proposed an in situ porosity prediction method for the DED process based on the printing melt pool images [145]. According to the melt pool's temperature distribution, the proposed self-organized maps model predicted the porosity position on a Ti-6Al-4V thin-wall specimen with an accuracy of 96%.

Semi-supervised learning

Table A.3: Semi-supervised ML algorithms utilized in defect detection for LBAM processes.

ML categories	ML algorithm	AM technology	Material type	Defect type	Dataset type	Accuracy	Reference
Semi-supervised machine learning	K-nearest neighbors and K-means clustering combination	DED	AlSi10Mg	Drift, no-drift	Optical tomography image	No specific accuracy	[331]
	Gaussian Mixture model	LPBF	Inconel 718	Faulty, acceptable	Photodiode data	77%	[220]
	Semi-supervised convolutional neural network	DED	Stainless steel 316L	Track average width, track continuity	Video data	No specific accuracy	[346]
		LBAM	ASTM F75 I CoCrMo	Under-melt, beautiful-weld, over-melt	Micrograph image	90%	[167]

As discussed before, supervised learning uses a dataset that includes labels to train the algorithm to understand which features are important to the problem at hand. However, the generation of labeled datasets is a costly process, especially when dealing with large volumes of data. On the other hand, unsupervised learning is trained on unlabeled data and must identify features and determine their importance by themselves based on inherent data patterns. The disadvantage of any unsupervised learning is the limited application range. To counter these disadvantages, the concept of semi-supervised learning was introduced. In semi-supervised learning, the algorithm is trained upon a combination of labeled and unlabeled data. Typically, this combination contains a limited amount of labeled data and a larger amount of unlabeled data. This is helpful for a few reasons. First, the process of labeling massive amounts of data for supervised learning is often prohibitively time-consuming and expensive, however, a small amount of labeled data can significantly improve the learning accuracy when paired with a large volume of unlabeled data. Furthermore, too much labeling can impose human biases on the model. That means including lots of unlabeled data during the training process tends to improve the final model's accuracy while reducing the time and cost spent building it. In such conditions, semi-supervised learning can be of great practical value. In the research situation, as shown in Table A.3, semi-supervised learning is also helpful as there is no need to label all data for training, which saves time and effort.

Semi-supervised ML has been studied by researchers for defect detection. Okaroa et al. introduced a semi-supervised ML algorithm for automatic defect detection in the LBAM printing process [220]. The result on the tensile test bars manufactured with Inconel 718 shows that the semi-supervised ML algorithm is a promising method for automatically identifying the LBAM product defects, which achieved a comparable result to a benchmark where all training data are labeled. To detect the inline drift in the LPBF process, Yadav et al. proposed a semi-supervised method, which uses certified product computer tomography as input data [331]. The supervised ML-based KNN algorithm is trained with the labeled input data from the unsupervised ML K-means clustering algorithm. Based on the case studies validated on cylindrical samples (diameter of 10 mm and height of 15 mm) with AlSi10Mg powder, the proposed semi-supervised ML algorithm had a reported 100% accuracy in predicting the exact drift layers.

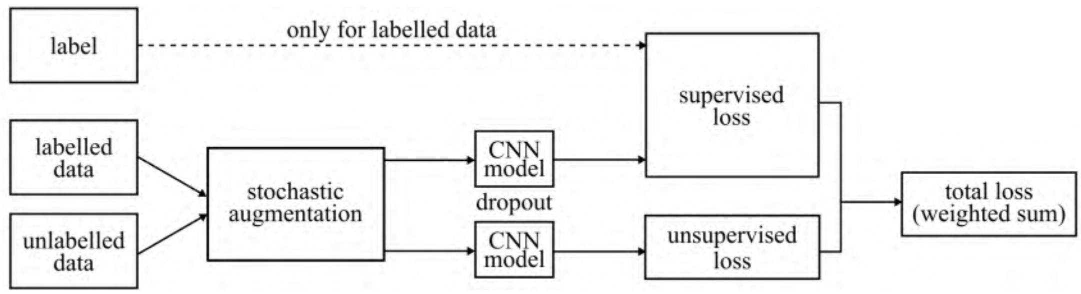


Figure A.13: Semi-supervised CNN architecture.

Semi-supervised CNNs eliminate the challenges of collecting large labeling datasets, making them more efficient for detecting defects in the LBAM processes. Yuan et al. developed a semi-supervised CNN, which only uses a limited amount of labeled data and a large amount of unlabeled data to monitor the LPBF printing process [346]. This approach used the videos from the single-track printing process with 316L stainless steel powder as input data. The result shows that the semi-supervised approach performs better than the fully supervised approach, no matter whether it is a regression or classification problem. The semi-supervised CNN model's architecture used in this research is shown in Fig. A.13. Li et al. proposed a DCNN model to analyze the product quality for the metal LBAM printing process [167]. In this method, the semi-supervised training data was used to mitigate the demand for a large amount of labeled image data. The algorithm proposed in this research had a 100% accuracy identification performance for under-melt, normal, and over-melt on the product made with ASTM F75 I CoCrMo powder.

Table A.4: Reinforcement learning algorithms utilized in defect detection for LBAM processes.

ML categories	ML algorithm	AM technology	Material type	Defect type	Dataset type	Accuracy	Reference
Reinforcement learning	Markov decision process	LBAM	Stainless steel 316L	Poor, medium, high quality	AE signal	74%, 79%, 82%	[312]
	Constrained Markov decision process	LBAM	Not mentioned	Defect states	Optical image	No specific accuracy	[338]

Reinforcement learning

Reinforcement learning (RL) is one of the three basic ML paradigms, alongside supervised learning and unsupervised learning. It is an area of ML, which is concerned with how agents ought to take actions in a specific environment to get the maximum cumulative reward. Therefore, RL also has been used as an approach for defect detection in LBAM, as shown in Table A.4. For example, Wasmer et al. integrated RL with acoustic data obtained from the printing process by acoustic emission to build quality monitoring in LBAM [312]. The classification accuracy obtained from a cuboid shape manufactured with stainless steel 316L shows that this RL approach had a high potential to be conducted in situ and in real-time. Furthermore, Yao et al. proposed a sensor-based product quality control model for the LBAM process through the constrained Markov decision process [338]. The experimental result shows that the proposed constrained Markov decision process provided an efficient policy for taking the right actions to remedy and mitigate incipient defects before the whole part is completed. Fig. A.14 shows the flow diagram of the in situ control sequential decision-making framework for the LBAM process through the constrained Markov decision process.

A.5 Outlook

An appropriate and effective defect detection system is a key driver for the development of next-generation technologies in LBAM. Their implementation will continue

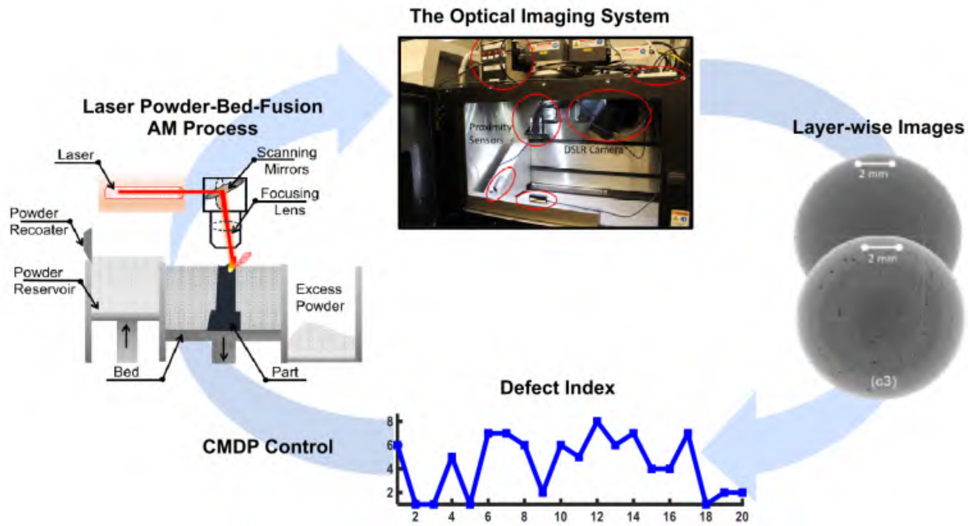


Figure A.14: Flow diagram of the sequential framework of constrained Markov decision process to control the quality of LBAM builds.

to increase the quality, efficiency, consistency, and sustainability of metal components manufactured using the LBAM processes [239, 306]. Future development for LBAM defect detection will likely be in the following four areas.

1) Integration of defect detection and product quality: The ultimate goal is to ensure the final product quality regardless of which algorithm is utilized for defect detection. In other words, the defect-to-property is the core for defect detection in LBAM. The defect detection should focus on detecting and evaluating the defect impact rather than just finding the defect. The decision-making system is also a critical part after defect impacting evaluation. Based on the evaluation result, the impacting defect level can be divided into negligible, marginal, and impactful. The printing process needs to be canceled to save energy, material, and time when an impactful defect occurs. For the decision-making system, the decision boundary from the machine learning algorithm has promising potential when combined with the defect impacting together.

2) Data fusion and advanced algorithms: Signals from the different processes vary significantly in terms of temporal and spatial resolution. Sensors with different ac-

curacies and resolutions need to be synchronized to handle different characteristics of the signal features, which provide a holistic understanding of the printing process. The improvement for the future advanced algorithms has two sides. (1) Fusion algorithms include in-process signal processing and defect detection. Most of the existing machine learning algorithms can only handle one type of input data. However, the defect has many influence factors and performance characteristics. Just detecting defects from only one dimension is not comprehensive and accurate. The advanced algorithm should have the ability to handle multi-type input at the same time. Hybrid machine learning and deep learning will play an increasingly important role in this field. (2) The development of the latest algorithms and their application. Hundreds and thousands of new algorithms are developed nowadays with new features and abilities. This development will significantly broaden the defect detection perspective in LBAM. For example, physics-informed ML will enable the integration of most recent understanding of defect formation mechanisms to reduce the data requirement and increase the detection accuracy [141].

3) Real-time feedback control and correction system that couples sensor data, computational models with advanced algorithms: The ways to mitigate the defects impacting can be done by correcting them when the defects are initially detected or when the defects are predicted before they occur. Therefore, real-time feedback control and correction systems and the predictive capability of the algorithms are essential, which should fuse various sensor data, computational models with advanced algorithms together to achieve this goal. Reconstruct 3D objects by the machine learning algorithm based on the different input signals, where physics-based computational models can supplement the experimental data with future status. Then mapping the detected or predicted defects to the printing code. According to the defect information, correct the correspondent code (e.g., laser speed, laser power, and the axis movement) to mitigate the next printing flaws. This may be a feasible way,

which has been proven on the other printing technology [6]. In the foreseeable future, the LBAM system will be more widely applied by adding an effective defect detection and feedback system to mitigate printing flaws [197, 250].

4) More compatible system: Active work has been performed in defect detection for LBAM; however, the algorithm is mostly specific to one defect and limited to one specific printer. As the development of a new detection system for another printing technology or another printer, even another material is time and effort consuming. Thus, a more unified and compatible framework that flexibly integrates available sensors, algorithms, and computational modeling can be transferred to the other system is important, which certainly will accelerate technology advancement. For example, transfer learning [344], which can boost the speed and accuracy of the training process, will play a significant role in compatible system development.

A.6 Conclusion

This paper summarizes recent research focused on the machine learning (ML) algorithms used in defect detection systems for the metal laser-based additive manufacturing (LBAM) processes. The comprehensive and exhaustive information listed in the paper provides a reference to help readers choose a suitable ML algorithm for detecting defects based on different printing technologies, material types, defect types, and data types.

Choosing a suitable ML algorithm is critical to achieving the appropriate level of defect detection in any LBAM system. A summary of commonly chosen ML algorithms is presented in this work. Convolutional neural networks (CNN) are the first choice when dealing with image data due to their unique advantages of processing image features. Using powder bed images, product layerwise optical images, or melt pool images, CNNs can be utilized for detecting various defects or product quality levels. Support vector machines using the alternative kernel function are a good choice for classification and are capable of handling both sensor signal and image data to perform both binary and multi-class classification. K-means clustering is a widely utilized algorithm in unsupervised and semi-supervised ML to partition the observations into different defect clusters. The Markov decision process is the most common algorithm for reinforcement learning, which performs well on both sensor signal and image data to detect defects and to evaluate product quality.

From the listed algorithms utilized in defect detection for LBAM processes, supervised machine learning algorithms are the most common. However, as supervised machine learning requires that all data be labeled it is time and effort-consuming. Algorithms that leverage unsupervised and semi-supervised learning have started to gain traction in the field and have demonstrated great potential to expand the field of fault detection in LBAM processes. Moreover, reinforcement learning is also being explored for defect detection in the LBAM process and offers the potential to develop accurate and highly efficient fault detection models.

While machine learning has its unique advantages; the application prerequisite is rigid. The small databases typically available for manufacturing systems usually results in over fit models that lead to poor fault detection accuracy. A considerable barrier for the real-world adoption of machine learning in defect detection of LBAM is the lack of a comprehensive, precise, and accessible databases for different materials, designs, and printing processes. A solution to this challenge is the development of a standard database with uniform design and fault criteria. Such a database would provide a large volume of accessible data for researchers. By leveraging transfer learning off of a massive database, reliable fault detection methodologies could be developed without the need of generating large specific datasets for each part or process.

It is worth pointing out the disproportionality in machine learning approaches for LBAM defect detection. Most of the literature's machine learning applications focus on directed energy deposition and laser powder bed fusion; no research has been done on laser-based wire-feed systems.

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In situ monitoring for fused filament fabrication process: A review

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Real-time structural validation for material extrusion additive manufacturing

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Machine learning algorithms for defect detection in metal laser-based additive manufacturing: A review

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